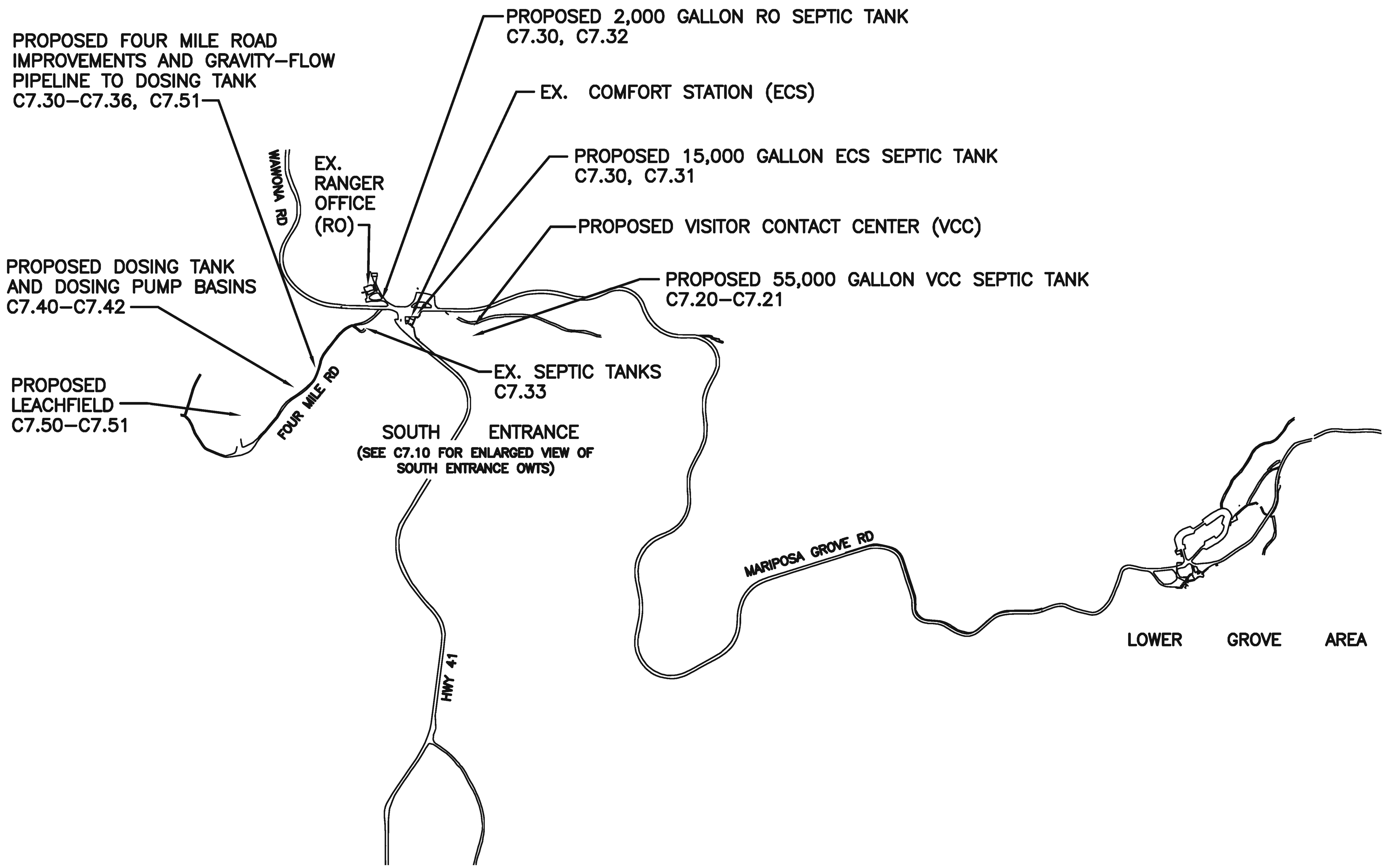

ONSITE WASTEWATER TREATMENT SYSTEM (OWTS) PLANS SHEET INDEX	
C7.00	KEY PLAN
C7.01	DESIGN CAPACITY
C7.02	DESIGN CAPACITY
C7.10	OVERALL SITE PLAN
C7.20	VISITOR CONTACT CENTER (VCC) SITE PLAN
C7.21	VISITOR CONTACT CENTER (VCC) SEPTIC TANK
C7.30	EXISTING COMFORT STATION (ECS) & RANGER OFFICE (RO) SITE PLAN
C7.31	EXISTING COMFORT STATION (ECS) SEPTIC TANK
C7.32	RANGER OFFICE (RO) SEPTIC TANK
C7.33	FOUR MILE ROAD
C7.34	FOUR MILE ROAD
C7.35	FOUR MILE ROAD
C7.36	FOUR MILE ROAD
C7.40	PUMP & TANK YARD
C7.41	DOSING TANK
C7.42	DOSING PUMP BASINS
C7.50	LEACHFIELD
C7.51	LEACHFIELD
C7.52	LEACHFIELD DETAILS
C7.53	LEACHFIELD DETAILS
C7.60	GENERAL NOTES
C7.61	GENERAL NOTES
C7.62	GENERAL NOTES
C7.63	GENERAL NOTES
C7.64	GENERAL NOTES
C7.70	SOILS INFO
C7.71	SOILS INFO
C7.72	SOILS INFO
C7.80	MONITORING WELLS

ONSITE WASTEWATER TREATMENT SYSTEM PLAN ABBREVIATIONS	
CVRWQCB	CENTRAL VALLEY REGIONAL WATER QUALITY CONTROL BOARD (CAL–EPA, FRESNO OFFICE)
ECS	EXISTING COMFORT STATION
EHS	ENVIRONMENTAL HEALTH SERVICES (MARIPOSA COUNTY)
FMR	FOUR MILE ROAD
MRC	MINIMUM RELATIVE COMPACTION
OWTS	ONSITE WASTEWATER TREATMENT SYSTEM
RO	RANGER OFFICE
VCC	VISITOR CONTACT CENTER
WW	WASTEWATER

500 0 500 1000

SCALE OF FEET

PHASE 5

Thomas G. Horgan

REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA

MARCH 14, 2016

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PRIME:
DAVID EVANS AND ASSOCIATES, INC.
BELLEVUE, WA

SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO.
C7.00

TITLE OF SHEET
SOUTH ENTRANCE
OWTS PLANS
KEY PLAN
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662

PMIS/PKG NO. 206431

SHEET 92 OF 291

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TABLE 1

DESIGN WASTEWATER FLOWS

YOSEMITE NATIONAL PARK — MARIPOSA GROVE RESTORATION PROJECT

	SITE	VISITORS PER 8-HR DAY	VISITOR CONTACT CENTER AND COMFORT STATION USERS PER 8-HR DAY	DESIGN CAPACITY TOTAL GPD
1	**SOUTH ENTRANCE STATION:**			
	A. RANGER OFFICE			200
	B. PEOPLE ENTERING PARK FROM SOUTH USING REHABILITATED EXISTING COMFORT STATION (2)	6,800 (3)	1,643 (4)	3,890
	C. NEW VISITOR CONTACT CENTER	5,202 (5)	5,202 (6)	15,830
	SITE TOTAL (13)			**19,920**
2	**LOWER GROVE AREA:**			
	NEW COMFORT STATION	5,202 (7)	66% = 3,434 (7)	7,360
	SITE TOTAL (14)			**7,360**
3	**UPPER GROVE AREA:**			
	REHABILITATED COMFORT STATION (10)	472 (8)	472 (8)	1,117
	SITE TOTAL (13)			**1,117**
TOTAL ALL SITES (13)				**28,397**

TABLE 2

NOTES & ASSUMPTIONS FOR TABLES 1, 3 AND 4:	
(1)	PER 2013 CALIFORNIA GREEN BUILDING STANDARDS CODE.
(2)	ALL EXISTING COMFORT STATION FIXTURES TO BE REPLACED WITH LOW-FLOW FIXTURES. SEE NOTE 1.
(3)	ESTIMATE OF PEAK DAY VISITORS ENTERING THE PARK FROM THE SOUTH IS BASED ON RECORD OF 5 BUSIEST DAYS IN 2011: MAY 28 = 6,799; JUNE 25 = 6,804; AUGUST 6 = 6,699; AUGUST 13 = 6,734; AND AUGUST 26 = 6,799.
(4)	PER JULY 2015 INSTRUCTIONS FROM YNP, 1,643 OF THE VISITORS ENTERING THE PARK FROM THE SOUTH WILL MAKE A BRIEF STOP TO USE THE EXISTING COMFORT STATION AND THEN CONTINUE TRAVELING ON TOWARD WAWONA.
(5)	MARIPOSA GROVE (MG) PEAK DAY VISITATION PER YNP INSTRUCTIONS.
(6)	ALL MG VISITORS ARE ASSUMED TO PASS THROUGH THIS GROVE ENTRANCE AND TO USE THIS VISITOR CONTACT CENTER AN AVERAGE OF 1.5 TIMES DURING THEIR STAY IN MG. USE OF THIS VISITOR CONTACT CENTER WILL BE ENCOURAGED BY SIGNAGE.
(7)	ALL MG VISITORS ARE ASSUMED TO VISIT THE LOWER GROVE AREA AND 66% OF THEM ARE ASSUMED TO USE THIS COMFORT STATION AN AVERAGE OF ONE TIME DURING THEIR VISIT.
(8)	PER JULY 2015 INSTRUCTIONS FROM YNP, 472 VISITORS (40% MEN, 60% WOMEN) ARE ASSUMED TO VISIT UPPER GROVE AREA AND ALL OF THEM ARE ASSUMED TO USE THIS COMFORT STATION AN AVERAGE OF ONE TIME DURING THEIR VISIT.
(9)	THE YNP STAFF ON 03/06/13 RECOMMENDED, BASED ON ITS HISTORICAL RECORD OF ACTUAL UNIT WASTFLOWS, THAT THE MINIMUM RECOMMENDED DESIGN WASTEFLOW BE BASED ON 5 GPD PER MG VISITOR TO ALLOW FOR WATER WASTE BY MALFUNCTIONING TOILETS AND URINALS. 5,202 PEAK DAY VISITORS @ 5 GPD = 26,010 GPD.
(10)	UGA COMFORT STATION RETROFIT OF EXISTING FIXTURES: ALL SINKS TO BE REMOVED; WATERLESS URINALS AND LOW-FLOW TOILETS TO BE INSTALLED.
(11)	THE FRACTION OF THE TOTAL WASTEWATER VOLUME COLLECTED AT VAULT TOILETS AND TRUCKED TO THE WAWONA WWTP IS CONSERVATIVELY OMITTED FOR OWTS DESIGN PURPOSES.
(12)	ALTHOUGH URINALS WILL BE WATERLESS TYPE, YNP AGREED TO THIS CONSERVATIVE ESTIMATE.
(13)	THESE DESIGN WASTEWATER FLOWS ARE BASED ON YNP PROJECTED VISITOR COUNTS AND DISTRIBUTION OF VISITORS TO THE FLUSH TOILET FACILITIES. SEE DTL. 1/C7.02 FOR PEAK-DAY HOURLY DISTRIBUTION OF VISITORS TO THE SOUTH ENTRANCE EXISTING COMFORT STATION (ECS) AND TO THE PROPOSED VISITOR CONTACT CENTER (VCC).
(14)	PREVIOUS YNP-APPROVED DESIGN CAPACITY. REFER TO MG PH. 4 LGA OWTS DWGS. C6.01 AND C6.02.

PHASE 5

Thomas G. Horgan

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TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS DESIGN CAPACITY
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 / 127662
PMIS/PKG NO. 206431
SHEET 93 OF 291

24-HOUR UNIFORM DOSING (GAL) (1)(3)						
HOUR	BEGINNING TANK VOLUME	SEPTIC TANK EFFLUENT % (4)	SEPTIC TANK EFFLUENT GALLONS	BEGINNING VOL. + EFFLUENT	VOLUME DOSED	ENDING TANK VOLUME
1:00	5332	0.15%	30	5362	830	4532
2:00	4532	0.15%	30	4562	830	3732
3:00	3732	0.15%	30	3762	830	2932
4:00	2932	0.15%	30	2962	830	2132
5:00	2132	0.20%	40	2171	830	1341
6:00	1341	0.30%	60	1401	830	571
7:00	571	1.30%	259	830	830	0
8:00	0	5.30%	1056	1056	830	226
9:00	226	13.90%	2769	2995	830	2165
10:00	2165	10.40%	2072	4236	830	3406
11:00	3406	9.70%	1932	5339	830	4509
12:00	4509	9.10%	1813	6321	830	5491
13:00	5491	8.80%	1753	7244	830	6414
14:00	6414	9.70%	1932	8347	830	7517
15:00	7517	10.70%	2131	9648	830	8818
16:00	8818	9.70%	1932	10750	830	9920
17:00	9920	7.30%	1454	11374(5)	830	10544
18:00	10544	2.10%	418	10963	830	10133
19:00	10133	0.15%	30	10163	830	9333
20:00	9333	0.15%	30	9362	830	8532
21:00	8532	0.15%	30	8562	830	7732
22:00	7732	0.15%	30	7762	830	6932
23:00	6932	0.15%	30	6962	830	6132
24:00	6132	0.15%	30	6162	830	5332
Total/day		100.0%	(2)19920		19920	

NOTES

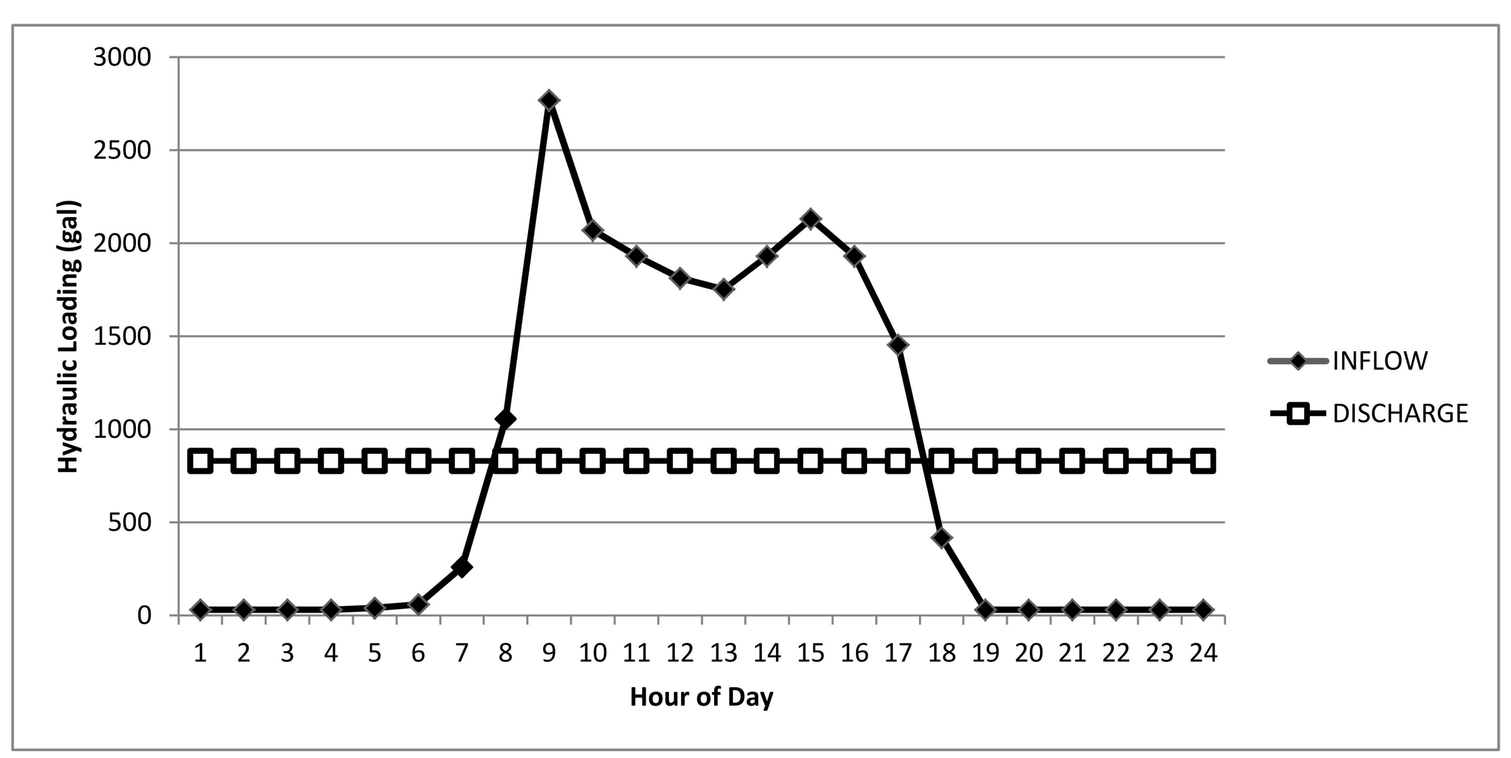
(1) ALL VOLUMES ARE ROUNDED TO THE NEAREST GALLON, SO SUM OF INDIVIDUAL VOLUMES MAY NOT EQUAL TOTAL.

(2) REVISED PEAK DAY DESIGN FLOW APPROVED BY YNP ON AUGUST 26, 2015.

(3) OPERATIONAL VOLUME – TANK DEAD VOLUME NOT INCLUDED.

(4) HOURLY VISITOR DISTRIBUTION PER YNP'S FEB. 20, 2015 FURNISHED DATA. PER YNP INSTRUCTIONS, THIS DISTRIBUTION IS TO BE USED TO SIZE THE DOSING SYSTEM'S OPERATIONAL VOLUME.

(5) DOSING SYSTEM OPERATIONAL STORAGE VOLUME REQUIRED FOR FLOW EQUALIZATION IS 11,374 GALLONS. SEE CONSTRUCTION NOTE 24/C7.41.



PEAK DAY DOSE CALCULATIONS

(1) 19,920 GPD ÷ 24 ZONES = 830 GPD/ZONE.

(2) 830 GPD/ZONE ÷ 2 DOSES/DAY = 415 GAL/DOSE PER ZONE.

(3) 830 GAL/HOUR UNIFORM DISTRIBUTION ÷ 415 GAL/DOSE = 2 DOSES/HOUR.

(4) 24 ZONES X 2 DOSES/DAY-ZONE = 48 TOTAL DOSES/DAY.

(5) 415 GAL/DOSE ÷ 55 GPM = 7.5± MINUTES/DOSE PUMPING

1 – DOSING TANK & PUMP DESIGN INFO
NOT TO SCALE

TABLE 3

YOSEMITE NATIONAL PARK — WASTEWATER FLOW ESTIMATES — METHOD 1: VISITOR FIXTURE USE — MARIPOSA GROVE RESTORATION PROJECT

SITE	VISITORS PER 8-HR DAY	VISITOR CONTACT CENTER AND COMFORT STATION USERS PER 8-HR DAY	URINALS: USES PER STAY	URINALS: GALLONS @ 0.5 GAL/USE (1)(12)	TOILETS: USES PER STAY	TOILETS: GALLONS @ 1.28 GAL/USE (1)	SINKS: USES PER STAY	SINKS: GALLONS @ 0.2 GAL/USE (1)	SUBTOTAL GPD	TOTAL GPD
1. SOUTH ENTRANCE STATION:										
D. RANGER OFFICE									200	
E. PEOPLE ENTERING PARK FROM SOUTH USING REHABILITATED EXISTING COMFORT STATION (2)	3,400 MEN	756	0.5	189	0.5	484	1	151	824	
	3,400 WOMEN	887			1	1,135	1	177	1,312	
	6,800 TOTAL (3)	1,643 (4)							2,136	
F. NEW VISITOR CONTACT CENTER	2,601 MEN	100% = 2,601	1	1,300	0.5	1,665	1.5	780	3,745	
	2,601 WOMEN	100% = 2,601			1.5	4,994	1.5	780	5,774	
	5,202 TOTAL (5)	5,202 (6)							9,519	
SITE TOTAL										11,855
2. LOWER GROVE AREA:										
NEW COMFORT STATION	2,601 MEN	66% = 1,717	0.5	429	0.5	1,099	1	343	1,871	
	2,601 WOMEN	66% = 1,717			1	2,198	1	343	2,541	
	5,202 TOTAL (7)	3,434 (7)							4,412	
SITE TOTAL										4,412
3. UPPER GROVE AREA:										
REHABILITATED COMFORT STATION (2)	472 X 40% = 189 MEN	100% = 189	0.5	47	0.5	121	1	38	206	
	472 X 60% = 283 WOMEN	100% = 283			1	362	1	57	419	
	472 TOTAL (8)	472 (8)							625	
SITE TOTAL										625
TOTAL METHOD 1										16,892

TABLE 4

YOSEMITE NATIONAL PARK — WASTEWATER FLOW ESTIMATES — METHOD 2: 5 GPD PER VISITOR — MARIPOSA GROVE RESTORATION PROJECT

SITE	VISITORS PER 8-HR DAY	VCC & COMFORT STATION USERS PER 8-HR DAY	GPD	TOTAL GPD
MARIPOSA GROVE SE VCC & LGA COMFORT STATION ONLY (EXCLUDES SE RANGER OFFICE AND EXISTING SE COMFORT STATION)	5,202 (5)	5,202	5,202 VISITORS @ 5 GPD (9)	26,010
TOTAL METHOD 2				26,010

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BELLEVUE, WA

SUBCONTRACTOR
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WOODLAND, CA

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DATE: 2/26/2016

SUB SHEET NO. C7.02

PHASE 5

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS DESIGN CAPACITY
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 94 OF 291

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CONSTRUCTION NOTES

1. THE CONTRACTOR SHALL INITIATE COMMUNICATIONS WITH PG&E CONCERNING PROPOSED CONSTRUCTION IN THE VICINITY OF THE WAWONA ROAD - SH41 INTERSECTION AND IN THE WESTERLY LEACHFIELD AREA, NEAR THE EXISTING PG&E POLE LINE. COORDINATE WITH THE COR.
2. COORDINATE TREE PRESERVATION AND TREE REMOVAL WITH THE COR.
3. YNP IS PLANNING TO THIN THE TREES ALONG THE SOUTHEASTERLY SIDE OF FOUR MILE ROAD IN THE VICINITY OF THE PROPOSED LEACHFIELD.

PHASE 5

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SUB SHEET NO. C7.10

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS OVERALL SITE PLAN
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 95 OF 291

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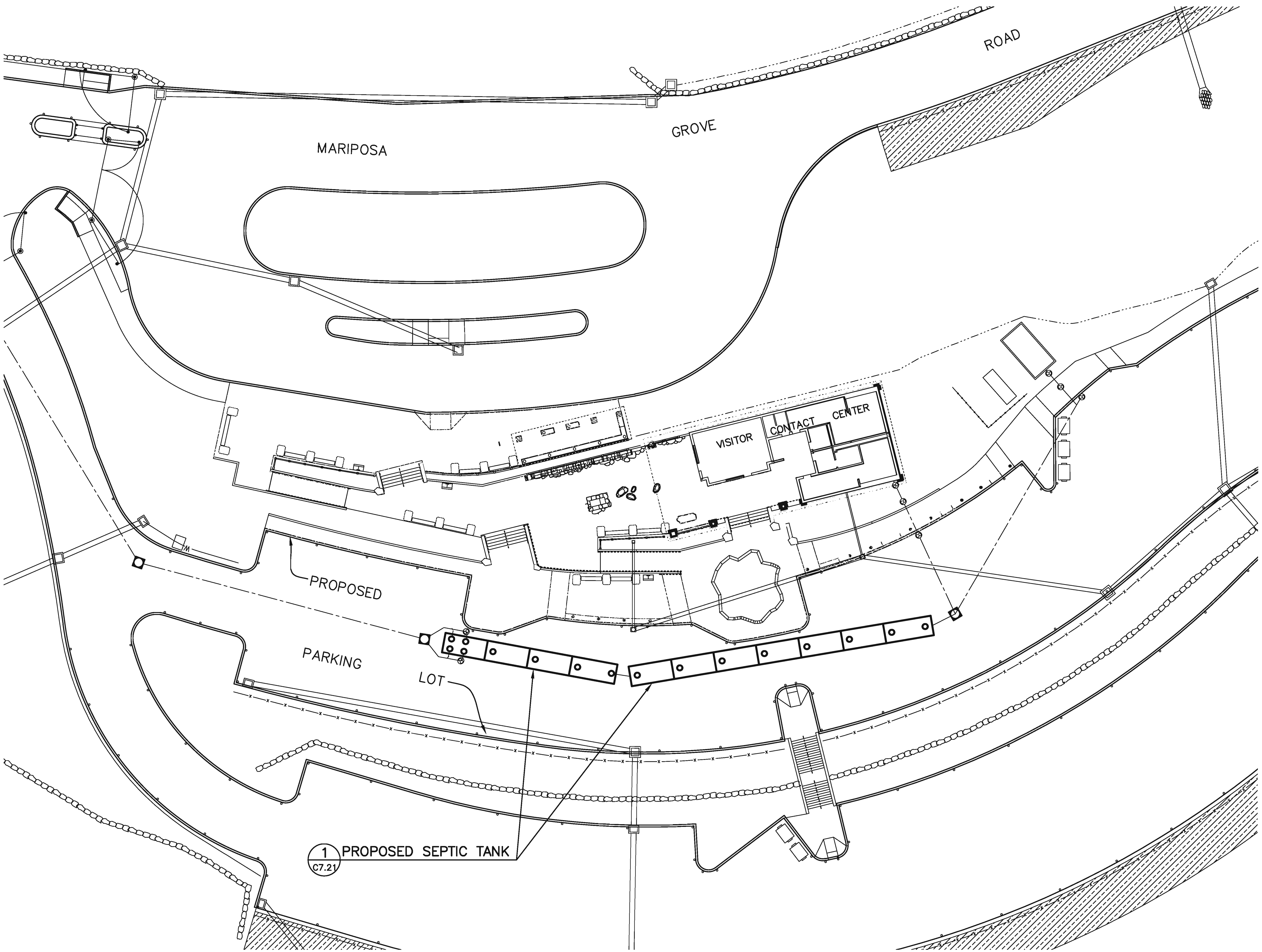

GENERAL NOTES

1. SEE SHEET C7.21 FOR SEPTIC TANK CONSTRUCTION DETAILS.
2. SEE SHEET C5.53 FOR OTHER VCC AREA PROPOSED UTILITIES.

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WOODLAND, CA

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SUB SHEET NO. C7.20

TITLE OF SHEET

SOUTH ENTRANCE OWTS PLANS VCC SITE PLAN

YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662

PMIS/PKG NO. 206431

SHEET 96 OF 291

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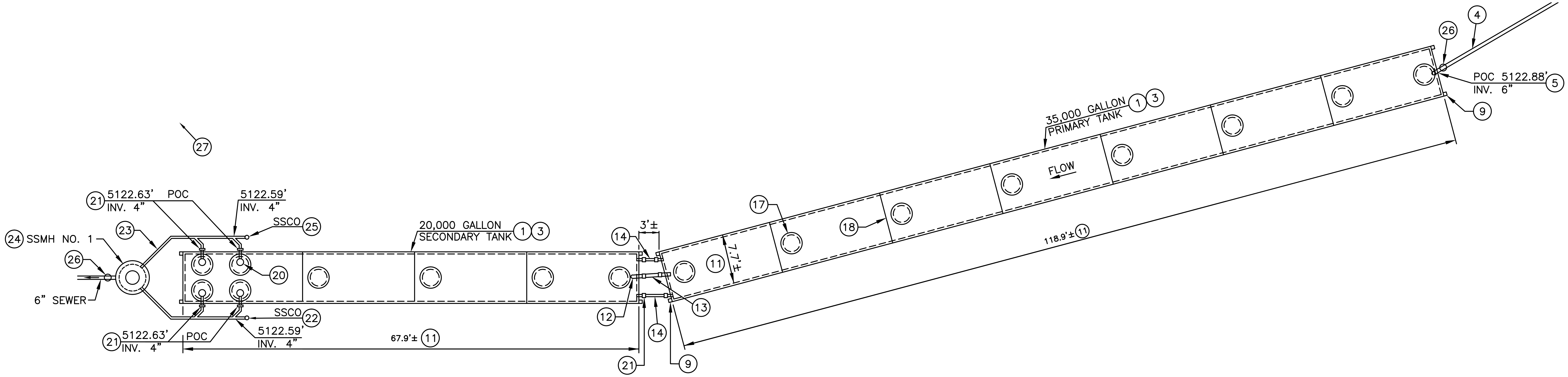
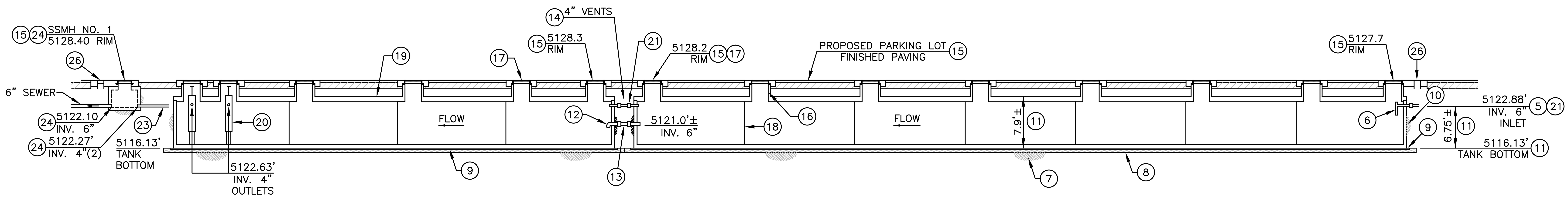


PLAN VIEW (2)(3)



SIDE VIEW

CONSTRUCTION NOTES

(1) SEE THE OWTS TANKS GENERAL NOTE 21 ON SUB-SHEETS NOS. C7.62 AND C7.63 FOR ADDITIONAL SEPTIC TANK CONSTRUCTION AND INSTALLATION REQUIREMENTS, INCLUDING INTERIOR PROTECTIVE COATING AND BUOYANCY COUNTERACTION.

(2) SEE SUB-SHEET C5.53 FOR VISITOR CONTACT CENTER AREA OVERALL UTILITY PLAN.

(3) INSTALL 55,000 GALLON PRECAST CONCRETE SEPTIC TANK. TANK CONFIGURATION SHOWN ON THIS PLAN IS BASED ON (1) A 35,000 GALLON PRIMARY TANK CONSISTING OF A BATTERY OF SEVEN JENSEN PRECAST MODEL JZ5000 TANKS, AND (2) A 20,000 GALLON SECONDARY TANK CONSISTING OF A BATTERY OF FOUR JENSEN PRECAST MODEL JZ5000 TANKS. AN ENGINEER-APPROVED EQUIVALENT SEPTIC TANK COMPLYING WITH THESE PLANS IS ACCEPTABLE FOR INSTALLATION.

(4) SEE SUB-SHEETS C5.53 AND C7.20 FOR 6" SDR 26 PVC SEWER PIPELINE FROM RESTROOMS. VERIFY PIPELINE SIZE, MATERIAL, LOCATION AND DEPTH.

(5) POINT OF CONNECTION (POC) TO TANK 6" INLET OPENING.

(6) INSTALL 6" PVC SANITARY INLET TEE. ALL PIPE-TO-TANK CONNECTIONS TO BE WATERTIGHT.

(7) EARTH SUBGRADE SHALL BE CONSTRUCTED TO PROPER LINE AND GRADE IN FIRM UNDISTURBED SOIL. IF SOIL IS NOT FIRM AND UNDISTURBED, PREPARE SUBGRADE AND COMPACT TO 95% MINIMUM RELATIVE COMPACTION (MRC) – SEE PROJECT EARTHWORK SPECIFICATIONS.

(8) LEVEL, MINIMUM 8" LAYER OF 3/4" CLASS 2 AB AT 95% MRC.

(9) IF REQUIRED BY THE TANK MANUFACTURER, INSTALL 2" X 6" REDWOOD GRADE BOARDS LENGTHWISE UNDER SIDES OF TANKS. EXTEND BOARDS 1'± BEYOND TANK ENDS.

(10) COMPACTED EARTH BACKFILL AT 95% MINIMUM RELATIVE COMPACTION. INSTALL TANK IN ACCORD WITH MANUFACTURER'S RECOMMENDATIONS.

(11) VERIFY ALL TANK DIMENSIONS WITH TANK MANUFACTURER AND COORDINATE WITH THE COR AND THE DESIGN ENGINEER.

(12) 6" IPS ASTM D2466 SCH. 40 PVC PIPE INVERTED ELBOW FITTING.

(13) INSTALL 6" IPS ASTM D1785 SCH. 40 SOLVENT-CEMENTED OR SDR 26 PVC PIPE WITH 6" COUPLINGS – SEE NOTE 21.

(14) INSTALL 4" SDR 26 PVC SEWER PIPE FOR TWO VENT PIPES.

(15) SEE SUB-SHEETS C5.02, C5.32 AND C5.42 FOR PROPOSED PAVING STRUCTURAL SECTION THICKNESS AND TO VERIFY FINISHED PAVING GRADES. CAREFULLY CONSTRUCT PAVING STRUCTURAL SECTION AND SUBGRADE TO PREVENT DAMAGE TO THE TANK AND PIPING.

(16) INSTALL AASHTO HS20-44 HEAVY-TRAFFIC RATED, 30" INSIDE DIAMETER, PRECAST CONCRETE RISERS, CAST-IRON FRAMES AND COVERS (TYP.). SEE GENERAL NOTE 21.J/C7.63 FOR RISER INTERIOR PROTECTIVE COATING. SEE GENERAL NOTE 21.G/C7.63 FOR FRAME AND COVER REQUIREMENTS, INCL. PROTECTIVE COATING.

(17) EACH FRAME AND COVER ASSEMBLY IS TO BE INSTALLED WITH A CAST-IN-PLACE, HEAVY-DUTY, REINFORCED CONCRETE SNOW PLOW PROTECTIVE APRON. SEE DETAILS 1/D5.21 AND 2/D5.21. SEE NOTE 26.

(18) JOB SITE JOINT SEALING OF SECTIONAL TANKS SHALL COMPLY WITH MANUFACTURER'S RECOMMENDATIONS.

(19) THICKEN/REINFORCE TOP OF THIS TANK SECTION AS MAY BE REQUIRED TO ACCOMMODATE THE MULTIPLE OPENINGS FOR EFFLUENT FILTERS.

(20) INSTALL FOUR EFFLUENT FILTERS – SEE GEN. NOTE 22/C7.63. WHEN FILTERS ARE CLEANED, THE "WASH-OFF" SHALL BE DRAINED INTO THE PRIMARY COMPARTMENT OF THIS SEPTIC TANK.

(21) INSTALL MISSION RUBBER CO. FLEX-SEAL ADJUSTABLE REPAIR COUPLING, OR APPROVED EQUIVALENT, WITH SS SHEAR RING (TYP.).

(22) INSTALL ONE-WAY SSCO PER DTL. 2/D5.22 IN AASHTO HS20-44 HEAVY-TRAFFIC RATED UTILITY BOX WITH CAST-IRON FRAME AND COVER. COVER TO BE LABELED "SEWER". INSTALL WITH CAST-IN-PLACE HEAVY-DUTY, REINFORCED CONCRETE SNOW PLOW PROTECTIVE APRON PER DTL 1/D5.21, SET TO GRADE.

(23) INSTALL 4" SDR 26 PVC SEWER PIPELINE TO SSMH NO. 1. NO FITTING ANGLES GREATER THAN 45° PERMITTED. SEE SUB-SHEET C7.20 AND DTL. 3/D5.11.

(24) SEE SUB-SHEET C5.53 FOR SSMH NO. 1.

(25) INSTALL ONE-WAY SSCO. SEE NOTE 22 AND COORDINATE SSCO LOCATION AND BOX INSTALLATION WITH PROPOSED CONCRETE WALKWAY LAYOUT AND CONSTRUCTION.

(26) INSTALL TRACER WIRE BOX PER DTL. 4/D5.21. SEE DTL. 2/D5.21 FOR ELONGATED CONCRETE COLLAR AND DTL. 1/D5.21 FOR SNOW PLOW PROTECTIVE APRON.

(27) SEE C5.53 FOR HOSE BIBB AND BACKFLOW PREVENTER INSTALLATION. HOSE BIBB IS FOR EFFLUENT FILTER CLEANING.

(1) 55,000 GAL. SEPTIC TANK FOR VISITOR CONTACT CENTER
NOT TO SCALE

PHASE 5

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STATE OF CALIFORNIA
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SOUTH ENTRANCE
OWTS PLANS
VCC SEPTIC TANK
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 97 OF 291

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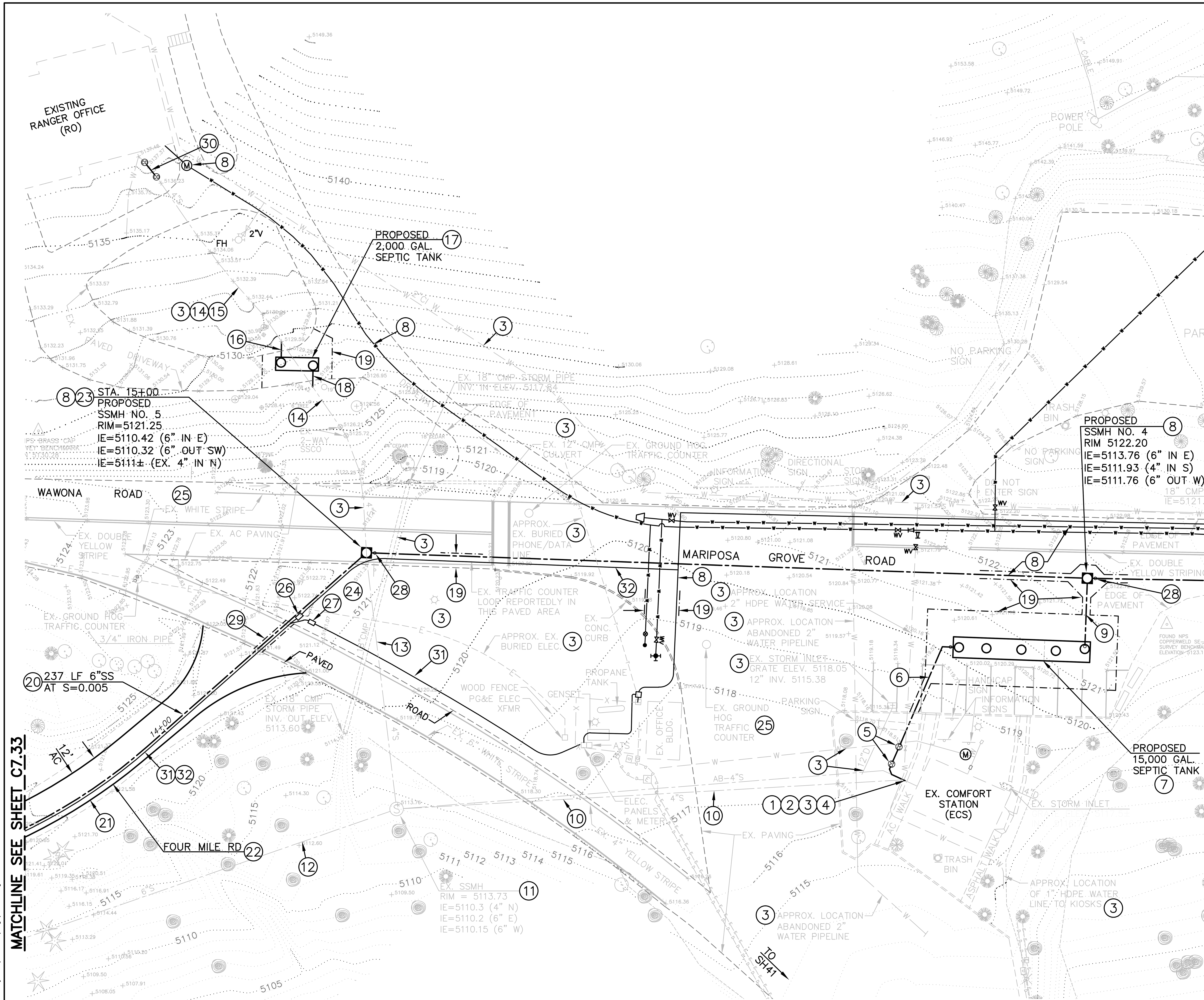

CONSTRUCTION NOTES

1. APPROX. POC LOCATION TO EX. 4" SEWER FROM COMFORT STATION. SEE SUB-SHEET C7.31.
2. MINIMIZE DISRUPTION OF COMFORT STATION SERVICE – SEE GENERAL NOTE 2/C7.60.
3. SEE GENERAL NOTE 3/C7.60 CONCERNING EX. UTILITIES.
4. COORDINATE ALL WORK WITH THE PROPOSED ACCESSIBLE ROUTE IMPROVEMENTS – SEE LANDSCAPE PLANS.
5. CONSTRUCT 2-WAY SSCO AT COMFORT STATION – SEE C7.31.
6. INSTALL 4" SEWER – SEE C7.31.
7. SEPTIC TANK – SEE C7.31.
8. SEE SUB-SHEETS C5.51–C5.53 FOR OTHER PROPOSED UTILITIES AND IMPROVEMENTS.
9. 4" SEWER – SEE C7.31.
10. EXISTING 4" HDPE SEWER WAS INSTALLED FROM COMFORT STATION TO SSMH IN FEB. 2012 PER PARK RECORDS – JAN. 10, 2012 DATED DRAWING (DWG. NO. YOS/5301B) HANDWRITTEN NOTES. LOCATION SHOWN IS APPROXIMATE. PREVIOUSLY ABANDONED 4" SEWER IS REPORTEDLY JUST NORTH OF THIS NEWER 4" HDPE PIPELINE. THESE PIPELINES TO BE ABANDONED–SEE GEN. NOTE 2/C7.60.
11. REMOVE EX. SSMH. SEE GENERAL NOTE 2.B/C7.60.
12. EX. LOCATION OF EX. 6" VITRIFIED CLAY PIPE SEWER FROM EX. SSMH TO EX. SEPTIC TANK. REMOVE THIS PIPELINE PER GENERAL NOTE 2.B/C7.60.
13. APPROX. LOCATION OF EX. 4" HDPE SEWER INSTALLED IN 2012 FROM RANGER OFFICE TO EX. SSMH PER PARK STAFF. THIS SECTION OF PIPELINE, SOUTH OF NEW SSMH NO. 5, IS TO BE ABANDONED PER GENERAL NOTE 2.B/C7.60.
14. APPROX. LOCATION OF EX. 4" HDPE SEWER INSTALLED IN 2012 FROM RANGER OFFICE TO EX. SSMH PER PARK STAFF.
15. MINIMIZE DISRUPTION OF OFFICE SEWER SERVICE – SEE GENERAL NOTE 2/C7.60.
16. INSTALL 4" SEWER CONNECTION TO EX. 4" PIPELINE – SEE SUB-SHEET C7.32.
17. SEPTIC TANK – SEE C7.32.
18. INSTALL 4" SEWER CONNECTION TO EX. 4" PIPELINE – SEE SUB-SHEET C7.32.
19. SAW-CUT AND REMOVE EXISTING PAVEMENT STRUCTURAL SECTION AS REQUIRED FOR UTILITY INSTALLATION, INCLUDING SHORING & BACKFILLING. REPAVE PER C5.31 AND DTL. 1/D5.11.
20. INSTALL 6" SDR 26 PVC SEWER IN JOINT-UTILITY TRENCH WITH ELEC. SEE DTL. 5/D5.11.
21. RECONSTRUCT FOUR MILE ROAD WITH A 12' MIN. WIDE PAVED ROADWAY, INCLUDING: DEMO; REGRADING AS REQUIRED FOR DRAINAGE; 8" MIN. SUBGRADE SCARIFICATION, MOISTURE CONDITIONING AND COMPACTION TO 95% MRC; 5" OF 3/4" CLASS 2 AGGREGRATE BASE (METHOD II, 95% MRC); AND 3" OF HOT ASPHALT CONCRETE PAVEMENT.

    CROSS SLOPE THE ROAD AT 2% TO THE SOUTHEAST TO DRAIN. FINISHED GRADE OF THE ROAD TO BE A MINIMUM OF 3" ABOVE EXISTING GRADE ON EITHER SIDE OF THE ROAD. SEE NOTE 22, AND DTL. 1/C7.51.
22. FOUR MILE ROAD: THIS EXISTING SERVICE ROAD VARIES IN WIDTH AND CONDITION. PORTIONS ARE DIRT ROAD, DIRT/GRAVEL ROAD, AND DIRT/GRAVEL OVER REMNANTS OF AC PAVING.
23. THE ELEVATION OF THE EX. 4" HDPE SEWER INVERT IS UNKNOWN. IF IT'S MORE THAN 2' ABOVE THE NEW SSMH 6" OUTLET INVERT, CONSTRUCT SSMH NO. 5 WITH INSIDE DROP CONNECTION PER DTL. 4/D5.22.
24. REMOVE AND REPLACE DECORATIVE BOULDERS ON EX. EARTH MOUNDING IN MEDIAN AS REQUIRED TO ACCOMMODATE PROPOSED 6" PIPELINE CONSTRUCTION.
25. SEE NOTE 1/C7.10.
26. INSTALL OLDCASTLE MODEL B1730, OR APPROVED EQUIVALENT, 17"x30" I.D. TRAFFIC-RATED UTILITY BOX WITH HOT-DIPPED GALV. STEEL FRAME AND BOLT-DOWN CHECKER PLATE COVER, REINFORCED CONC. BOX EXTENSION. LABEL COVER "CONTROLS". SET BOX ON MIN. 4 SOLID CONC. BUILDING BLOCKS (2"x4"x8" MIN. SIZE) AND 6" MIN. LAYER OF 3/4" DRAIN ROCK. ROCK TO EXTEND AT LEAST 6" BEYOND PERIMETER OF BOX. SEAL CONDUIT ENDS WATERTIGHT. SEE E5.02 AND DTL. 2/D5.71.
27. INSTALL TRAFFIC-RATED BOX FOR PCP POWER SUPPLY. LABEL COVER "ELECTRICAL". SEE NOTE 26.
28. INSTALL TRACER WIRE BOX PER DTL. 4/D5.21 WITH ELONGATED CONCRETE COLLAR PER DTL. 2/D5.21.
29. SAW-CUT AND REMOVE EXISTING PAVEMENT STRUCTURAL SECTION AS REQUIRED FOR UTILITY INSTALLATION, INCLUDING SHORING & BACKFILLING. REPAVE PER DTL. 1/D5.11, EXCEPT USE MIN. 4" AC & MIN. 20" AB.
30. CONSTRUCT 2-WAY SSCO AT RANGER OFFICE – SEE C7.32.
31. 2" CONDUIT. SEE E5.02 FOR OWTS PUMPING EQUIPMENT POWER SUPPLY CONNECTION TO EX. PANEL "A" & EX. GENSET.
32. 2" CONDUIT – SEE E5.02 FOR SCADA FIBER OPTIC CABLING FROM OWTS PUMP CONTROL PANEL TO SCADA POC AT EX. KIOSK.

PHASE 5

SCALE OF FEET: 20 0 20 40

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER — THOMAS G. HORGAN — No. 24772 — CIVIL — STATE OF CALIFORNIA
MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM
PRIME:
DAVID EVANS AND ASSOCIATES, INC.
BELLEVUE, WA

SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.30

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS ECS & RO SITE PLAN
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 98 OF 291

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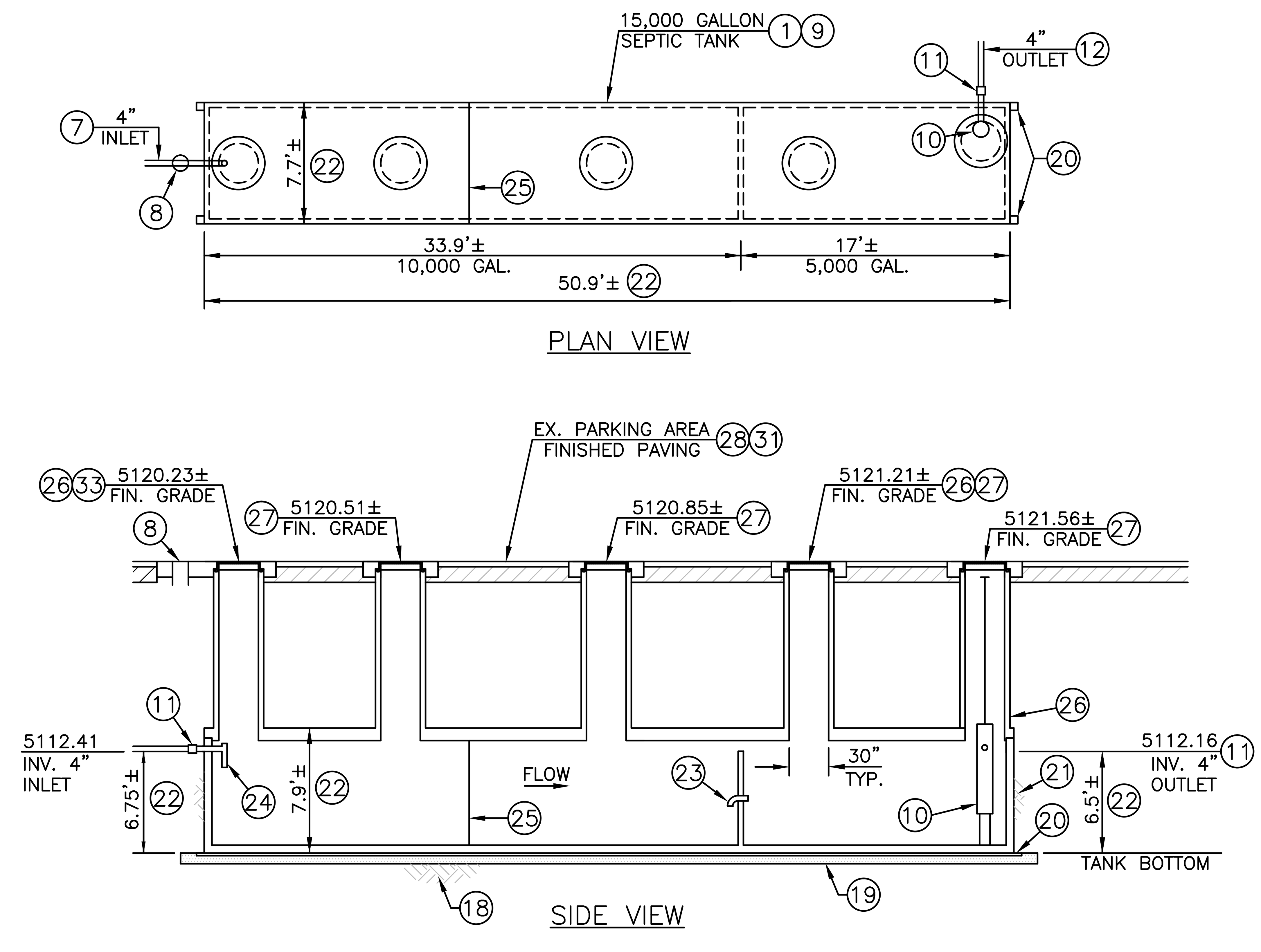
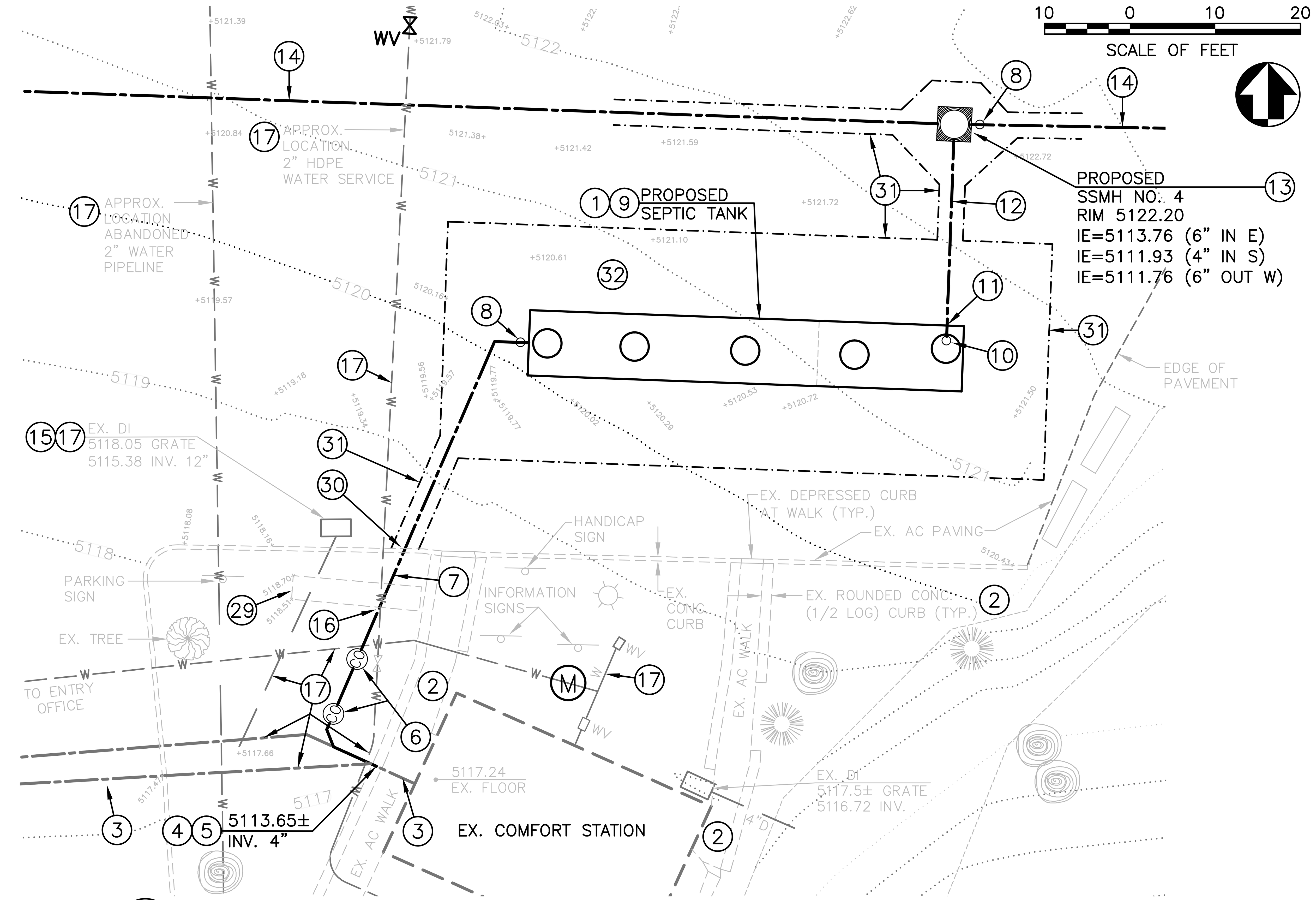


1 15,000 GAL. SEPTIC TANK FOR EX. COMFORT STATION (ECS)
- NOT TO SCALE



2 PLAN VIEW OF SEPTIC TANK WITH INLET & OUTLET PIPING AT EX. COMFORT STATION

CONSTRUCTION NOTES FOR DETAILS 1 & 2:

1. SEE THE OWTS TANKS GENERAL NOTE 21 ON SUB-SHEETS NOS. C7.62 AND C7.63 FOR ADDITIONAL SEPTIC TANK CONSTRUCTION AND INSTALLATION REQUIREMENTS, INCLUDING INTERIOR PROTECTIVE COATING AND BUOYANCY COUNTERACTION.

   TANK INLET AND OUTLET INVERTS TO BE VERIFIED TO ACCOMMODATE INLET PIPING POC INVERT NEAR BUILDING – SEE NOTE 4 AND COORDINATE WITH ENGINEER.

2. SEE SUB-SHEET C7.30 FOR SOUTH ENTRANCE EXISTING COMFORT STATION AREA OVERALL OWTS PLANS. COORDINATE ALL WORK WITH SITE UTILITY, EARTHWORK AND PAVING CONTRACTORS.

   COORDINATE WITH PROPOSED ACCESSIBLE ROUTE IMPROVEMENTS – SEE LANDSCAPE PLANS.

3. EXISTING 4" HDPE SEWER WAS INSTALLED FROM COMFORT STATION TO SSMH IN FEBRUARY 2012 PER PARK RECORDS – JANUARY 10, 2012 DATED DRAWING (DWG. NO. YOS/5301B) HANDWRITTEN NOTES. LOCATION SHOWN IS APPROXIMATE. ABANDONED 4" SEWER IS REPORTEDLY JUST NORTH OF THIS NEWER 4" HDPE PIPELINE. SEE C7.30.

4. APPROXIMATE POINT OF CONNECTION (POC) TO EXISTING 4" HDPE BUILDING SEWER AT EXISTING COMFORT STATION. INVERT ELEV. IS APPROX.

   FIELD VERIFY EXISTING PIPELINE SIZE, MATERIAL, LOCATION AND DEPTH. PRIOR TO INSTALLING SEPTIC TANK, VERIFY SEPTIC TANK GRADE USING 2% PIPELINE SLOPE FROM POC TO TANK. COORDINATE WITH ENGINEER AND COR TO MINIMIZE TANK DEPTH. SEE NOTES 2 & 17. SAW-CUT AND REMOVE SECTION OF EX. PIPELINE AS REQUIRED. SEE C7.30.

5. MINIMIZE DISRUPTION OF EXISTING SEWER SERVICE. SEE GENERAL NOTE 2.A/C7.60.

6. INSTALL 2-WAY SSCO PER DTL. 1/D5.22. SEE NOTE 2 ABOVE. EXTEND SEWER TRACER WIRE UP INTO SOUTHERLY SSCO BOX.

7. INSTALL 62 LF OF 4" SDR 26 PVC SEWER PIPE AT 2% SLOPE PER DTL. 3/D5.11 – SEE NOTE 4. NO SINGLE FITTING ANGLES GREATER THAN 45° PERMITTED.

8. INSTALL TRACER WIRE BOX PER DTL. 4/D5.21. SEE DTL. 2/D5.21 FOR ELONGATED CONCRETE COLLAR AND DTL. 1/D5.21 FOR SNOW PLOW PROTECTIVE APRON.

9. INSTALL 15,000 GALLON PRECAST CONCRETE SEPTIC TANK. TANK CONFIGURATION SHOWN ON THIS PLAN IS BASED ON A BATTERY OF THREE JENSEN PRECAST MODEL JZ5000 TANKS. AN ENGINEER-APPROVED EQUIVALENT SEPTIC TANK COMPLYING WITH THESE PLANS IS ACCEPTABLE FOR INSTALLATION.

   NOTE DEPTH OF TANK IN DETAIL ABOVE. SEE NOTE 4.

10. INSTALL EFFLUENT FILTER – SEE GEN. NOTE 22/C7.63. EXTEND FILTER HANDLE TO WITHIN 12" BELOW RISER COVER. WHEN FILTER IS CLEANED, THE "WASH-OFF" SHALL BE DRAINED INTO THE PRIMARY COMPARTMENT OF THIS SEPTIC TANK.

11. INSTALL MISSION RUBBER CO. FLEX-SEAL 4"x4" ADJUSTABLE REPAIR COUPLING, OR APPROVED EQUIVALENT, WITH SS SHEAR RING (TYP.).

12. INSTALL 23 LF OF 4" SDR 26 PVC SEWER PIPE AT 1% SLOPE PER DTL. 3/D5.11, EXCEPT NO TRACER WIRE REQUIRED.

13. PROPOSED SSMH NO. 4 – SEE SUB-SHEET C5.52 AND DTL. 5/D5.21.

14. SEE SUB-SHEETS C5.51–C5.53 FOR OTHER PROPOSED UTILITIES AND IMPROVEMENTS.

15. EXISTING DI. 12" OUTLET PIPING ACTUAL ROUTE IS NOT KNOWN.

16. APPROX. LOCATION WHERE NEW 4" SEWER CROSSES EX. WATER SERVICE. IF EX. WATER HAS LESS THAN 12" SEPARATION FROM SEWER, ENCASE WATER IN CONCRETE PER COR.

17. SEE GENERAL NOTE 3/C7.60 CONCERNING EX. UTILITIES. LOCATIONS SHOWN ARE APPROXIMATE. CONTRACTOR TO MODIFY EX. UTILITIES IN CONFLICT WITH THE PROPOSED WORK TO THE SATISFACTION OF THE COR AND THE ENGINEER.

18. EARTH SUBGRADE SHALL BE CONSTRUCTED TO PROPER LINE AND GRADE IN FIRM UNDISTURBED SOIL. IF SOIL IS NOT FIRM AND UNDISTURBED, PREPARE SUBGRADE AND COMPACT TO 95% MINIMUM RELATIVE COMPACTION (MRC) – SEE PROJECT EARTHWORK SPECIFICATIONS.

19. LEVEL, MINIMUM 8" LAYER OF ¾" CLASS 2 AB AT 95% MRC.

20. IF REQUIRED BY THE TANK MANUFACTURER, INSTALL 2" X 6" REDWOOD GRADE BOARDS LENGTHWISE UNDER SIDES OF TANKS. EXTEND BOARDS 1'± BEYOND TANK ENDS.

21. COMPACTED EARTH BACKFILL AT 95% MINIMUM RELATIVE COMPACTION. INSTALL TANK IN ACCORD WITH MANUFACTURER'S RECOMMENDATIONS.

22. VERIFY ALL TANK DIMENSIONS WITH TANK MANUFACTURER.

23. 4" SCH. 40 PVC PIPE INVERTED ELBOW FITTING.

24. INSTALL 4" PVC SANITARY INLET TEE. ALL PIPE-TO-TANK CONNECTIONS TO BE WATERTIGHT. SEE GENERAL NOTE 21.F/C7.63.

25. JOB SITE JOINT SEALING OF SECTIONAL TANKS SHALL BE SECURE AND WATERTIGHT AND SHALL COMPLY WITH TANK MANUFACTURER'S RECOMMENDATIONS.

26. INSTALL AASHTO HS20-44 HEAVY-TRAFFIC RATED, 30" INSIDE DIAMETER, PRECAST CONCRETE RISERS, CAST-IRON FRAMES AND COVERS (TYP.). SEE GENERAL NOTE 21.J/C7.63 FOR RISER INTERIOR PROTECTIVE COATING. SEE GENERAL NOTE 21.G/C7.63 FOR FRAME AND COVER REQUIREMENTS, INCL. PROTECTIVE COATING.

27. EACH FRAME AND COVER ASSEMBLY IS TO BE INSTALLED WITH A CAST-IN-PLACE, HEAVY-DUTY, REINFORCED CONCRETE SNOW PLOW PROTECTIVE APRON. SEE DETAILS 1/D5.21 AND 2/D5.21.

28. FINISHED PAVING LINE SHOWN IS APPROXIMATE. REPLACEMENT PAVEMENT TO CONFORM TO EXISTING PAVING WITH UNIFORM CROSS-SLOPE FOR SURFACE DRAINAGE. SEE C5.31.

29. REMOVE AND REPLACE EXISTING TRASH AND RECYCLE CONTAINERS AND CONCRETE PADS AS REQUIRED TO ACCOMMODATE PROPOSED PIPELINE WORK.

30. SAW-CUT AND REMOVE AND REPLACE EXISTING CONCRETE CURB AS REQUIRED TO ACCOMMODATE PROPOSED WORK. USE DOWELLED CONNECTIONS TO EXISTING CURB.

31. SAW-CUT AND REMOVE EXISTING PAVEMENT STRUCTURAL SECTION AS REQUIRED FOR PIPING AND TANK INSTALLATION, INCLUDING SHORING & BACKFILLING. REMOVAL AREA SHOWN IS APPROXIMATE. REPAVE PER DTL. 1/D5.11, EXCEPT USE 4.5" AC & 16" AB.

32. SUBSURFACE EXPLORATION IN THIS AREA BY THE PARK DISCOVERED SOME COBBLES.

33. INSTALL ELONGATED CONCRETE COLLAR PER DTL. 2/D5.21. SEE NOTE 8.

PHASE 5

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER
THOMAS G. HORGAN
No. 24772
CIVIL
STATE OF CALIFORNIA
MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM	DESIGNED: TGH	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO. 104 / 127662
PRIME: DAVID EVANS AND ASSOCIATES, INC. BELLEVUE, WA	CADD MSW	C7.31	SOUTH ENTRANCE OWTS PLANS ECS SEPTIC TANK	PMIS/PKG NO. 206431
SUBCONTRACTOR LAUGENOUR AND MEIKLE WOODLAND, CA	TECH. REVIEW: TGH		YOSEMITE NATIONAL PARK	SHEET
	DATE: 2/26/2016		MARIPOSA GROVE RESTORATION	99 OF 291

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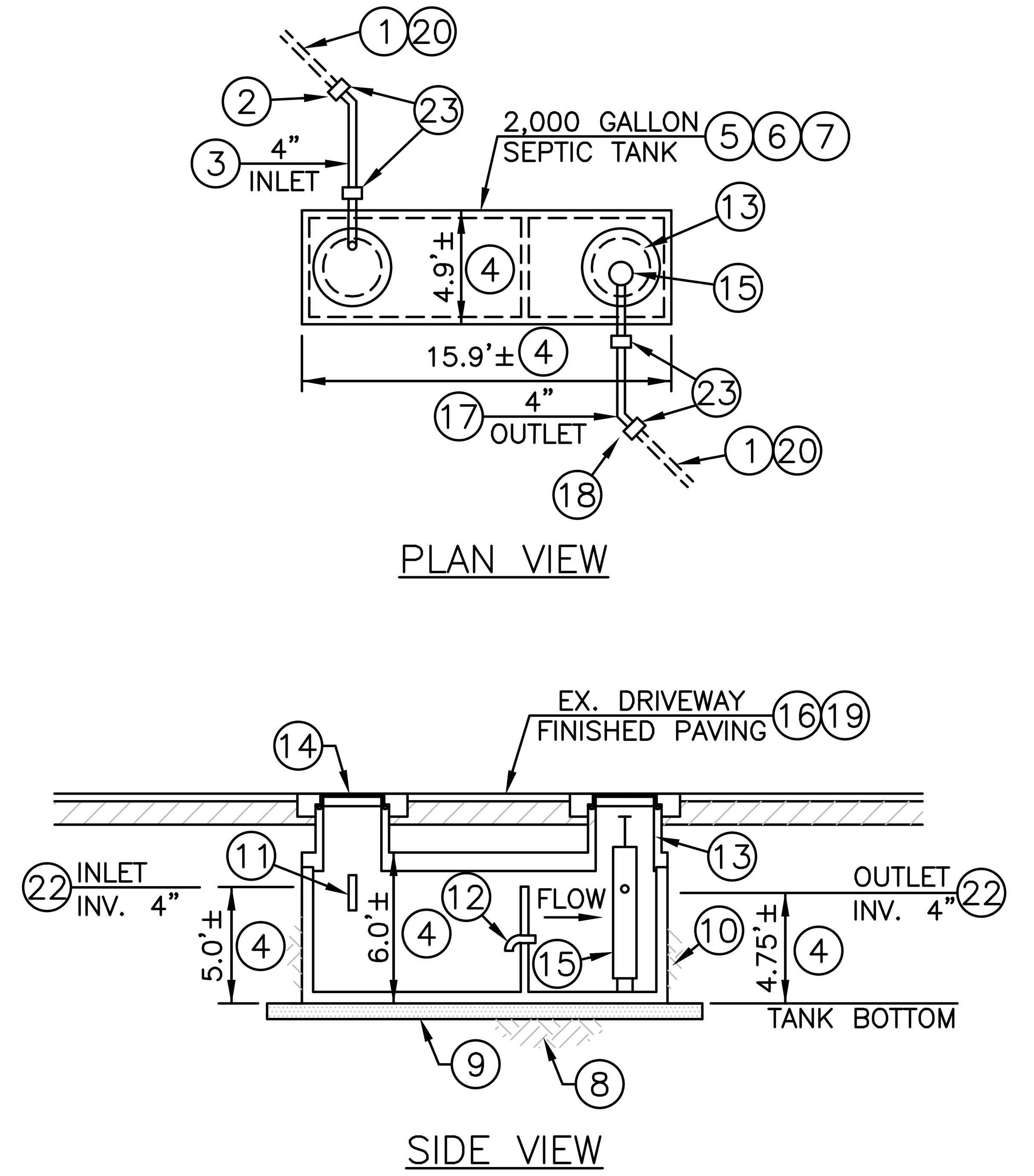
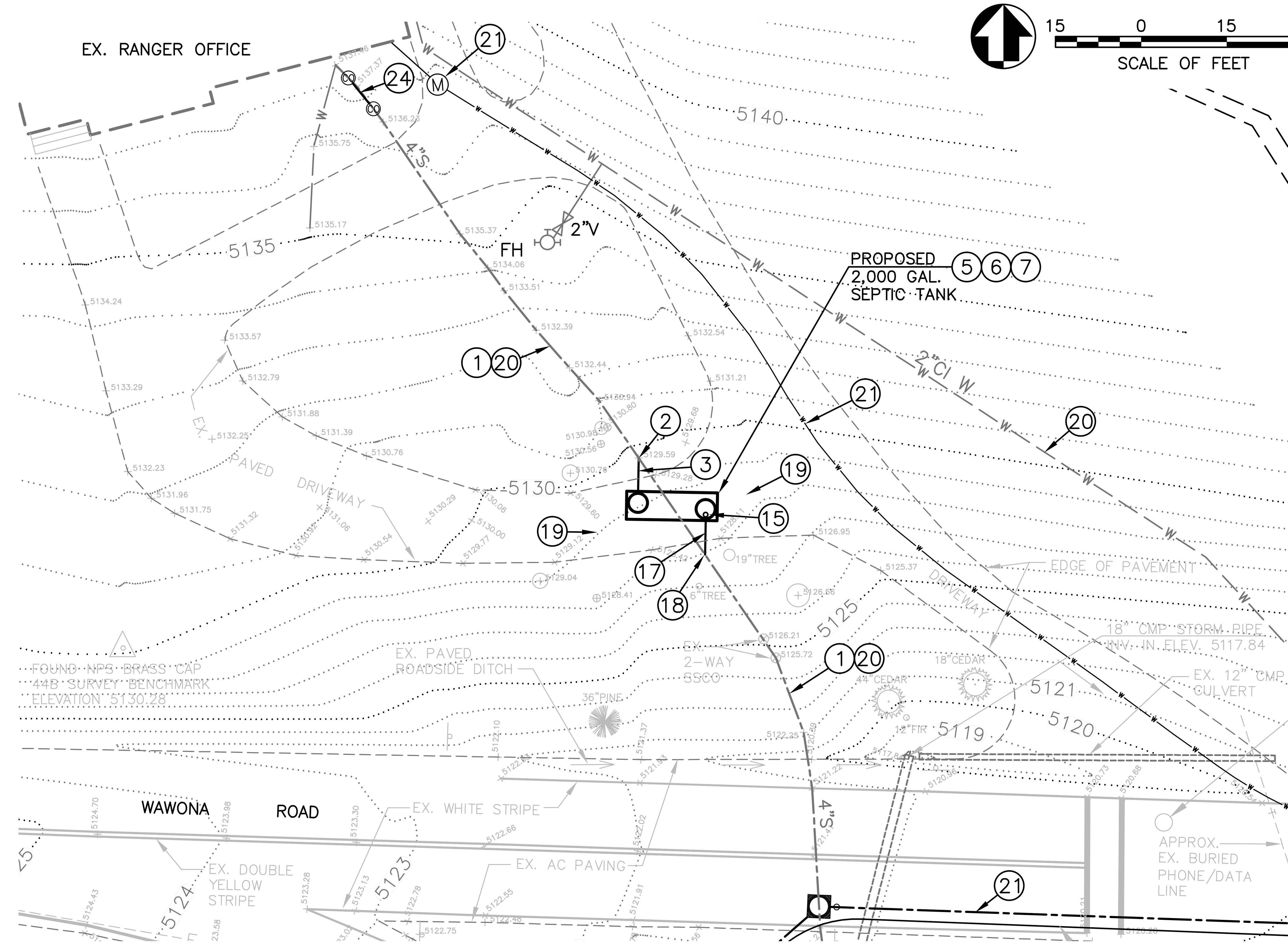


1 2,000 GAL. SEPTIC TANK FOR EX. RANGER OFFICE
– NOT TO SCALE



2 PLAN VIEW OF SEPTIC TANK INLET & OUTLET PIPING AT EX. RANGER OFFICE
–

CONSTRUCTION NOTES FOR DETAILS 1 & 2

1. APPROX. LOCATION OF EX. 4" HDPE SEWER PIPE (INSTALLED IN 2012) BASED ON LOCATION MARKS BY PARK.

2. POINT OF CONNECTION TO EX. 4" HDPE BUILDING SEWER. VERIFY EX. PIPELINE SIZE, MATERIAL, LOCATION AND DEPTH. SAW-CUT AND REMOVE SECTION OF EX. 4" PIPE AS NEEDED.

   INSTALL MISSION RUBBER COMPANY FLEX-SEAL 4" X 4" ADJUSTABLE REPAIR COUPLING, OR APPROVED EQUIVALENT, WITH SS SHEAR RING (TYP.).

3. INSTALL 4" SDR 26 PVC SEWER PIPE AT 2% MINIMUM SLOPE – SEE NOTE 22. NO SINGLE FITTING ANGLE GREATER THAN 45° PERMITTED. INSTALL FITTINGS AS REQUIRED TO PROPERLY ALIGN WITH SEPTIC TANK INLET OPENING.

4. VERIFY ALL TANK DIMENSIONS WITH TANK MANUFACTURER.

5. INSTALL 2,000 GALLON PRECAST CONCRETE SEPTIC TANK. SEE THE OWTS TANKS GENERAL NOTE 21 ON SUB-SHEETS NOS. C7.62 AND C7.63 FOR ADDITIONAL SEPTIC TANK REQUIREMENTS.

   THE 2,000 GALLON SEPTIC TANK CONFIGURATION SHOWN ON THIS PLAN IS BASED ON A JENSEN PRECAST MODEL JP2000 TANK. AN ENGINEER-APPROVED EQUIVALENT SEPTIC TANK COMPLYING WITH THESE PLANS IS ACCEPTABLE FOR INSTALLATION.

   PRECAST OPENINGS IN TOP OF TANK TO BE CAREFULLY POSITIONED TO FIT SIDE INLET TEE AND SIDE OUTLET FILTER POSITIONS.

6. SEE SUB-SHEET C7.30 FOR SOUTH ENTRANCE EX. COMFORT STATION AREA OVERALL UTILITY PLANS. COORDINATE ALL WORK WITH SITE UTILITY, EARTHWORK AND PAVING CONTRACTORS.

7. MINIMIZE DISRUPTION OF EX. SEWER SERVICE. SEE GENERAL NOTE 2.A/C7.60.

8. EARTH SUBGRADE SHALL BE CONSTRUCTED TO PROPER LINE AND GRADE IN FIRM UNDISTURBED SOIL. IF SOIL IS NOT FIRM AND/OR UNDISTURBED, PREPARE SUBGRADE AND COMPACT TO 95% MINIMUM RELATIVE COMPACTION (MRC) – SEE PROJECT EARTHWORK SPECIFICATIONS.

9. LEVEL, MINIMUM 8" LAYER OF 3/4" CLASS 2 AB AT 95% MRC.

10. COMPACTED EARTH BACKFILL AT 95% MINIMUM RELATIVE COMPACTION. INSTALL TANK IN ACCORD WITH MANUFACTURER'S RECOMMENDATIONS.

11. INSTALL 4" PVC SANITARY TEE.

12. 4" PVC PIPE INVERTED ELBOW FITTING.

13. INSTALL AASHTO HS20-44 HEAVY-TRAFFIC RATED, 30" INSIDE DIAMETER, PRECAST CONCRETE RISERS, CAST-IRON FRAMES AND COVERS (TYP.). SEE GENERAL NOTE 21.J/C7.63 FOR RISER INTERIOR PROTECTIVE COATING. SEE GENERAL NOTE 21.G/C7.63 FOR FRAME AND COVER REQUIREMENTS, INCL. PROTECTIVE COATING.

14. EACH FRAME AND COVER ASSEMBLY IS TO BE INSTALLED WITH A CAST-IN-PLACE, HEAVY-DUTY, REINFORCED CONCRETE SNOW PLOW PROTECTIVE APRON. SEE DETAILS 1/D5.21 AND 2/D5.21.

15. INSTALL EFFLUENT FILTER – SEE GEN. NOTE 22/C7.63. EXTEND FILTER HANDLE TO WITHIN 12" BELOW RISER COVER. WHEN FILTER IS CLEANED, THE "WASH-OFF" SHALL BE DRAINED INTO THE PRIMARY COMPARTMENT OF THIS SEPTIC TANK.

16. FINISHED PAVING LINE SHOWN IS APPROXIMATE. REPLACEMENT PAVEMENT GRADE TO CONFORM TO EX. PAVING AT CUT LINES AND TO HAVE UNIFORM CROSS-SLOPE FOR POSITIVE DRAINAGE.

17. INSTALL 4" SDR 26 PVC SEWER PIPE AT 2% MINIMUM SLOPE – SEE NOTE 22. NO SINGLE FITTING ANGLE GREATER THAN 45° PERMITTED. INSTALL FITTINGS AS REQUIRED TO PROPERLY ALIGN WITH SEPTIC TANK OUTLET OPENING.

18. POINT OF CONNECTION TO EX. 4" HDPE BUILDING SEWER. VERIFY EX. PIPELINE SIZE, MATERIAL, LOCATION AND DEPTH. SAW-CUT AND REMOVE SECTION OF EX. 4" PIPE AS NEEDED.

    INSTALL MISSION RUBBER COMPANY FLEX-SEAL 4" X 4" ADJUSTABLE REPAIR COUPLING, OR APPROVED EQUIVALENT, WITH SS SHEAR RING (TYP.).

19. SAW-CUT AND REMOVE EX. PAVING STRUCTURAL SECTION AS REQUIRED FOR TANK AND PIPING INSTALLATION. REPAVE PER DTL. 1/D5.11.

20. SEE GENERAL NOTE 3/C7.60 CONCERNING EX. UTILITIES.

21. SEE SUB-SHEET C5.51 FOR OTHER PROPOSED UTILITIES.

22. SET SEPTIC TANK ELEVATION TO INTERCEPT EX. 4" SEWER SUCH THAT THE NEW SECTIONS OF 4" INLET AND OUTLET PIPING (NOTES 3 AND 17, RESPECTIVELY) ARE INSTALLED WITH APPROXIMATELY EQUAL SLOPES.

23. INSTALL MISSION RUBBER CO. FLEX-SEAL 4"x4" ADJUSTABLE REPAIR COUPLING, OR APPROVED EQUIVALENT, WITH SS SHEAR RING (TYP.).

24. APPROX. LOCATION OF PROPOSED 2-WAY SSCO INSTALLATION IN EX. 4" HDPE BUILDING SEWER. FIELD VERIFY SSCO POSITION WITH COR. VERIFY EX. PIPELINE SIZE, MATERIAL, LOCATION AND DEPTH. SAW-CUT AND REMOVE SECTION OF EX. 4" PIPE AS NEEDED. CONSTRUCT 2-WAY SSCO PER DTL. 1/D5.22. MINIMIZE DISRUPTION OF RANGER OFFICE SEWER SERVICE.

PHASE 5

Thomas G. Horgan

REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA

MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM	
PRIME: DAVID EVANS AND ASSOCIATES, INC. BELLEVUE, WA	DESIGNED: TGH
	CADD: MSW
SUBCONTRACTOR LAUGENOUR AND MEIKLE WOODLAND, CA	TECH. REVIEW: TGH
	DATE: 2/26/2016

SUB SHEET NO.
C7.32

TITLE OF SHEET
SOUTH ENTRANCE
OWTS PLANS
RO SEPTIC TANK
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104
127662
PMIS/PKG NO. 206431
SHEET
100 OF 291

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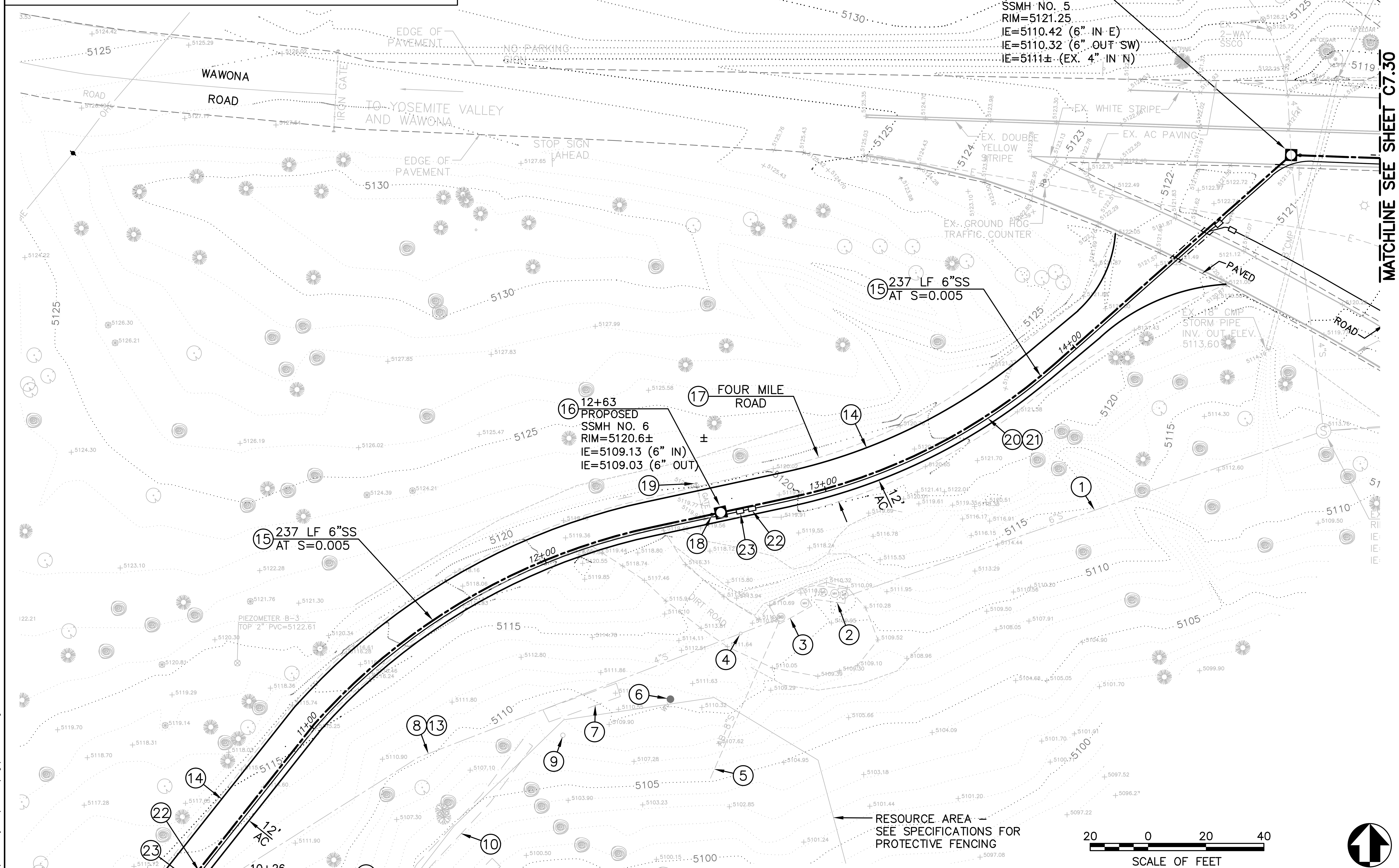
(21) INSTALL 2" SCH. 80 PVC SCADA FIBER OPTIC CABLE CONDUIT TO PCP FROM KIOSK – SEE JOINT UTILITY TRENCH DTL. 5/D5.11, E5.02 AND E6.01.

(22) INSTALL OLDCASTLE MODEL B1730, OR APPROVED EQUIVALENT, 17"x30" I.D. TRAFFIC-RATED UTILITY BOX WITH HOT-DIPPED GALV. STEEL FRAME AND BOLT-DOWN CHECKER PLATE COVER, REINFORCED CONC. BOX EXTENSION. LABEL COVER "CONTROLS". SET BOX ON MIN. 4 SOLID CONC. BUILDING BLOCKS (2"x4"x8" MIN. SIZE) AND 6" MIN. LAYER OF 3/4" DRAIN ROCK. ROCK TO EXTEND AT LEAST 6" BEYOND PERIMETER OF BOX. SEAL CONDUIT ENDS WATERTIGHT. SEE E5.02 AND DTL. 2/D5.71.

(23) INSTALL TRAFFIC-RATED BOX FOR PCP POWER SUPPLY. LABEL COVER "ELECTRICAL". SEE NOTE 22.



(16) 10+26 PROPOSED SSMH NO. 7 RIM=5114.3± IE=5107.84 (6" IN) IE=5107.74 (6" OUT)

MATCHLINE SEE SHEET C7.34

SCALE OF FEET: 20 0 20 40

CONSTRUCTION NOTES

(1) APPROX. LOCATION OF EX. 6" VITRIFIED CLAY PIPE SEWER FROM SSMH TO EX. SEPTIC TANK. REMOVE THIS PIPELINE PER GENERAL NOTE 2.B/C7.60.

(2) EXISTING 1,000± GALLON SEPTIC TANK INSTALLED IN 1986 PER YNP DWG. NO. 104/60,409 (TANK CAPACITY NOT VERIFIED).
EXISTING MANWAY RIMS: 5110.20 EAST INLET INV. 5108.7
5110.11 MIDDLE TANK FLOOR: 5104.6±
5110.11 WEST
THIS TANK IS TO BE REMOVED – SEE GENERAL NOTE 2.B/C7.60.

(3) EXISTING SEPTIC TANK (INSTALLATION DATE IS UNKNOWN. SHOWN AS EXISTING ON 1986 PLANS). CAPACITY IS UNKNOWN.
EXISTING MANWAY RIMS: 5110.75 EAST TANK FLOOR: 5103±
5110.55 WEST
THIS TANK IS TO BE REMOVED – SEE GENERAL NOTE 2/C7.60.

(4) APPROXIMATE LOCATION OF DAMAGED SSCO IN DIRT ROADWAY. THIS SSCO IS TO BE REMOVED.

(5) APPROXIMATE LOCATION OF ABANDONED 8" VITRIFIED CLAY PIPELINE. THIS PIPELINE IS TO REMAIN IN PLACE. PLUG CUT END AT TANK.

(6) ABANDONED VALVE IN 8" VITRIFIED CLAY PIPE RISER TO REMAIN.

(7) ABANDONED REDWOOD-LINED WASTEWATER DISPOSAL TRENCH (CESSPOOL), APPROXIMATE 4' WIDE AND 4' DEEP, TO REMAIN.

(8) APPROXIMATE LOCATION OF 346± LF OF EXISTING 4" PVC PIPELINE INSTALLED AT APPROX. 1% SLOPE IN 1986, PER YNP DWG NO. 104/60,409, FROM EXISTING SEPTIC TANK TO VALVE NEAR DOSING SIPHON.

THIS PIPELINE TO BE REMOVED – SEE GENERAL NOTE 2.B/C7.60.

(9) ABANDONED 8" VITRIFIED CLAY PIPE TO REMAIN.

(10) ABANDONED REDWOOD-LINED WASTEWATER DISPOSAL TRENCH (CESSPOOL), APPROXIMATE 4' WIDE AND 4' DEEP, TO REMAIN.

(11) ABANDONED REDWOOD-LINED WASTEWATER DISPOSAL TRENCH (CESSPOOL), APPROXIMATE 4' WIDE AND 4' DEEP, TO REMAIN.

(12) ABANDONED 6" VITRIFIED CLAY PIPE TO REMAIN.

(13) SEE GENERAL NOTE 3/C7.60 CONCERNING EX. UTILITIES.

(14) RECONSTRUCT FOUR MILE ROAD WITH A 12' MIN. WIDE PAVED ROADWAY, INCLUDING: DEMO; REGRADING AS REQUIRED FOR DRAINAGE AND SMOOTHING OF ROADWAY LONGITUDINAL AND CROSS SLOPES; 8" MIN. SUBGRADE SCARIFICATION, MOISTURE CONDITIONING AND COMPACTION TO 95% MRC; INSTALLATION OF TENSAR BX1100, OR APPROVED EQUIVALENT, GEOGRID PER MFR'S INSTRUCTIONS; 5" OF 3/4" CLASS 2 AGGREGRATE BASE (METHOD II, 95% MRC); AND 3" OF HOT ASPHALT CONCRETE PAVEMENT.

CROSS SLOPE THE ROAD AT 2% TO THE SOUTHEAST TO DRAIN. FINISHED GRADE OF THE ROAD TO BE A MINIMUM OF 3" ABOVE EXISTING GRADE ON EITHER SIDE OF THE ROAD.

SEE CONSTRUCTION NOTE 17 BELOW, DWG. C5.13 AND DTL. 1/C7.51 FOR ROAD REGRADING. COORDINATE WITH COR.

(15) INSTALL 6" SDR 26 PVC SEWER IN JOINT-UTILITY TRENCH WITH ELEC AND CONTROLS CONDUITS. SEE E5.02, E6.07 AND DTL. 5/D5.11. COORDINATE ALIGNMENT WITH COR.

(16) SEE DTLS. 2 & 5/D5.21 FOR SSMH CONSTRUCTION. COORDINATE FRAME & COVER (RIM) GRADE WITH PROPOSED GRADING AND PAVING. SEE NOTE 18.

(17) FOUR MILE ROAD: THIS EXISTING SERVICE ROAD VARIES IN WIDTH AND CONDITION. PORTIONS ARE DIRT ROAD, DIRT/GRAVEL ROAD, AND DIRT/GRAVEL OVER REMNANTS OF AC PAVING.

(18) INSTALL TRACER WIRE BOX PER DTL. 4/D5.21. SEE DTL. 2/D5.21 FOR ELONGATED CONCRETE COLLAR AND DTL. 1/D5.21 FOR SNOW PLOW PROTECTIVE APRON.

(19) REMOVE EX. GATE. REINSTALL AT LOCATION AS DIRECTED BY COR.

(20) INSTALL 2" SCH. 80 PVC ELEC. POWER CONDUIT TO OWTS PUMP CONTROL PANEL (PCP) FROM EX. PANEL "A" (NEAR KIOSK) – SEE E5.02, E5.40 AND DTL. 5/D5.11.

PHASE 5

REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA
MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

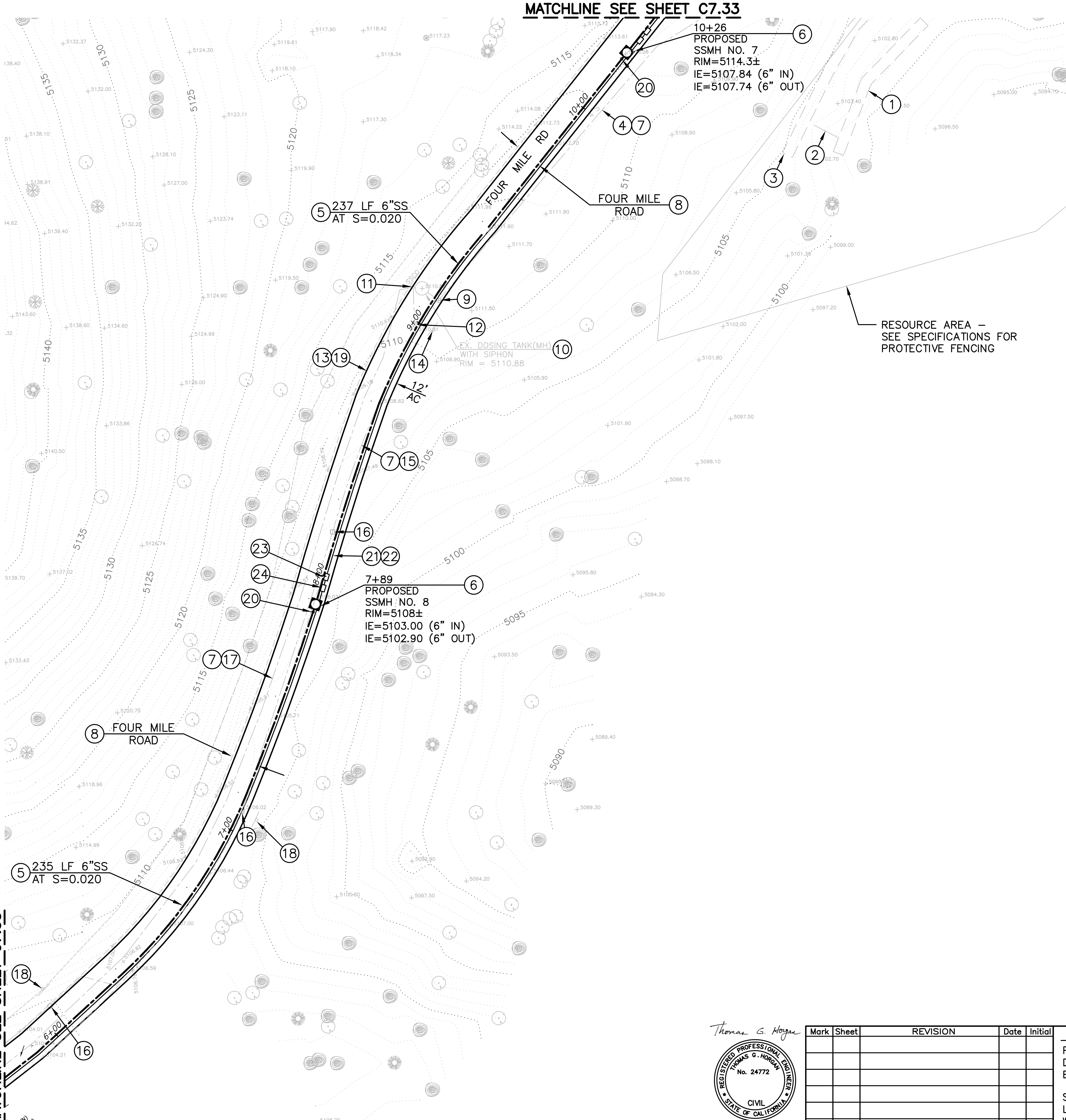
A/E FIRM
PRIME: DAVID EVANS AND ASSOCIATES, INC. BELLEVUE, WA
SUBCONTRACTOR: LAUGENOUR AND MEIKLE WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.33

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS FOUR MILE ROAD
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 101 OF 291



CONSTRUCTION NOTES

1. ABANDONED REDWOOD–LINED WASTEWATER DISPOSAL TRENCH (CESSPOOL), APPROXIMATE 4' WIDE AND 4' DEEP, TO REMAIN.
2. ABANDONED 6" VITRIFIED CLAY PIPE TO REMAIN.
3. ABANDONED REDWOOD–LINED WASTEWATER DISPOSAL TRENCH (CESSPOOL), APPROXIMATE 4' WIDE AND 4' DEEP, TO REMAIN.
4. APPROXIMATE LOCATION OF 346± LF OF EXISTING 4" PVC PIPELINE INSTALLED AT APPROX. 1% SLOPE IN 1986, PER YNP DWG NO. 104/60,409, FROM EXISTING SEPTIC TANK TO VALVE NEAR DOSING SIPHON.

   THIS PIPELINE TO BE REMOVED – SEE GENERAL NOTE 2.B/C7.60.
5. INSTALL 6" SDR 26 PVC SEWER IN JOINT–UTILITY TRENCH WITH ELEC AND CONTROLS CONDUITS. SEE E5.02, E6.07 AND DTL. 5/D5.11. COORDINATE ALIGNMENT OF 6" SEWER, RELATIVE TO EX. PIPELINES, WITH COR.
6. SEE DTLS. 2 & 5/D5.21 FOR SSMH CONSTRUCTION. COORDINATE FRAME AND COVER (RIM) GRADE WITH PROPOSED GRADING & PAVING. SEE NOTE 20.
7. SEE GENERAL NOTE 3/C7.60 CONCERNING EX. UTILITIES.
8. FOUR MILE ROAD: THIS EXISTING SERVICE ROAD VARIES IN WIDTH AND CONDITION. PORTIONS ARE DIRT ROAD, DIRT/GRAVEL ROAD, AND DIRT/GRAVEL OVER REMNANTS OF AC PAVING. SEE NOTE 13.
9. EXISTING EFFLUENT SHUT–OFF VALVE (APPROXIMATE LOCATION) TO BE REMOVED.
10. REMOVE EXISTING DOSING TANK (48" MH) WITH DOSING SIPHON. BACKFILL WITH 3/4" CL. 2 AB COMPACTED IN PLACE (95% MRC).
11. REMOVE EXISTING DIVERTER VALVE SERVING THE 1986 AND 2005 LEACHFIELDS. BACKFILL WITH 3/4" CL. 2 AB COMPACTED IN PLACE (95% MRC).
12. REMOVE EXISTING SHUT–OFF VALVE (APPROXIMATE LOCATION) PER GENERAL NOTE 2.B/C7.60.
13. RECONSTRUCT FOUR MILE ROAD WITH A 12' MIN. WIDE PAVED ROADWAY, INCLUDING: DEMO; REGRADING AS REQUIRED FOR DRAINAGE AND SMOOTHING OF ROADWAY LONGITUDINAL AND CROSS SLOPES; 8" MIN. SUBGRADE SCARIFICATION, MOISTURE CONDITIONING AND COMPACTION TO 95% MRC; INSTALLATION OF TENSAR BX1100, OR APPROVED EQUIVALENT, GEOGRID PER MFR'S INSTRUCTIONS; 5" OF 3/4" CLASS 2 AGGREGRATE BASE (METHOD II, 95% MRC); AND 3" OF HOT ASPHALT CONCRETE PAVEMENT.

    CROSS SLOPE THE ROAD AT 2% TO THE SOUTHEAST TO DRAIN. FINISHED GRADE OF THE ROAD TO BE A MINIMUM OF 3" ABOVE EXISTING GRADE ON EITHER SIDE OF THE ROAD.

    FOR ROAD CONSTRUCTION OVER THE EXISTING LEACHFIELD TRENCHES, AFTER THE EXISTING LEACH PIPE IS REMOVED FROM THE TRENCHES, COVER THE TOP OF THE EXISTING LEACH TRENCH DRAIN ROCK WITH MIRAFI 600X, OR APPROVED EQUIVALENT, GEOTEXTILE FABRIC. COORDINATE WITH 6" SEWER CONSTRUCTION.

    SEE CONSTRUCTION NOTES 8 AND 19 AND DTL. 1/C7.51.
14. REMOVE SEWER VALVE WITNESS POST (2005).
15. APPROXIMATE LOCATION OF 800± LF OF EXISTING 4" PVC PERFORATED LEACHFIELD PIPING INSTALLED IN NOVEMBER 2005.

    THIS LEACHFIELD PIPING IS TO BE REMOVED PER GENERAL NOTE 2.B/C7.60.
16. REMOVE EXISTING 4" PVC PIPE INSTALLED VERTICALLY AND CAPPED AS A LEACHFIELD INSPECTION PORT.
17. APPROXIMATE LOCATION OF 800± LF OF LEACHFIELD TRENCH AND 4" PERFORATED PIPING INSTALLED IN 1986 PER YNP DWG. NO. 104/60,409 WITH CAPPED 4" PVC PIPE RISER INSPECTION PORTS INSTALLED TO GROUND SURFACE AT THE END OF EACH OF THE EIGHT 100'± LONG LEACH TRENCH SECTIONS. MANY OF THESE PORTS HAVE BEEN DAMAGED BY VEHICLE TRAFFIC. TRENCH DEPTH VARIES FROM APPROXIMATELY 5' TO 9'.

    THIS LEACHFIELD PIPING IS TO BE REMOVED PER GENERAL NOTE 2.B/C7.60.
18. REMOVE LEACHFIELD INSPECTION PORT WITNESS POST (2005).
19. SEE DTL. 1/C7.51 FOR ROAD REGRADING. COORDINATE WITH COR.
20. INSTALL TRACER WIRE BOX PER DTL. 4/D5.21. SEE DTL. 2/D5.21 FOR ELONGATED CONCRETE COLLAR AND DTL. 1/D5.21 FOR SNOW PLOW PROTECTIVE APRON.
21. INSTALL 2" SCH. 80 PVC ELEC. POWER CONDUIT TO OWTS PUMP CONTROL PANEL (PCP) – SEE E5.02, E5.40 AND DTL. 5/D5.11.
22. INSTALL 2" SCH. 80 PVC SCADA FIBER OPTIC CABLE CONDUIT TO PCP – SEE E5.02, E6.01 AND DTL. 5/D5.11.
23. INSTALL OLDCASTLE MODEL B1730, OR APPROVED EQUIVALENT, 17"x30" I.D. TRAFFIC–RATED UTILITY BOX WITH HOT–DIPPED GALV. STEEL FRAME AND BOLT–DOWN CHECKER PLATE COVER, REINFORCED CONC. BOX EXTENSION. LABEL COVER "CONTROLS". SET BOX ON MIN. 4 SOLID CONC. BUILDING BLOCKS (2"x4"x8" MIN. SIZE) AND 6" MIN. LAYER OF 3/4" DRAIN ROCK. ROCK TO EXTEND AT LEAST 6" BEYOND PERIMETER OF BOX. SEAL CONDUIT ENDS WATERTIGHT. SEE E5.02 AND DTL. 2/D5.71.
24. INSTALL TRAFFIC–RATED BOX FOR PCP POWER SUPPLY. LABEL COVER "ELECTRICAL". SEE NOTE 23.

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SCALE OF FEET

PHASE 5

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA
MARCH 14, 2016

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PRIME:
DAVID EVANS AND ASSOCIATES, INC.
BELLEVUE, WA

SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.34

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS FOUR MILE ROAD
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 102 OF 291

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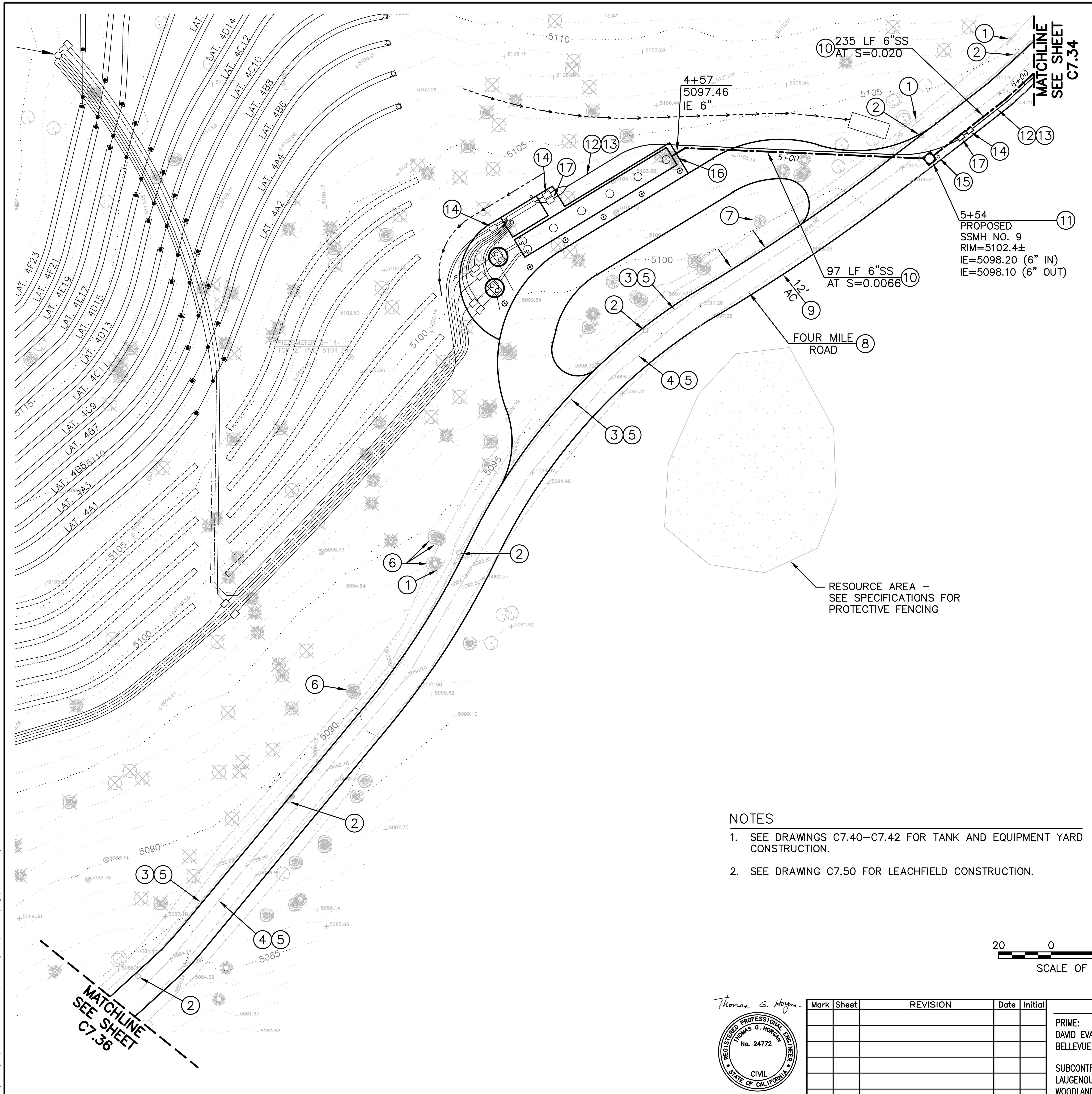

CONSTRUCTION NOTES

1. REMOVE LEACHFIELD INSPECTION PORT WITNESS POST (2005) (TYP.).
2. REMOVE EXISTING 4" PVC PIPE INSTALLED VERTICALLY AND CAPPED AS A LEACHFIELD INSPECTION PORT (TYP.).
3. APPROXIMATE LOCATION OF 800± LF OF EXISTING LEACHFIELD TRENCH AND 4" PVC PERFORATED PIPING INSTALLED IN NOVEMBER 2005.

   THIS EXISTING LEACHFIELD PIPING IS TO BE REMOVED PER GENERAL NOTE 2.B/C7.60.
4. APPROXIMATE LOCATION OF 800± LF OF LEACHFIELD TRENCH AND 4" PERFORATED PIPING INSTALLED IN 1986 PER YNP DWG. NO. 104/60,409 WITH CAPPED 4" PVC PIPE RISER INSPECTION PORTS INSTALLED TO GROUND SURFACE AT THE END OF EACH OF THE EIGHT 100'± LONG LEACH TRENCH SECTIONS. MANY OF THESE PORTS HAVE BEEN DAMAGED BY VEHICLE TRAFFIC. TRENCH DEPTH REPORTEDLY VARIES FROM APPROXIMATELY 5' TO 9', AND TRENCH LOCATION IS REPORTEDLY IN THE "MIDDLE" OF THE ROADWAY.

   THIS EXISTING LEACHFIELD PIPING IS TO BE REMOVED PER GENERAL NOTE 2.B/C7.60.
5. SEE GENERAL NOTE 3/C7.60 CONCERNING EX. UTILITIES.
6. PRESERVE EXISTING TREE. SEE C5.11–C5.12.
7. PRESERVE EXISTING OAK TREE.
8. FOUR MILE ROAD: THIS EXISTING SERVICE ROAD VARIES IN WIDTH AND CONDITION. PORTIONS ARE DIRT ROAD, DIRT/GRAVEL ROAD, AND DIRT/GRAVEL OVER REMNANTS OF AC PAVING.
9. RECONSTRUCT FOUR MILE ROAD WITH A 12' MIN. WIDE PAVED ROADWAY, INCLUDING: DEMO; REGRADING AS REQUIRED FOR DRAINAGE AND SMOOTHING OF ROADWAY LONGITUDINAL AND CROSS SLOPES; 8" MIN. SUBGRADE SCARIFICATION, MOISTURE CONDITIONING AND COMPACTION TO 95% MRC; INSTALLATION OF TENSAR BX1100, OR APPROVED EQUIVALENT, GEOGRID PER MFR'S INSTRUCTIONS; 5" OF 3/4" CLASS 2 AGGREGATE BASE (METHOD II, 95% MRC); AND 3" OF HOT ASPHALT CONCRETE PAVEMENT.

   CROSS SLOPE THE ROAD AT 2% TO THE SOUTHEAST TO DRAIN. FINISHED GRADE OF THE ROAD TO BE A MINIMUM OF 3" ABOVE EXISTING GRADE ON EITHER SIDE OF THE ROAD.

   FOR ROAD CONSTRUCTION OVER THE EXISTING LEACHFIELD TRENCHES, AFTER THE EXISTING LEACH PIPE IS REMOVED FROM THE TRENCHES, COVER THE TOP OF THE EXISTING LEACH TRENCH DRAIN ROCK WITH MIRAFI 600X, OR APPROVED EQUIVALENT, GEOTEXTILE FABRIC. COORDINATE WITH 6" SEWER CONSTRUCTION.

   SEE CONSTRUCTION NOTE 8
10. INSTALL 6" GRAVITY–FLOW SDR 26 PVC SEWER PIPELINE – SEE JOINT UTILITY TRENCH DTL. 5/D5.11. COORDINATE ALIGNMENT OF 6" SEWER, RELATIVE TO EX. PIPELINES, WITH COR.
11. CONSTRUCT SSMH NO. 9 WITH FLAT SLAB, TRAFFIC–RATED TOP. SEE DTL. 5/D5.21 FOR TYP. SSMH CONSTRUCTION. COORDINATE FRAME & COVER (RIM) GRADE WITH PROPOSED GRADING AND PAVING. SEE NOTE 15.
12. INSTALL 2" SCH. 80 PVC ELEC. POWER CONDUIT AND WIRING TO OWTS PUMP CONTROL PANEL (PCP) FROM EX. PANEL "A" NEAR KIOSK – SEE E5.02, E5.40 AND DTL. 5/D5.11.
13. INSTALL 2" SCH. 80 PVC SCADA FIBER OPTIC CABLE CONDUIT TO PCP – SEE E5.02, E6.01 AND DTL. 5/D5.11.
14. INSTALL OLDCASTLE MODEL B1730, OR APPROVED EQUIVALENT, 17"x30" I.D. TRAFFIC–RATED UTILITY BOX WITH HOT–DIPPED GALV. STEEL FRAME AND BOLT–DOWN CHECKER PLATE COVER, REINFORCED CONC. BOX EXTENSION. LABEL COVER "CONTROLS". SET BOX ON MIN. 4 SOLID CONC. BUILDING BLOCKS (2"x4"x8" MIN. SIZE) AND 6" MIN. LAYER OF 3/4" DRAIN ROCK. ROCK TO EXTEND AT LEAST 6" BEYOND PERIMETER OF BOX. SEAL CONDUIT ENDS WATERTIGHT. SEE E5.02 AND DTL. 2/D5.71.
15. INSTALL TRACER WIRE BOX PER DTL. 4/D5.21 FOR ELONGATED CONCRETE COLLAR AND DTL. 1/D5.21 FOR SNOW PLOW PROTECTIVE APRON.
16. INSTALL TRACER WIRE BOX PER DTL. 4/D5.21 FOR ELONGATED CONCRETE COLLAR AT TANK RISER FRAME AND COVER.
17. INSTALL TRAFFIC–RATED BOX FOR PCP POWER SUPPLY. LABEL COVER "ELECTRICAL". SEE NOTE 14.

NOTES

1. SEE DRAWINGS C7.40–C7.42 FOR TANK AND EQUIPMENT YARD CONSTRUCTION.
2. SEE DRAWING C7.50 FOR LEACHFIELD CONSTRUCTION.

PHASE 5

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA
MARCH 14, 2016

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A/E FIRM
PRIME: DAVID EVANS AND ASSOCIATES, INC. BELLEVUE, WA
SUBCONTRACTOR LAUGENOUR AND MEIKLE WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.35

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS FOUR MILE ROAD
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 103 OF 291

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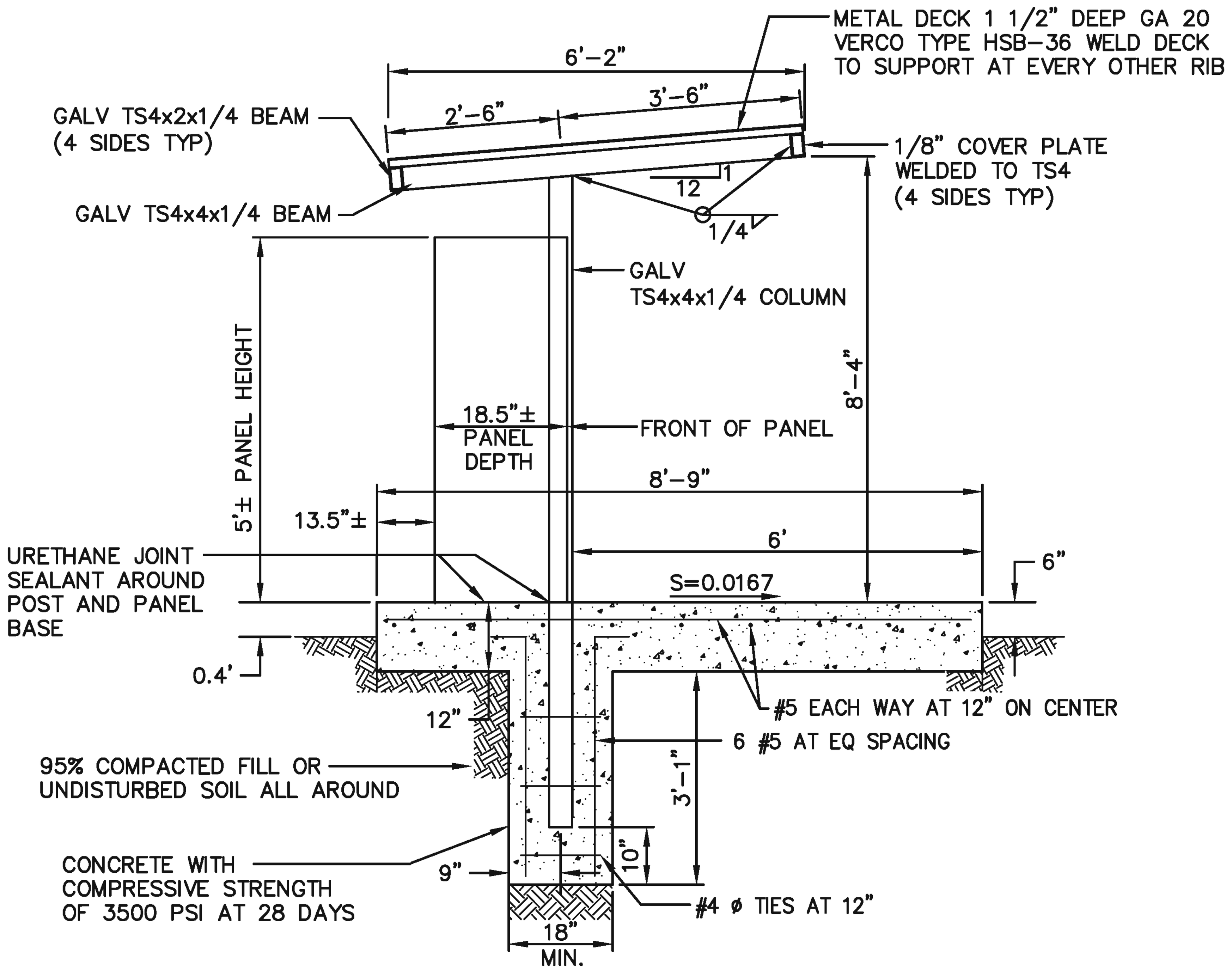
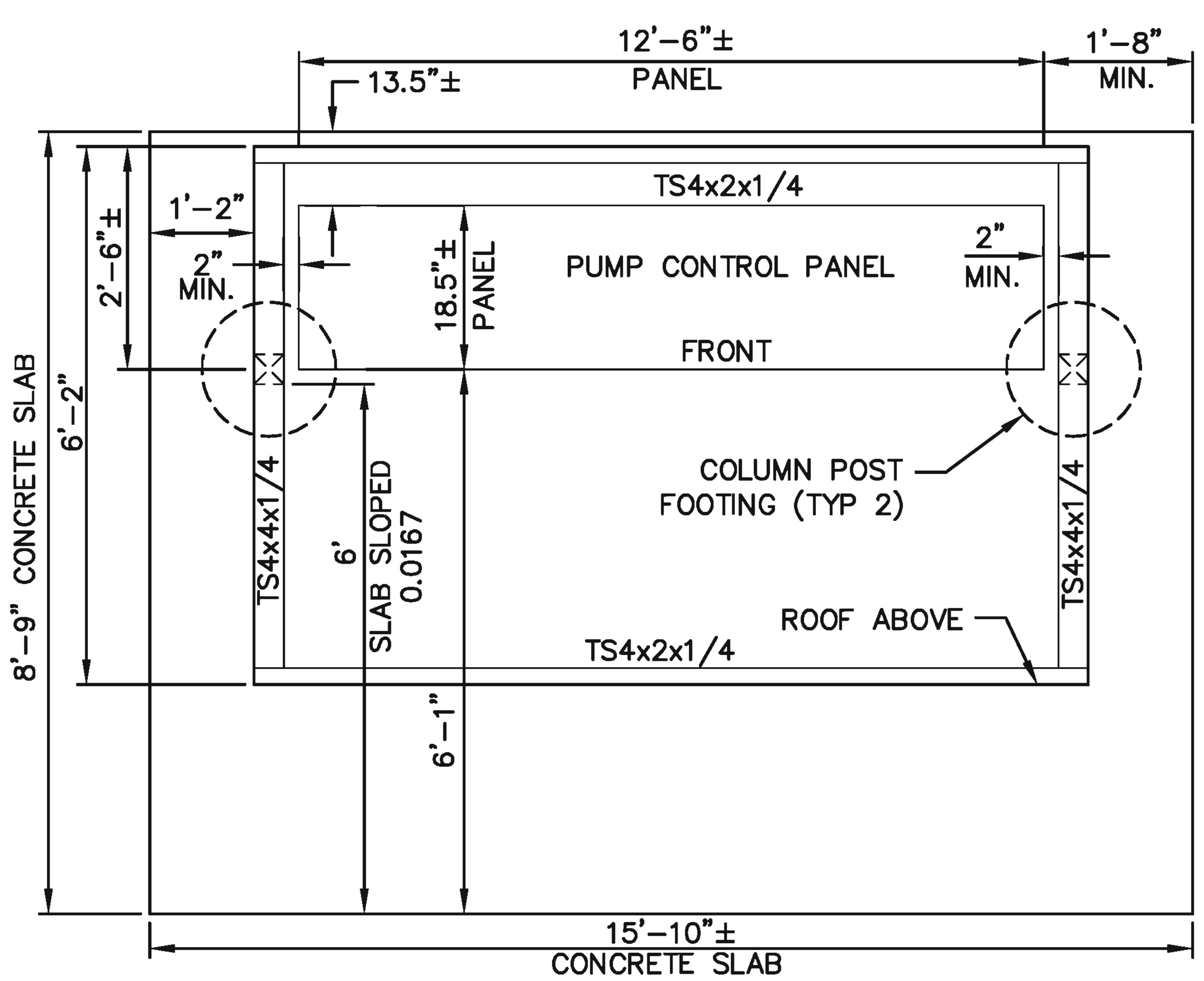
NOTES

1. VERIFY ALL PUMP CONTROL PANEL DIMENSIONS WITH TESCO.
2. SEE C7.40 FOR CONCRETE SLAB GRADES.
3. PAINT ALL METAL EXPOSED SURFACES SEAL BROWN-VERIFY WITH COR. SEE SPECIFICATIONS.



SECTION AT COLUMN



PLAN

1 DOSING PUMP CONTROL PANEL SUNSHADE DETAIL

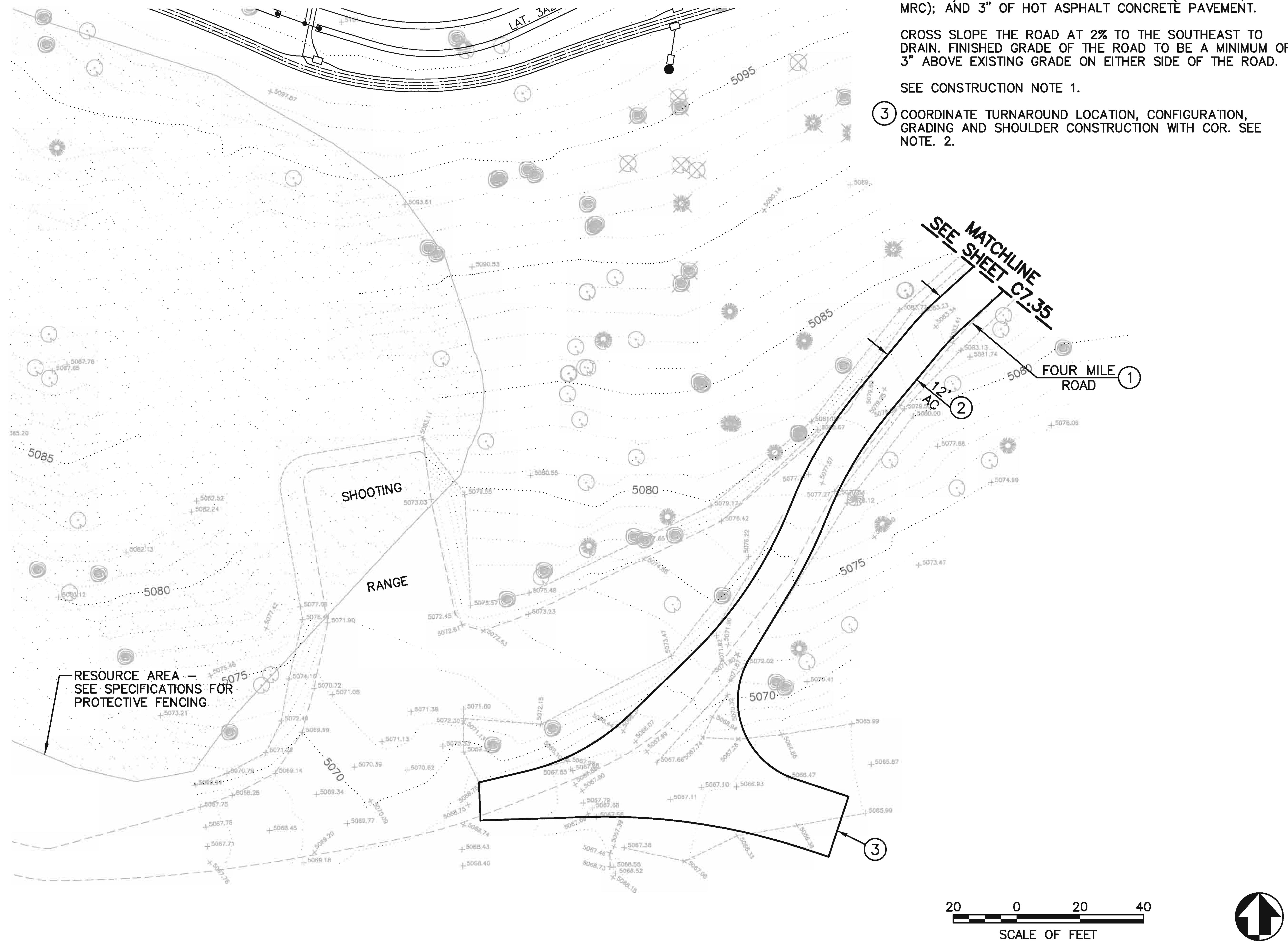
NOT TO SCALE

CONSTRUCTION NOTES

1. FOUR MILE ROAD: THIS EXISTING SERVICE ROAD VARIES IN WIDTH AND CONDITION. PORTIONS ARE DIRT ROAD, DIRT/GRAVEL ROAD, AND DIRT/GRAVEL OVER REMNANTS OF AC PAVING.
2. RECONSTRUCT FOUR MILE ROAD WITH A 12' MIN. WIDE PAVED ROADWAY, INCLUDING: DEMO; REGRADING AS REQUIRED FOR DRAINAGE AND SMOOTHING OF ROADWAY LONGITUDINAL AND CROSS SLOPES; 8" MIN. SUBGRADE SCARIFICATION, MOISTURE CONDITIONING AND COMPACTION TO 95% MRC; INSTALLATION OF TENSAR BX1100, OR APPROVED EQUIVALENT, GEOGRID PER MFR'S INSTRUCTIONS; 5" OF 3/4" CLASS 2 AGGREGRATE BASE (METHOD II, 95% MRC); AND 3" OF HOT ASPHALT CONCRETE PAVEMENT.

   CROSS SLOPE THE ROAD AT 2% TO THE SOUTHEAST TO DRAIN. FINISHED GRADE OF THE ROAD TO BE A MINIMUM OF 3" ABOVE EXISTING GRADE ON EITHER SIDE OF THE ROAD.

   SEE CONSTRUCTION NOTE 1.
3. COORDINATE TURNAROUND LOCATION, CONFIGURATION, GRADING AND SHOULDER CONSTRUCTION WITH COR. SEE NOTE. 2.



PHASE 5

THOMAS G. HORGAN No. 24772 CIVIL REGISTERED PROFESSIONAL ENGINEER STATE OF CALIFORNIA

MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM

PRIME:
DAVID EVANS AND ASSOCIATES, INC.
BELLEVUE, WA

SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.36

TITLE OF SHEET

SOUTH ENTRANCE OWTS PLANS FOUR MILE ROAD

YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 104 OF 291

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CONSTRUCTION NOTES

1. SEE DWGS. C7.33 – C7.35 FOR 2" SCH. 80 PVC POWER AND CONTROLS (SCADA FIBER OPTIC CABLE) CONDUITS TO PUMP CONTROL PANEL.
2. SEE DWG. C7.35 FOR 6-INCH SEWER PIPING TO DOSING TANK, INCLUDING SSMH NO. 9, AND POWER AND SCADA CONDUITS.
3. SEE DWG. C7.35 FOR PROPOSED FOUR MILE ROAD GRADING AND PAVING.
4. SEE DWG. C7.35 FOR EXISTING LEACH LINES TO BE REMOVED.
5. EXISTING ROADWAY WIDTH VARIES. SEE DWG. C7.35.
6. PRESERVE EXISTING OAK TREE. COORDINATE WITH COR.
7. CONSTRUCT SERVICE ROAD, INCLUDING: DEMOLITION; EXCAVATION AND GRADING; 8-INCH MINIMUM SUBGRADE SCARIFICATION, MOISTURE CONDITIONING AND COMPACTION TO 95% MRC; INSTALLATION OF TENSAR BX1100, OR APPROVED EQUIVALENT, GEOGRID PER MANUFACTURER'S INSTRUCTIONS; AND 6 INCHES OF 3/4-INCH CLASS 2 AGGREGATE BASE (METHOD II, 95% MRC).

   AT SERVICE ROAD CONNECTIONS TO FOUR MILE ROAD, COORDINATE SERVICE ROAD GRADING WITH FOUR MILE ROAD REGRADING – SEE COR.
8. COORDINATE TREE REMOVAL AND GRADING OF THE AREA BETWEEN FOUR MILE ROAD AND THE SERVICE ROAD WITH THE ENGINEER AND COR. INTERIM GRADING OF THIS AREA MAY BE NEEDED TO ACCOMMODATE DOSING TANK DELIVERY AND INSTALLATION. SEE NOTE 6.
9. CONSTRUCT YARD AREA, INCLUDING: DEMOLITION; EXCAVATION AND GRADING; 8-INCH MINIMUM SUBGRADE SCARIFICATION, MOISTURE CONDITIONING AND COMPACTION TO 95% MRC; AND 6 INCHES OF 3/4-INCH CLASS 2 AGGREGATE BASE (METHOD II, 95% MRC). COORDINATE WITH THE ENGINEER AND COR.
10. CONSTRUCT LINED SURFACE DRAINAGE INTERCEPTOR DITCH PER DTL. 3/D5.42. COORDINATE WITH COR AND ENGINEER.
11. CONSTRUCT 15-FOOT LONG EROSION CONTROL APRON AT END OF LINED DITCH USING DITCH LINING MATERIAL. COORDINATE WITH COR AND ENGINEER.
12. INSTALL 20,000 GALLON DOSING TANK PER C7.41.
13. INSTALL PAD-MOUNTED TESCO PUMP CONTROL PANEL (PCP) INCLUDING CONDUIT, WIRING AND CONNECTIONS TO PUMPS, BASINS, METERS, PRESSURE TRANSDUCERS, AND LEACHFIELD DISTRIBUTING VALVE MONITORS. SEE DWGS. E6.02 – E6.07. THE SCADA MODULE IS TO BE POSITIONED INSIDE THIS PCP.

    CONSTRUCT 12-INCH MINIMUM THICK REINFORCED CONCRETE PAD WITH SUNSHADE FOR PCP PER DTL. 1/C7.36. SECURELY ANCHOR PANEL TO PAD WITH EXPANSION TYPE ANCHOR BOLTS. USE HEAVY-DUTY URETHANE TYPE SEALANT BETWEEN PANEL AND PAD FOR DURABLE WATERTIGHT CONNECTION. COORDINATE CONFIGURATION OF UNDERGROUND CONDUITS ENTERING PANEL THROUGH CONCRETE SLAB TO FACILITATE CONDUCTOR AND CABLE CONNECTIONS INSIDE PCP. SUNSHADE COLUMNS AND FOOTINGS ARE NOT SHOWN – SEE DTL. 1/C7.36.
14. INSTALL CODE-APPROVED GROUNDING, INCLUDING PRECAST CONCRETE BOX AND CAST IRON FRAME AND COVER ("GROUND ROD" MARKING), GROUND ROD, CONDUCTOR, CONNECTOR, CONDUIT WITH BUSHING, LOCKNUT AND JUMPER. SEE ELECTRICAL PLAN.
15. SEE C7.35 FOR CONTROLS (SCADA) UTILITY BOX.
16. SEE C7.35 FOR ELECTRICAL UTILITY BOX.
17. SEE C7.35 FOR CONTROLS UTILITY BOX. SEE C7.50 FOR UTILITIES EXTENDING TO LEACHFIELD.
18. INSTALL TWO (ONLY 1 IS SHOWN FOR EACH METER & PRESSURE TRANSMITTER ASSEMBLY) 1-INCH MINIMUM SIZE, SCH. 40 PVC ELECTRICAL CONDUITS WITH METER AND PRESSURE TRANSMITTER MANUFACTURERS' RECOMMENDED CABLES. USE WATERTIGHT CONDUIT END SEALS. SEE C7.42. CONDUITS HORIZONTAL SPACING IS SHOWN EXAGGERATED FOR CLARITY.
19. INSTALL 2-INCH FLOW METER, PRESSURE TRANSDUCER, VALVING AND ENCLOSURE PER C7.42.
20. FOR EACH DUPLEX PUMP UNIT INSTALL TWO 1-INCH SCH. 40 PVC ELECTRICAL CONDUITS (ONLY ONE IS SHOWN FOR CLARITY) AND WIRING FROM PUMP CONTROL PANEL TO POWER AND CONTROLS SPLICE BOXES INSIDE BASIN RISER. CONTROLS CONDUIT TO INCLUDE FLOATS WIRING AND BASIN WATER LEVEL TRANSMITTER MFR'S CABLING. SEE C7.42. CONDUITS HORIZONTAL SPACING IS SHOWN EXAGGERATED FOR CLARITY. SEE ELEC. PLANS.
21. INSTALL TWO IDENTICAL 72-INCH (I.D.) PRECAST REINFORCED CONCRETE DOSING PUMP BASINS, PUMP BASIN 1-2 AND BASIN 3-4, WITH PUMPS AND DOUBLE-LEAF ACCESS HATCHES PER C7.42.
22. INSTALL DOSING TANK 4-INCH OUTLET PIPING PER C7.41.
23. INSTALL DOSING TANK OUTLET PIPE VALVES PER C7.41.
24. INSTALL DOSING PUMP BASIN VENT PIPING PER C7.41.
25. INSTALL TRACER WIRE BOX FOR 6-INCH SEWER PER C7.35.
26. INSTALL 6 BOLLARDS PER DTL. 3/D5.18, EXCEPT THAT CONC. BASES ARE TO BE 18" MIN. DIAMETER AND 4' MIN. DEPTH. COORDINATE WITH COR.
27. SEE C7.50 – C7.51 FOR EMPTY 3-INCH PRESSURE PIPELINE FOR POSSIBLE FUTURE USE. MARK PIPE END WITH 5-FOOT LONG, 3-INCH PIPE STUB INSTALLED VERTICALLY, WITH TOP 2 FEET ABOVE GRADE.
28. SEE C7.50 – C7.51 FOR CONDUIT AND WIRING FROM DISTRIBUTING VALVE MONITORS AND MONITORING WELL TRANSDUCERS.
29. SEE C7.50 – C7.51 FOR FOUR 3-INCH HDPE PUMP DISCHARGE PIPELINES FROM DUPLEX PUMP UNITS TO DISTRIBUTING VALVES. PIPELINES HORIZONTAL SPACING IS SHOWN EXAGGERATED FOR CLARITY. INSTALL PIPING AND CONDUITS IN COMMON TRENCH.
30. REGRADE SLOPE – COORDINATE WITH COR AND ENGINEER.
31. GRADE DRAINAGE SWALE TO DRAIN. COORDINATE WITH COR AND ENGINEER.
32. GRADE YARD AREA TO DRAIN. COORDINATE WITH COR AND ENGINEER.
33. SEE DWG. C7.42 FOR HATCH FRAME DRAIN.
34. TO THE EXTENT PRACTICAL, PRESERVE EXISTING TREE (TYP.). COORDINATE TREE PRESERVATION AND TREE REMOVAL WITH COR.

PHASE 5

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER
THOMAS G. HORGAN
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CIVIL
STATE OF CALIFORNIA
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TITLE OF SHEET
SOUTH ENTRANCE
OWTS PLANS
PUMP AND TANK YARD
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104
127662
PMIS/PKG NO. 206431
SHEET 105 OF 291

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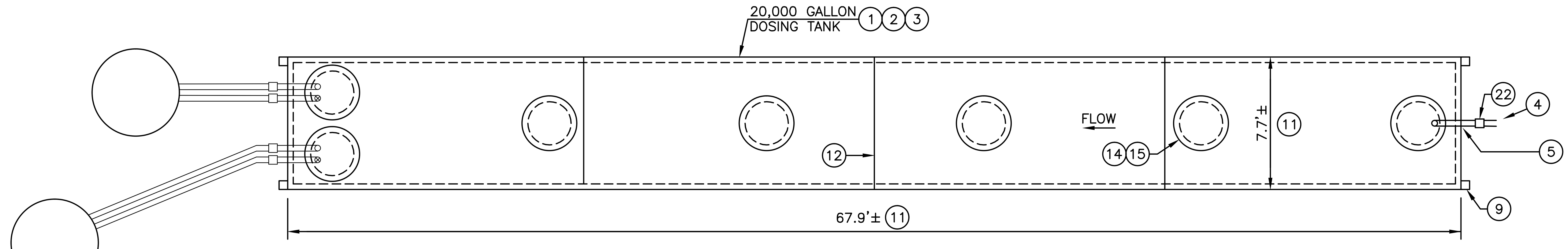
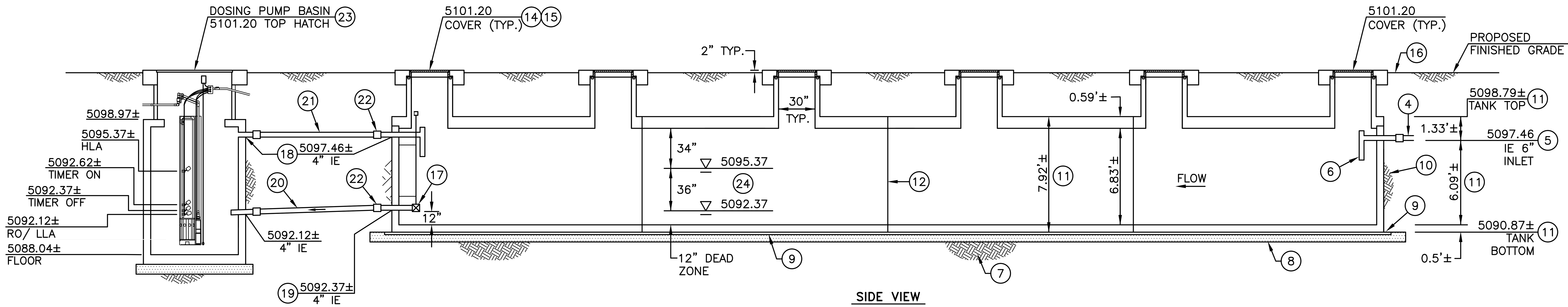


PLAN VIEW



SIDE VIEW

CONSTRUCTION NOTES

1. SEE OWTS TANK GENERAL NOTE 21 ON SUB-SHEETS NOS. C7.62–C7.63 FOR ADDITIONAL TANK CONSTRUCTION AND INSTALLATION REQUIREMENTS, INCLUDING PIPE-TO-TANK CONNECTIONS, INTERIOR PROTECTIVE COATING AND BUOYANCY COUNTERACTION.
2. SEE SUB-SHEET C7.10 FOR OVERALL OWTS PLAN.
3. INSTALL 20,000 GALLON PRECAST REINFORCED CONCRETE DOSING TANK. TANK CONFIGURATION SHOWN ON THIS PLAN IS BASED A BATTERY OF FOUR JENSEN PRECAST MODEL JZ5000 TANKS. AN ENGINEER-APPROVED EQUIVALENT TANK COMPLYING WITH THESE PLANS IS ACCEPTABLE FOR INSTALLATION. APPROXIMATELY 285 GALLONS PER INCH OF WATER-DEPTH.
4. SEE SUB-SHEETS C7.35 AND C7.40 FOR 6" SDR 26 PVC SEWER PIPELINE FROM SOUTH ENTRANCE AREA.
5. POINT OF CONNECTION (POC) TO TANK 6" INLET OPENING.
6. INSTALL 6" PVC SANITARY INLET TEE.
7. EARTH SUBGRADE SHALL BE CONSTRUCTED TO PROPER LINE AND GRADE IN FIRM UNDISTURBED SOIL. IF SOIL IS NOT FIRM AND UNDISTURBED, PREPARE SUBGRADE AND COMPACT TO 95% MINIMUM RELATIVE COMPACTION – SEE PROJECT EARTHWORK SPECIFICATIONS.
8. INSTALL LEVEL, MINIMUM 8" LAYER OF 3/4" CLASS 2 AB AT 95% MRC.
9. IF REQUIRED BY THE TANK MANUFACTURER, INSTALL 2'x6" REDWOOD GRADE BOARDS LENGTHWISE UNDER SIDES OF TANK. EXTEND BOARDS 1'± BEYOND TANK ENDS.
10. COMPACTED EARTH BACKFILL AT 95% MINIMUM RELATIVE COMPACTION. INSTALL TANK IN ACCORD WITH MANUFACTURER'S RECOMMENDATIONS.
11. VERIFY ALL TANK DIMENSIONS WITH TANK MANUFACTURER AND COORDINATE WITH THE COR AND THE DESIGN ENGINEER.
12. JOB SITE JOINT SEALING OF SECTIONAL TANKS SHALL COMPLY WITH MANUFACTURER'S RECOMMENDATIONS.
13. SEE SUB-SHEET C7.40 FOR PROPOSED GRADING AND YARD SURFACING STRUCTURAL SECTION THICKNESS AND TO VERIFY FINISHED GRADES. CAREFULLY CONSTRUCT GRADING AND YARD SURFACING STRUCTURAL SECTION AND SUBGRADE TO PREVENT DAMAGE TO THE TANK AND PIPING.
14. INSTALL AASHTO HS20-44 HEAVY-TRAFFIC RATED, 30" INSIDE DIAMETER, PRECAST CONCRETE RISERS, CAST-IRON FRAMES AND COVERS (TYP.). COVERS NOT SHOWN IN PLAN VIEW.
15. EACH FRAME AND COVER ASSEMBLY IS TO BE INSTALLED WITH A CAST-IN-PLACE, HEAVY-DUTY, REINFORCED CONCRETE COLLAR. SEE DETAIL 3/D5.21.
16. SEE SUB-SHEET C7.35 FOR TRACER WIRE BOX WITH ELONGATED CONCRETE COLLAR.
17. INSTALL 4" STAINLESS STEEL PLUG VALVE WITH SECURELY BRACED, EXTENDED STEM AND 2" SQUARE OPERATING NUT. SEE GENERAL NOTE 29/C7.64. ATTACH VALVE TO SHORT LENGTH OF 4" SCH. 80 PVC PIPE THAT EXTENDS TO COUPLING OUTSIDE OF TANK – SEE NOTE 22.
18. NOTE THAT 4" VENT INVERT IS SAME ELEVATION AS 6" SEWER INVERT (NOTE 5).
19. POC TO TANK 4" OUTLET OPENING.
20. INSTALL 4" SDR 26 PVC PIPELINE TO DOSING PUMP BASIN. SEE SUB-SHEET C7.40 AND DTL. 3/D5.11.
21. INSTALL 4" SDR 26 PVC VENT PIPING TO DOSING PUMP BASIN.
22. INSTALL MISSION RUBBER CO. FLEX-SEAL ADJUSTABLE REPAIR COUPLING, OR APPROVED EQUIVALENT, WITH SS SHEAR RING (TYP.).
23. SEE DWG. C7.42 FOR DOSING PUMP BASINS. ONLY ONE BASIN IS SHOWN IN THIS SIDE VIEW FOR CLARITY. BOTH BASINS ARE IDENTICAL.
24. A. DOSING SYSTEM OPERATIONAL VOLUME AND STORAGE:

    2 PUMP BASINS = 35 GAL/INCH
    DOSING TANK = 285
    TOTAL = 320 GAL/INCH

    B. OPERATIONAL VOLUME REQUIRED:

    11,400 GAL (NOTE 5/DTL. 1/C7.02).
    11,400 GAL ÷ 320 GAL/INCH = 36". 36" IS PROVIDED.

    C. THE PARK'S PROPOSED PREPARATIONS FOR AN EMERGENCY EVENT AT THE DOSING SYSTEM INCLUDE THE FOLLOWING ITEMS:

    1. MARIPOSA CO. EHS EMERGENCY STORAGE REQUIREMENT:

       19,920 GAL (PEAK DAY) x 0.45 = 9,000 GAL.
       9,000 GAL ÷ 320 GAL/INCH = 28". 34" IS AVAILABLE TO FLOOD THE TANK TO SOFFIT LEVEL.

    2. PERMANENT GENSET FOR BACK-UP POWER SUPPLY. THIS GENSET IS ALREADY IN PLACE AT THE SOUTH ENTRANCE STATION AND IT'LL BE CONNECTED TO THE DOSING PUMP SYSTEM POWER SUPPLY.
    3. POWER PLUG ON THE PUMP CONTROL PANEL FOR CONNECTION OF PORTABLE GENSET.
    4. PUMPER TRUCK STATIONED AT WAWONA.
    5. CAPABILITY TO CLOSE RESTROOMS AT THE SOUTH ENTRANCE WHEN NEEDED.
    6. SPARE DUPLEX DOSING PUMP ASSEMBLY STORED AT WAWONA.

1 20,000 GAL. DOSING TANK
NOT TO SCALE

PHASE 5

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA
MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

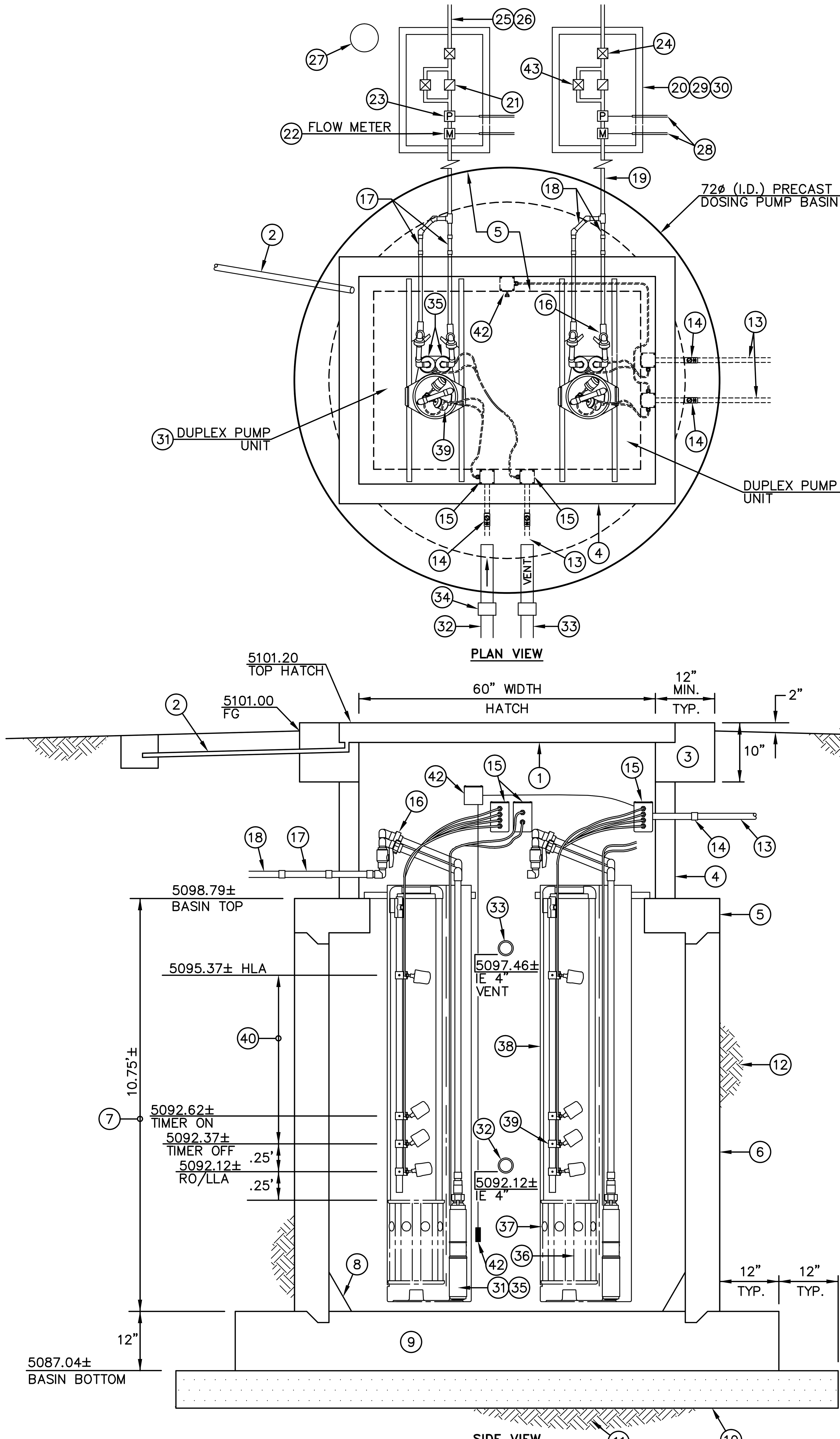
A/E FIRM
PRIME: DAVID EVANS AND ASSOCIATES, INC. BELLEVUE, WA
SUBCONTRACTOR LAUGENOUR AND MEIKLE WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.41

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS DOSING TANK
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 / 127662
PMIS/PKG NO. 206431
SHEET 106 OF 291



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DOSING PUMP BASIN AND PUMPS

NTS

CONSTRUCTION NOTES

1. INSTALL BILCO MDL. JD–H20–R, OR APPROVED EQUIVALENT, 42" LENGTH (HINGE SIDE), 60" WIDTH, DOUBLE–LEAF ACCESS HATCH. SEE GENERAL NOTE 28/C7.64.
2. INSTALL 1.5" SCH. 80 GALV. STEEL GRAVITY–FLOW DRAIN PIPE FROM HATCH FRAME DRAIN COUPLING TO DAY–LIGHT. WRAP PIPE WITH TWO WRAPS OF 10 MIL PVC TAPE. SET PIPE OUTLET IN 2'x2'x2' DRY WELL FILLED WITH 3/4" DRAIN ROCK, COORDINATE DRAIN PIPE INSTALLATION WITH COR.
3. CONSTRUCT CAST–IN–PLACE, FULL PERIMETER, REINFORCED CONCRETE COLLAR FOR HATCH FRAME ANCHORAGE. MIN. COLLAR SIZE IS 12" WIDE BY 10" DEEP, WITH TWO NO. 4 REBARS AT TOP AND BOTTOM. (NOT SHOWN IN PLAN VIEW.) SEE NOTE 2.
4. CONSTRUCT RECTANGULAR, PRECAST OR CAST–IN–PLACE, REINFORCED CONCRETE RISER WITH SECURE, WATERTIGHT CONNECTION TO FLAT–SLAB TOP SECTION BELOW AND TO CONCRETE COLLAR ABOVE. FIELD VERIFY RISER HEIGHT. COORDINATE WITH ENGINEER.
5. CONSTRUCT PRECAST, REINFORCED CONCRETE, FLAT–SLAB SSMH TOP SECTION WITH RECTANGULAR OPENING FOR RISER. COORDINATE OPENING CONFIGURATION WITH PUMP VAULT AND HATCH SIZES AND WITH ENGINEER.
6. INSTALL 72"ø (I.D.) SSMH–TYPE PRECAST REINFORCED CONCRETE DOSING PUMP BASIN. SEE GENERAL NOTE 28/C7.64.
7. VERIFY ALL BASIN DIMENSIONS WITH BASIN MANUFACTURER.
8. PROVIDE GROUTED 60° SLOPED WALLS IN BOTTOM OF BASIN.
9. CONSTRUCT 12" THICK, CAST–IN–PLACE, CONCRETE BASIN BASE REINFORCED WITH NO. 5 REBARS AT 12" CC, TOP AND BOTTOM, AND WATERTIGHT CONNECTION TO BARRELL SECTION.
10. INSTALL LEVEL, MINIMUM 8" LAYER OF 3/4" CLASS 2 AB AT 95% MRC.
11. EARTH SUBGRADE SHALL BE CONSTRUCTED TO PROPER LINE AND GRADE IN FIRM UNDISTURBED SOIL. IF SOIL IS NOT FIRM AND UNDISTURBED, PREPARE SUBGRADE AND COMPACT TO 95% MINIMUM RELATIVE COMPACTION – SEE PROJECT EARTHWORK SPECIFICATIONS.
12. COMPACTED EARTH BACKFILL AT 95% MINIMUM RELATIVE COMPACTION. INSTALL BASIN IN ACCORD WITH MANUFACTURER'S RECOMMENDATIONS.
13. FOR EACH DUPLEX PUMP UNIT INSTALL TWO 1" SCH. 40 PVC ELECTRICAL CONDUITS AND WIRING FROM PUMP CONTROL PANEL TO SEPARATE POWER AND CONTROL SPLICE BOXES INSIDE RISER. USE FLEXIBLE,WATERTIGHT SEAL AT ALL RISER PENETRATIONS (TYP.). POWER CONDUIT TO HAVE FOUR #10 (MIN. GAUGE) CONDUCTORS AND TWO #10 GRD FOR PUMPS. CONTROLS CONDUIT TO HAVE EIGHT #14 CONDUCTORS AND ONE #14 GRD FOR FLOATS AND BASIN WATER LEVEL SENSOR MFR'S CABLING. INSTALL CONDUITS WITH 24" MIN. COVER WHERE PRACTICAL. ALL WORK TO COMPLY WITH LOCAL CODES. SEE E6.05–E6.06.
14. INSTALL ORENCO SBCS, OR APPROVED EQUIVALENT, CONDUIT SEAL KIT ON EACH OF THE TWO CONDUITS (TYP.).
15. INSTALL INTERNAL SPLICE BOX. USE ORENCO MODEL SB, OR APPROVED EQUIVALENT, NEMA 4X, PVC INTERNAL SPLICE BOX WITH GROMMET (INSTALLED WITH ADHESIVE/SEALANT) AND WATERPROOF WIRE NUTS AND W/CORD GRIPS FOR PUMPS & FLOATS. POSITION BOXES SECURELY ONTO RISER SIDEWALL SO FILTER CARTRIDGE MAY BE EASILY REMOVED FOR MAINTENANCE. COORDINATE BOX PLACEMENT WITH ENGINEER.
16. INSTALL TWO ORENCO MODEL HV 200 BCXQPR, OR APPROVED EQUIVALENT, 2" DISCHARGE ASSEMBLIES WITH BALL VALVE AND CHECK VALVE, MODIFIED WITH THE ADDITION OF A 2" BALL VALVE AND PIPING LOCATED INSIDE THE RISER & POSITIONED TO FACILITATE DRAINING (TO THE EXTENT PRACTICAL) OF THE 3" TRANSMISSION PIPELINE, THAT EXTENDS TO THE LEACHFIELD, DURING MAINTENANCE SHUT–DOWN OF THE OWTS. POSITION PIPING SO FILTER CARTRIDGE MAY BE EASILY REMOVED FOR MAINTENANCE. COORDINATE WITH ENGINEER.
17. INSTALL TWO ORENCO MODEL HVX200PR, OR APPROVED EQUIVALENT, 2" PVC EXTERNAL FLEXIBLE HOSES, SHOWN IN ROTATED VIEW FOR CLARITY – SEE PLAN VIEW.
18. INSTALL 2" SCH. 40 PVC PUMP DISCHARGE PIPING. SEE NOTE 16.
19. INSTALL 2" SCH. 40 PVC PUMP DISCHARGE PIPING. SEE NOTE 16.
20. INSTALL OLDCASTLE CHRISTY N52 (30"x60" I.D.) PRECAST REINFORCED CONCRETE UTILITY BOX WITH BOX EXTENSIONS AND 2–PIECE GALV. STEEL CHECKER PLATE COVER WITH HOLD DOWN BOLTS. LEGIBLY BEAD–WELD METER NO. TO COVER PRIOR TO HOT–DIP GALVANIZING. VERIFY FIT–UP OF PIPING COMPONENTS INSIDE BOX. COORDINATE WITH ENGINEER.
21. INSTALL ORENCO KSC 2000, OR APPROVED EQUIVALENT, (REDUNDANT) SWING CHECK VALVE WITH UNIONS.
22. INSTALL 2" FLOW METER WITH TRANSMITTER AND DISPLAY LOCATED INSIDE PCP – SEE GENERAL NOTE 25/C7.63.
23. INSTALL DISCHARGE PIPELINE PRESSURE TRANSMITTER PER GENERAL NOTE 31/C7.64.
24. INSTALL SCH. 80 PVC TRUE UNION GATE VALVE TO EQUALIZE FLOWS TO LEACHFIELD CELLS AS MAY BE NEEDED.
25. INSTALL SMOOTH, HEAVY–DUTY TRANSITION FROM 2" PVC PIPE TO 3" HDPE PIPING.
26. SEE C7.50 FOR 3" HDPE PIPING TO DISTRIBUTING VALVES.
27. INSTALL TRACER WIRE BOX – SEE C7.40.
28. INSTALL TWO 1" SCH. 40 PVC ELECTRICAL CONDUITS WITH METER AND PRESSURE TRANSMITTER MFRS' RECOMMENDED CABLES, AND CONNECTIONS IN PCP. USE WATERTIGHT CONDUIT END SEALS. SEE NOTES 22 & 23.
29. INSTALL 12" MIN. THICKNESS OF 3/4" DRAIN ROCK UNDER BOX. ROCK TO EXTEND AT LEAST 6" BEYOND EXTERIOR OF BOX. INSTALL WIRE RODENT BARRIER – SEE DTL. 1/C7.53.
30. INSTALL DURABLE R30 INSULATION PLUG WITH STRAP FOR EASY REMOVAL UNDER BOX COVER.
31. PUMPING UNIT ASSEMBLIES SHOWN ARE REFERENCE SPECIFIED AS BEING ORENCO SYSTEMS PRODUCTS. AN ENGINEER AND COR APPROVED EQUIVALENT PUMPING UNIT ASSEMBLY COMPLYING WITH THESE PLANS IS ACCEPTABLE FOR INSTALLATION. SOME EQUIPMENT POSITIONS ARE SHOWN ROTATED OR SHIFTED FOR CLARITY.

    DUPLEX PUMP UNITS NOS. 1, 2, 3 AND 4 ARE TO BE IDENTICAL.
32. SEE SUB–SHEET C7.41 FOR 4" EFFLUENT PIPELINE FROM DOSING TANK.
33. SEE DWG. C7.41 FOR 4" AIR VENT TO DOSING TANK.
34. INSTALL MISSION RUBBER CO. FLEX–SEAL 4"x4" ADJUSTABLE REPAIR COUPLING, OR APPROVED EQUIVALENT, WITH SS SHEAR RING (TYP.).
35. INSTALL ORENCO MODEL PF501512 (240 VOLT), OR APPROVED EQUIVALENT, TIMED–DOSE, 55± GPM, 1.5 HP, 12.6 FLA, HIGH–HEAD, TURBINE TYPE DUPLEX EFFLUENT PUMPS.
36. ORENCO, OR APPROVED EQUIVALENT, FILTER CARTRIDGE.
37. ORENCO, OR APPROVED EQUIVALENT, PUMP VAULT INLET PORTS.
38. INSTALL ORENCO CUSTOM MODEL PVUP 126–3625, OR APPROVED EQUIVALENT, BIOTUBE PUMP VAULT WITH 1/16" MESH FILTRATION. COORDINATE VAULT SIZE PER NOTES 7 AND 31. VAULTS ARE SHOWN ROTATED FOR CLARITY.
39. INSTALL ORENCO MODEL MF4V, OR APPROVED EQUIVALENT, LEVEL CONTROL FLOAT ASSEMBLY, ALL NORMALLY OPEN, WITH 2" TETHER LENGTH. USE 1" SCH. 40 PVC PIPE FLOAT STEM. VERIFY FLOAT SETTINGS WITH ACTUAL PUMP AND BASIN DIMENSIONS. COORDINATE WITH ENGINEER AND COR. FLOAT MEASUREMENTS SHOWN ARE TO CENTER OF ADJUSTABLE FLOAT COLLAR.
40. SEE C7.41 FOR DOSING SYSTEM OPERATIONAL VOLUME.
41. SEE SUB–SHEETS C7.40 AND C7.41 FOR PROPOSED YARD GRADING AND AB SURFACING. GRADE AROUND ACCESS OPENINGS TO DIRECT SURFACE DRAINAGE AWAY FROM ACCESS OPENINGS (TYP.). COORDINATE WITH COR AND ENGINEER.
42. INSTALL BASIN WATER LEVEL TRANSMITTER ASSEMBLY INCL. 1" SCH. 40 PVC CONDUIT AND CABLING. SEE GEN. NOTE 30/C7.64.
43. INSTALL SCH. 80 PVC TRUE UNION 1.5" BALL VALVE AND PIPING TO BYPASS CHECK VALVE FOR DRAINING OF THE 3" TRANSMISSION PIPELINE.

PHASE 5

Thomas G. Horgan

REGISTERED PROFESSIONAL ENGINEER
THOMAS G. HORGAN
No. 24772
CIVIL
STATE OF CALIFORNIA

MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM	
PRIME: DAVID EVANS AND ASSOCIATES, INC. BELLEVUE, WA	DESIGNED: TGH
	CADD: MSW
SUBCONTRACTOR LAUGENOUR AND MEIKLE WOODLAND, CA	TECH. REVIEW: TGH
	DATE: 2/26/2016

SUB SHEET NO. C7.42

TITLE OF SHEET

SOUTH ENTRANCE OWTS PLANS DOSING PUMP BASINS

YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662

PMIS/PKG NO. 206431

SHEET 107 OF 291

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PHASE 5

Thomas G. Horgan

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WOODLAND, CA

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DATE: 2/26/2016

SUB SHEET NO. C7.50

TITLE OF SHEET

SOUTH ENTRANCE OWTS PLANS LEACHFIELD

YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662

PMIS/PKG NO. 206431

SHEET 108 OF 291

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C7.50 LEACHFIELD CONSTRUCTION NOTES

1. THE CONTRACTOR SHALL INITIATE COMMUNICATIONS WITH PG&E CONCERNING PROPOSED CONSTRUCTION IN THE VICINITY OF THE WAWONA ROAD – SH41 INTERSECTION AND IN THE WESTERLY LEACHFIELD AREA, NEAR THE EXISTING PG&E POLE LINE. COORDINATE WITH THE COR.

2. SEE DTL. 1/C7.80. COORDINATE MONITORING WELL LOCATION WITH ENGINEER AND COR.

3. EX. PIEZOMETER B–14. SEE TABLE 2/C7.72 FOR RECORD OF GROUNDWATER LEVEL MEASUREMENTS.

   SEE CONSTRUCTION NOTE E/C7.80 FOR PARK'S REMOVAL OF EXISTING DATA LOGGER.

   PIEZOMETER B–14 SHALL BE DESTROYED BY A C–57 LICENSED DRILLING CONTRACTOR IN ACCORD WITH GENERAL NOTE 2.C/C7.60.

4. EX. ROCKY AREA SHOWN IS APPROXIMATE. FIELD VERIFY ROCKY LOCATIONS NEAR PROPOSED UNDERGROUND CONSTRUCTION. REMOVE ROCKS AS REQUIRED TO ACCOMMODATE PROPOSED CONSTRUCTION. COORDINATE WITH ENGINEER AND COR.

5. MINIMIZE DISTURBANCES OF DUFF AND TOPSOIL IN LEACHFIELD AREA, INCLUDING SOIL COMPACTION BY CONSTRUCTION EQUIPMENT AND ACTIVITIES.

   NOTE THAT ALL EXISTING TREES WERE NOT INCLUDED IN THE TOPOGRAPHIC SURVEY AND, THEREFORE, NOT ALL TREES AND STUMPS ARE SHOWN ON THESE PLANS.

   SEE GENERAL NOTE 17/C7.61–C7.62 FOR SITE PREPARATION, CLEARING, EARTHWORK AND GRADING.

6. CONSTRUCT 4–CELL PRESSURE DISTRIBUTION LEACHFIELD. SEE C7.10, AND DTL. 1/C7.52 FOR TYPICAL LEACH TRENCH DETAILS.

   THE LEACHFIELD TRENCH ALIGNMENTS SHOWN ARE INTENDED TO GENERALLY FOLLOW THE GROUND SURFACE CONTOURS THAT ARE PLOTTED BASED ON THE TOPOGRAPHIC SURVEY THAT WAS PERFORMED PRIOR TO TREE, STUMP AND DUFF REMOVAL. COORDINATE ANY PROPOSED MODIFICATIONS TO THE TRENCH LAYOUT SHOWN WITH THE ENGINEER AND COR.

   PER DTL. 1/C7.52, PD PIPING AND TRENCH BOTTOMS IN EACH PAIR OF LATERALS ARE TO BE LEVEL WITHIN A TOLERANCE OF ±1" (2" MAX. TOTAL VARIATION) PER 100'; AND THE DEPTH OF DRAIN ROCK COVERING THE PD PIPING ORIFICE SHIELDS WILL NEED TO INCREASE ABOVE THE 2" MIN. AS MAY NEEDED TO LIMIT THE SOIL COVER TO 18" MAX.

   SEE GENERAL NOTES 26 AND 27/C7.63 AND C7.64 FOR ADDITIONAL DOSING AND PD SYSTEM REQUIREMENTS.

7. INSTALL 1.5" SCH. 40 PVC LATERAL PIPING (TYP.). DISTANCE BETWEEN TRENCH CENTERLINES VARIES AS SHOWN, WITH 7' MIN. DIMENSION.

8. INSTALL 2" SCH. 40 PVC MANIFOLD PIPING FOR ALL 6 ZONES IN EACH CELL.

9. INSTALL AUTOMATIC MECHANICALLY (PRESSURE/FLOW) OPERATED DISTRIBUTING VALVE (DV) ASSEMBLY TO SEQUENTIALLY DISTRIBUTE PUMPED EFFLUENT TO THE 6 LEACHFIELD ZONES ("A" THROUGH "F") IN EACH CELL. INSTALL VALVE WITH VALVE MONITOR. SEE DTL. 1/C7.53.

10. INSTALL OLDCASTLE MODEL B1730, OR APPROVED EQUIVALENT, 17"x30" I.D. TRAFFIC–RATED UTILITY BOX WITH HOT–DIPPED GALV. STEEL FRAME AND BOLT–DOWN CHECKER PLATE COVER AND REINFORCED CONC. BOX EXTENSION. LABEL COVER "CONTROLS". SET BOX ON 6" MIN. LAYER OF 3/4" DRAIN ROCK. ROCK TO EXTEND AT LEAST 6" BEYOND PERIMETER OF BOX. SEAL CONDUIT ENDS WATERTIGHT.

11. FOR DISTRIBUTING VALVE MONITORS INSTALL 1" SCH. 40 PVC ELECTRICAL CONDUIT AND WIRING (2–#12 POWER, 1–#12 GRD, AND 2–#14 FOR EACH DV'S ALARMS) TO PCP IN COMMON TRENCH WITH 3" EFFLUENT TRANSPORT PIPELINES, SIMILAR TO DTL. 5/D5.11.

12. INSTALL 1" SCH. 40 PVC CONDUIT WITH MFR'S CABLING FOR MONITORING WELL TRANSDUCERS. SEE NOTE E, DTL. 1/C7.80.

13. INSTALL ONE 3" HDPE PRESSURE PIPELINE, WITH TRACER WIRE, FROM EACH DOSING PUMP TO EACH DISTRIBUTING VALVE WITH 2' MIN. COVER (TYP.) BELOW FINISHED GRADE AND WITHOUT ANY DEPRESSIONS OR HIGH POINTS THAT COULD TRAP AIR. SEE GENERAL NOTE 20/C7.62 FOR PIPING SPECIFICATIONS. TO FACILITATE POSSIBLE MAINTENANCE SHUTDOWN DRAINING, SLOPE PIPELINE –TO THE EXTENT PRACTICAL– TO GRAVITY DRAIN BACK INTO DOSING TANK. COORDINATE WITH ENGINEER AND COR.

14. INSTALL EMPTY 3" HDPE PRESSURE PIPELINE WITH LOCATING WIRE FOR POSSIBLE FUTURE PD CELL. COORDINATE PIPELINE END POINT LOCATIONS AND MARKINGS WITH ENGINEER AND COR.

15. UTILITIES IN COMMON TRENCH (TYP.).

16. CONSTRUCT "BROW" SURFACE DRAINAGE INTERCEPTOR DITCH PER DTL. 3/D5.42, EXCEPT THAT THE EARTH TRANSITION SLOPE ON THE UPHILL SIDE OF THE LINED DITCH SECTION SHALL BE 3:1. USE MIRAFI 600X, OR APPROVED EQUIVALENT, WOVEN GEOTEXTILE. COORDINATE DITCH ROUTE AND CONFIGURATION WITH COR.

17. CONSTRUCT 15' MIN. LONG EROSION CONTROL APRON AT END OF LINED DITCH USING DITCH LINING MATERIAL. COORDINATE WITH COR.

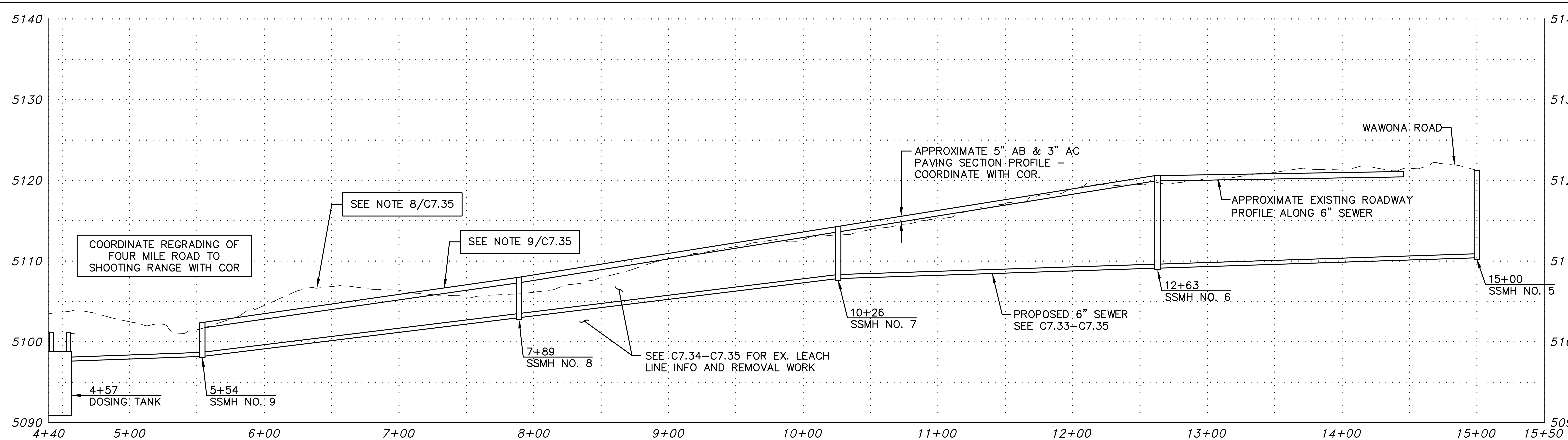
18. INSTALL 18" MIN. SIZE CULVERT ACROSS EX. DIRT ROADWAY. COORDINATE CULVERT LOCATION AND SIZE WITH COR.

19. SEE C7.70–C7.72 FOR SOILS INFO.

20. PROTECT EXISTING TREES (TYP.). COORDINATE WITH COR.

21. AS PART OF ITS OWTS O&M, THE PARK IS TO KEEP THE LEACHFIELD AREA CLEAR OF BURROWING RODENTS. NO RODENTICIDES ARE PERMITTED, ONLY MECHANICAL TRAPS APPROVED BY NPS WILDLIFE STAFF ARE TO BE USED.

    IN ACCORD WITH NPS REFERENCE MANUAL 83B1, "WASTEWATER SYSTEMS", THE LEACHFIELD SHALL BE "KEPT CLEAR OF TREES AND BUSHES, WHICH MAY SEND ROOTS INTO THE DRAIN FIELD PIPING SYSTEM RESULTING IN CLOGGING AND CAUSING PREMATURE FAILURE."



1 FOUR MILE ROAD PROFILE

NTS

PHASE 5

Thomas G. Horgan

REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA

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WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.51

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS LEACHFIELD
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 109 OF 291

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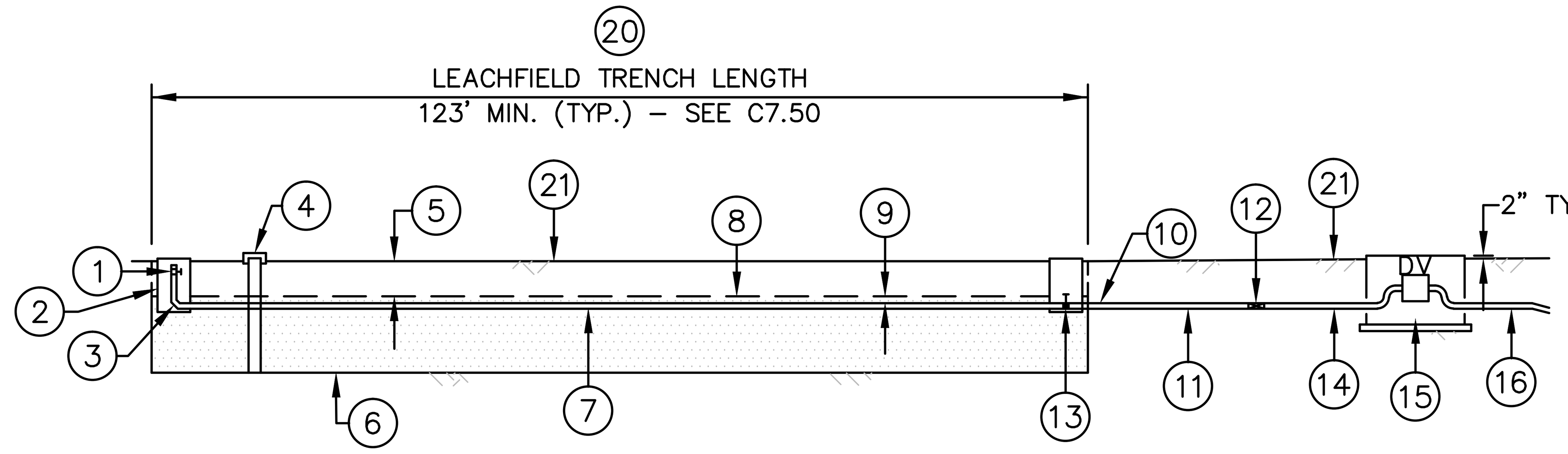

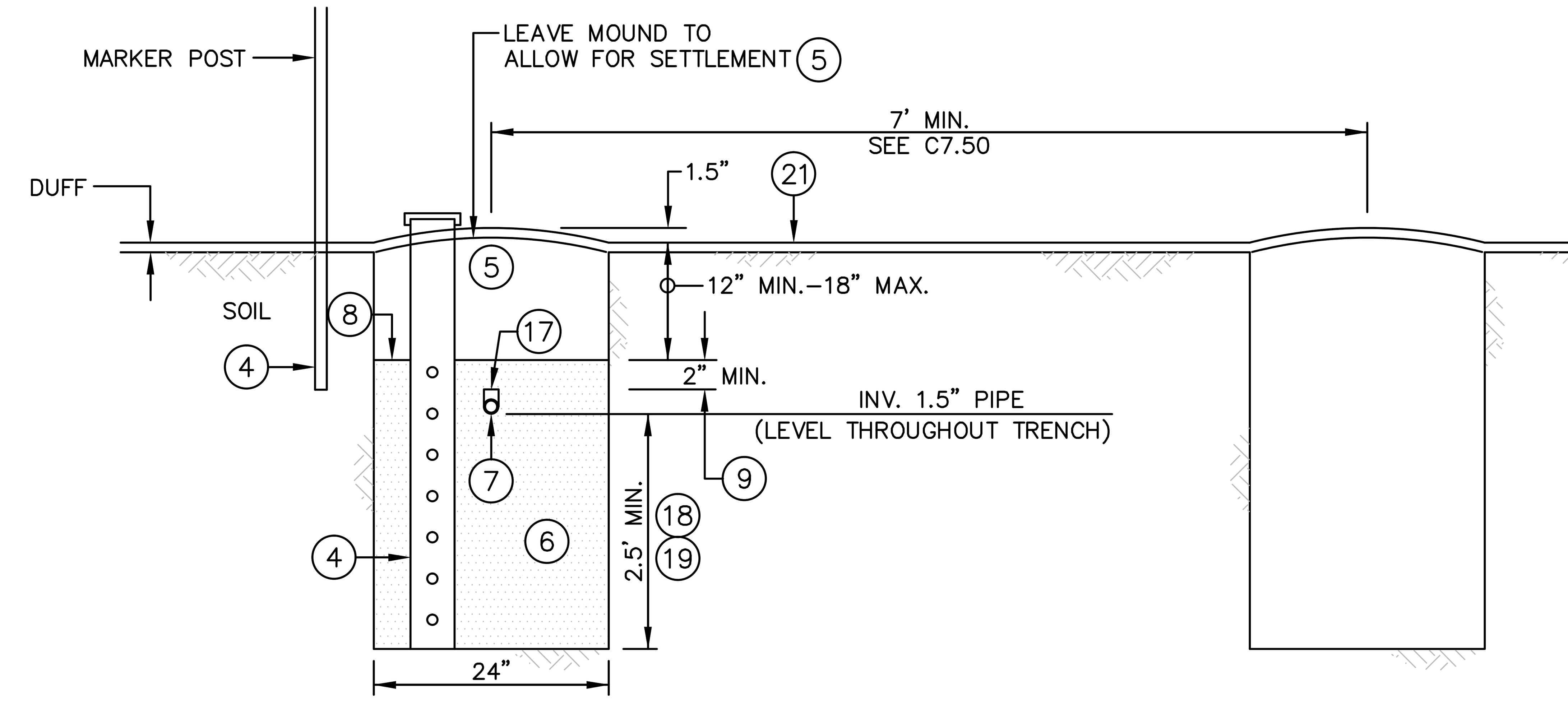

LEACH TRENCH TYP. SECTION

CONSTRUCTION NOTES

1. INSTALL 1.5" SCH. 40 PVC BALL–TYPE PURGE VALVE AT THE END OF EACH DISTRIBUTION LINE. VALVE OUTLET TO ACCOMMODATE A REMOVABLE FITTING THAT WILL ALLOW A SQUIRT TEST TO BE CONDUCTED. SEE GENERAL NOTE 27/C7.64.

2. INSTALL ORENCO MDL. 1217, OR APPROVED EQUIVALENT, VALVE BOX WITH BOX EXTENSION, SUITABLE FOR INSTALLATION OF A REMOVABLE FITTING AND STANDPIPE INTO THE VALVE OUTLET FOR CONDUCTING A SQUIRT TEST. BOX TO BE SECURELY FOUNDED ON THE DRAIN ROCK BELOW. SET TOP OF BOX 2" ABOVE FINISHED GRADE. EACH BOX IS TO HAVE AN EASY–TO–READ, HEAVY–DUTY, CORROSION–PROOF NAMEPLATE, WITH LATERAL NO., SECURELY ATTACHED TO BOX COVER. COORDINATE WITH COR AND THE ENGINEER.

3. INSTALL 1.5" SCH. 40 PVC LONG–SWEEP ELL FITTING.

4. OPEN BOTTOM INSPECTION PORT INSTALLED AT THE TERMINAL END OF EACH LATERAL PER PLAN. THE PIPE SHALL BE 4–INCH DIAMETER SCH. 40 PVC, PERFORATED IN THE HORIZON OF THE LEACH LINE, AND SHALL EXTEND VERTICALLY FROM THE BOTTOM OF THE LEACH LINE TRENCH TO 2" ABOVE GRADE. WRAP PERFORATED SECTION OF PIPE WITH FILTER FABRIC. A REMOVABLE, SECURE, INSECT–PROOF AND VERMIN–PROOF, WATERTIGHT CAP SHALL BE PROVIDED OVER EACH INSPECTION PORT.

   INSTALL A 6' STEEL "T" FENCE POST TO MARK THE LOCATION OF EACH PORT. INSTALL A 2" YELLOW HDPE PIPE SLEEVE ONTO EACH POST. INSTALL AN EASY–TO–READ, HEAVY–DUTY, CORROSION–PROOF NAMEPLATE, WITH LATERAL NO., SECURELY ATTACHED TO SLEEVE. COORDINATE POST LOCATION AND MARKING WITH ENGINEER AND COR.

5. BACKFILL WITH EXCAVATED SOIL.

   BACKFILL SHALL BE CAREFULLY PLACED TO MINIMIZE COMPACTION AND PREVENT DAMAGE TO THE SYSTEM. THE QUALITY OF THE BACKFILL SHALL BE CONSISTENT IN STRUCTURE AND TEXTURE WITH THE TOPSOIL ALREADY EXISTING ON THE SITE. BACKFILL SHALL BE FREE OF ANY MATERIALS THAT COULD DAMAGE THE SYSTEM, INCLUDING, BUT NOT LIMITED TO, ROCKS, CONSTRUCTION MATERIALS, AND WOOD.

   LEACH LINES SHALL BE BACKFILLED TO A LEVEL THAT WILL MATCH GRADE AFTER SETTLING OF THE BACKFILL.

   HEAVY EQUIPMENT OR VEHICULAR TRAFFIC SHALL NOT BE ALLOWED OVER THE ABSORPTION SYSTEM DURING CONSTRUCTION OR AFTER COMPLETION OF THE ONSITE SEWAGE DISPOSAL SYSTEM. INSTALL TEMPORARY TRAFFIC BARRIER OR FENCE AROUND THE DISPOSAL FIELD AND REPLACEMENT AREA TO PROTECT THEM FROM EQUIPMENT AND VEHICULAR TRAFFIC.

   SEE NOTE 21.

6. PLACE 3/4" – 2 1/2" CLEAN, WASHED RIVER ROCK, GRAVEL OR OTHER COUNTY EHS AND ENGINEER APPROVED HARD ROCK.

   INFORMATIONAL NOTE: IN PREVIOUS OWTS PROJECTS, THE 1 1/2"x3/4" CLEAN, WASHED "ROUND" AGGREGATE FROM VULCAN MATERIALS CO. FRESNO FACILITY WAS ACCEPTABLE FOR USE. ITS GRADATION: 1.5"–98%, 1"–29%, 3/4"–5%, 1/2"–2%, 3/8"–1%, NO. 4–1%, NO. 200–0%.

   BEFORE PLACING ROCK IN THE TRENCH, ALL SMEARED OR COMPACTED SOIL SURFACES IN THE SIDEWALLS AND BOTTOM OF EACH TRENCH SHALL BE SCARIFIED TO THE DEPTH OF SMEARING OR COMPACTION, AND THE LOOSE MATERIAL REMOVED, ALL TO THE SATISFACTION OF COUNTY EHS AND THE ENGINEER.

   THE BOTTOM OF THE TRENCH SHALL BE CONSTRUCTED LEVEL WITHIN A TOLERANCE OF 2" PER 100'.

7. TYP. 120' LONG, 1.5" DIA., SCH. 40 PVC PERFORATED PRESSURE DISTRIBUTION (PD) PIPE LATERAL INSTALLED IN CENTER OF TRENCH AND LEVEL WITHIN A TOLERANCE OF 2" PER 100'.

   THE PIPE SHALL BE INSTALLED WITH 1/8" DIA., NEATLY DRILLED ORIFICES POINTING UP AND SPACED AT 4', WITH THE FIRST AND LAST ORIFICES EQUALLY SPACED FROM EACH END OF THE DISTRIBUTION LINE. THE TYPICAL 120' LONG DISTRIBUTION LATERAL WILL HAVE 31 ORIFICES.

   HOLES SHALL BE CAREFULLY DRILLED PERPENDICULAR TO THE PIPE WITH NEW, SHARP, STRAIGHT DRILL BITS, METICULOUSLY AVOIDING ENLARGING THE HOLE BY WOBBLING THE BIT OR WHEN HOLE DEBURRING IS REQUIRED.

   THIS PD SYSTEM DESIGN IS BASED ON 0.43 GPM PER 1/8" ORIFICE AT A 5' RESIDUAL HEAD. RESEARCH HAS SHOWN THAT MISDRILLING OR ENLARGING THE HOLE SIZE BY ONLY 1/64" SIGNIFICANTLY INCREASES THE FLOW REQUIRED TO ACHIEVE THE SAME 5' RESIDUAL HEAD BY 27%. SO DRILLING TOOLS AND METHOD ARE CRITICAL FACTORS IN SATISFACTORY ORIFICE PERFORMANCE. SHOP–DRILLED HOLES USING A DRILL PRESS OR GUIDE ARE PREFERRED TO FIELD DRILLING.

   DISTRIBUTION TO AND THROUGH ALL LATERALS SHALL BE BALANCED SO THAT ALL EQUAL–LENGTH LATERALS RECEIVE EQUAL FLOW.

8. INSTALL PERMEABLE BARRIER, MIRAFI 140N, OR APPROVED EQUIVALENT, NONWOVEN GEOTEXTILE FABRIC. USE 12"–24" LAP SPLICES.

9. 2" MIN. DRAIN ROCK COVER OVER DISTRIBUTION PIPE ORIFICE SHIELDS. DEPTH OF DRAIN ROCK COVER VARIES AS NEEDED TO PROVIDE 18" MAX. SOIL COVER OVER DRAIN ROCK.

10. THE CROSS–SECTION OF THE TRANSMISSION LINE AND THE BEGINNING OF THE LEACH LINE TRENCH SHALL BE STEPPED SO AS TO PREVENT SEEPAGE OF EFFLUENT FROM TRENCH TO TRENCH.

11. INSTALL 1.5" DIA., SCH 40 PVC DISTRIBUTION PIPELINE.

12. SEE C7.50 FOR 2" SCH. 40 PVC TEE OR CROSS (MANIFOLD) LOCATION (TYP.).

13. 1.5" SCH. 40 PVC GATE TYPE BALANCING VALVE INSTALLED AT THE BEGINNING OF EACH DISTRIBUTION LINE IN A 10" MIN. I.D. ROUND OR RECTANGULAR PLASTIC OR CONCRETE BOX WITH ADEQUATE EXTENSIONS AND SECURE COVERS. BOX TO BE SECURELY FOUNDED ON THE DRAIN ROCK BELOW. SET TOP OF BOX 2" ABOVE FINISHED GRADE. EACH BOX IS TO HAVE AN EASY–TO–READ, HEAVY–DUTY, CORROSION–PROOF NAMEPLATE, WITH LATERAL NO., SECURELY ATTACHED TO BOX COVER. COORDINATE WITH COR AND THE ENGINEER.

14. INSTALL 2" DIA., SCH 40 PVC PIPE MANIFOLD TO EACH ZONE.

15. INSTALL 1.5" AUTOMATIC DISTRIBUTING VALVE ASSEMBLY, INCLUDING MONITOR, PER DTL. 1/C7.53.

16. INSTALL DISCHARGE PIPELINE TO DISTRIBUTING VALVE ASSEMBLY. SEE C7.50 AND DTL. 1/C7.53.

17. EACH ORIFICE IS TO BE PROTECTED BY AN ORENCO MODEL OS150, OR APPROVED EQUIVALENT, SNAP–ON PVC ORIFICE SHIELD.

18. DESIGN INFILTRATIVE AREA OF TRENCH IS 2.5 FEET OF SIDEWALL ON EACH SIDE OF TRENCH FOR TOTAL OF 5 SF/LF OF TRENCH.

    TO ACCOMMODATE LEVEL PD PIPING THROUGHOUT THE LEACHFIELD, TRENCH DEPTH WILL VARY. SEE NOTE 9.

19. DESIGN SOUTH ENTRANCE AREA PEAK–DAY WASTEWATER FLOW IS 19,920 GPD – SEE SUB–SHEET C7.01. HYDRAULIC LOADING RATE IS 19,920 GPD/(11,520 LF TRENCH x 5 SF/LF) = 0.35 GPD/SF. 0.4 TO 0.6 GPD/SF IS COUNTY'S MAX. ALLOWABLE APPLICATION RATE FOR TYPICAL DOMESTIC (RESIDENTIAL) WASTE SOURCE.

20. AS PART OF ITS OWTS O&M, THE PARK IS TO KEEP THE LEACHFIELD AREA CLEAR OF BURROWING RODENTS. NO RODENTICIDES ARE PERMITTED, ONLY MECHANICAL TRAPS APPROVED BY NPS WILDLIFE STAFF ARE TO BE USED.

    IN ACCORD WITH NPS REFERENCE MANUAL 83B1, "WASTEWATER SYSTEMS", THE LEACHFIELD SHALL BE "KEPT CLEAR OF TREES AND BUSHES, WHICH MAY SEND ROOTS INTO THE DRAIN FIELD PIPING SYSTEM RESULTING IN CLOGGING AND CAUSING PREMATURE FAILURE."

21. SEE GENERAL NOTE 17/C7.61–C7.62 CONCERNING TREES, TREE STUMPS AND ROOTS, DUFF, TOPSOIL, EXCAVATION AND GRADING.

1 LEACHFIELD SCHEMATIC SECTIONS
NOT TO SCALE

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER
THOMAS G. HORGAN
No. 24772
CIVIL
STATE OF CALIFORNIA
MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM
PRIME:
DAVID EVANS AND ASSOCIATES, INC.
BELLEVUE, WA

SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.52

PHASE 5

TITLE OF SHEET
SOUTH ENTRANCE
OWTS PLANS
LEACHFIELD DETAILS
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 / 127662
PMIS/PKG NO. 206431
SHEET 110 OF 291

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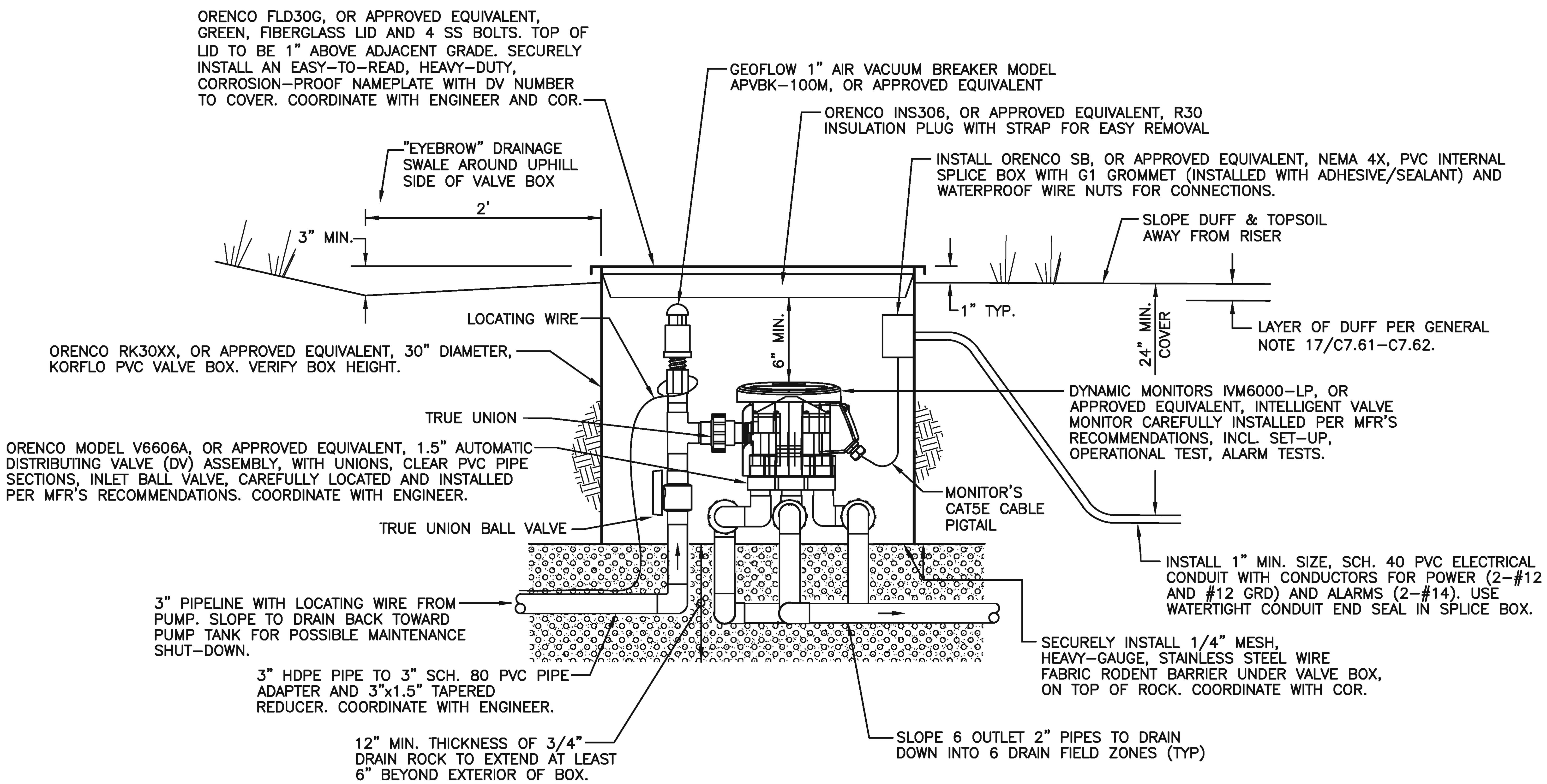
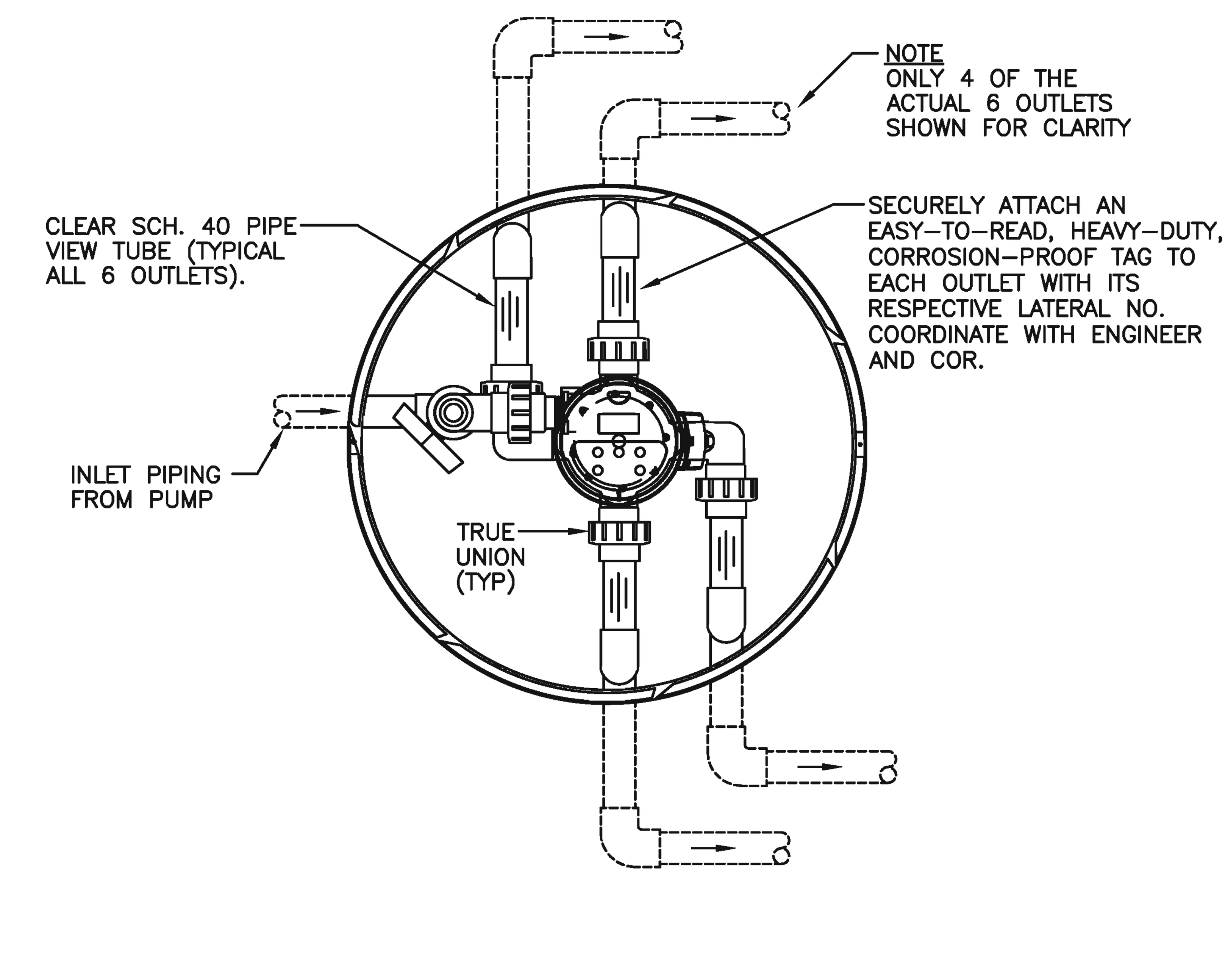
NOTE

THE CONTRACTOR SHALL FURNISH TO THE COR TWO COMPLETE SPARE DISTRIBUTING VALVE ASSEMBLIES, INCLUDING MONITORS, AND TWO DISTRIBUTING VALVE SPARE PARTS KITS.



SIDE VIEW



TOP VIEW

1 DISTRIBUTING VALVE ASSEMBLY

NOT TO SCALE

PHASE 5

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA
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SHEET 111 OF 291

GENERAL NOTES

1. STANDARDS, PLANS & SPECIFICATIONS

A. ALL ONSITE WASTEWATER TREATMENT SYSTEM (OWTS) CONSTRUCTION, INCLUDING MATERIALS AND INSTALLATION WORK, SHALL BE IN ACCORD WITH THE PROJECT CONTRACT DOCUMENTS, INCLUDING PLANS AND SPECIFICATIONS, AND WITH THE MARIPOSA COUNTY SEWAGE DISPOSAL REGULATIONS WHICH ARE AVAILABLE FROM THE MARIPOSA COUNTY HEALTH DEPARTMENT, ENVIRONMENTAL HEALTH SERVICES (EHS), 5100 BULLION STREET, MARIPOSA, CA 95338, PHONE: 209-966-2220.

MARIPOSA COUNTY WILL REVIEW AND APPROVE OWTS DESIGN, ADVISE THE CVRWQCB OF SUCH REVIEW AND APPROVAL, AND PERFORM OWTS CONSTRUCTION INSPECTION, BUT WILL NOT ISSUE AN OWTS PERMIT FOR THIS PROJECT AS THE STATE OF CALIFORNIA CENTRAL VALLEY REGIONAL WATER QUALITY CONTROL BOARD (CVRWQCB) FRESNO OFFICE HAS JURISDICTION AND PERMIT AUTHORITY.

B. THE CVRWQCB, FRESNO OFFICE, 1685 E STREET, SUITE 200, FRESNO, CA 93706, WILL NOT ISSUE A CONSTRUCTION PERMIT FOR THIS OWTS, BUT WILL PROVIDE THE GOVERNMENT WITH A LETTER AUTHORIZING OWTS CONSTRUCTION, AND, AT A LATER DATE, ISSUE AN OWTS OPERATIONAL PERMIT TO THE GOVERNMENT.

C. ANY FIELD MODIFICATIONS TO THESE PLANS AND SPECIFICATIONS MUST BE DISCUSSED WITH AND APPROVED BY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR), THE ENGINEER OF RECORD (ENGINEER), EHS AND CVRWQCB. ALL CONSTRUCTION MATERIALS AND METHODS WILL BE SUBJECT TO VERIFICATION BY BOTH THE ENGINEER AND EHS. ANY SUBSTITUTIONS OR MODIFICATIONS WILL REQUIRE THE PRIOR APPROVAL BY THE ENGINEER AND EHS.

D. ALL MATERIALS, FIXTURES, PIPE, VALVES, PUMPS, FITTINGS, DISTRIBUTION BOXES AND DEVICES USED TO CONSTRUCT THE OWTS MUST BE IAPMO APPROVED AND LISTED, OR BE APPROVED BY THE COUNTY EHS.

E. ELECTRICAL, PLUMBING AND MECHANICAL WORK AND MATERIALS SHALL MEET AND CONFORM TO THE CODES IDENTIFIED IN THE PROJECT SPECIFICATIONS AND THE NATIONAL SANITATION FOUNDATION, AS WELL AS CONFORM TO ALL APPLICABLE COUNTY AND STATE OWTS REGULATIONS.

F. THE CONTRACTOR SHALL CONFORM TO THE RULES AND REGULATIONS OF THE FEDERAL OSHA CONSTRUCTION SAFETY ORDINANCE PERTAINING TO EXCAVATIONS AND TRENCHES.

G. REFER TO GENERAL NOTES ON DRAWING G5.05 FOR ADDITIONAL REQUIREMENTS.

H. PROVIDE A COMPLETE AND PROPERLY OPERABLE ONSITE WASTEWATER TREATMENT SYSTEM WITHIN THE INTENT OF THE CONTRACT DOCUMENTS, INCLUDING, BUT NOT LIMITED TO, ALL CONSTRUCTION MATERIALS AND LABOR FOR COMPLETE INSTALLATION, TESTING, AND COMMISSIONING IN COMPLIANCE WITH APPLICABLE PROVISIONS AND RECOMMENDATIONS IN THESE REFERENCED STANDARDS.

I. FOR THIS PROJECT, THE TERMS "APPROVED EQUAL" OR "APPROVED EQUIVALENT" SHALL MEAN EQUAL OR EQUIVALENT IN THE JUDGEMENT OF THE ENGINEER.

2. EXISTING WASTEWATER FACILITIES

A. THE EXISTING SOUTH ENTRANCE OWTS SERVES THE RANGER OFFICE AND THE EXISTING COMFORT STATION. THIS OWTS IS TO REMAIN OPERATIONAL AS LONG AS IS PRACTICABLE, WITHOUT IMPAIRING PROPOSED CONSTRUCTION OF THE NEW OWTS. DURING THE PERIOD THAT THE EXISTING OWTS IS TAKEN OUT OF SERVICE AND UNTIL THE NEW OWTS IS APPROVED AND COMMISSIONED FOR SERVICE, THE CONTRACTOR SHALL PROVIDE, POSITION AND MAINTAIN, TO THE SATISFACTION OF THE COR, PORTABLE TOILET FACILITIES, INCLUDING HAND WASH STATIONS, AT BOTH THE RANGER OFFICE AND EXISTING COMFORT STATION AREAS:

PLACE NEAR THE RANGER OFFICE:
2 NONACCESSIBLE PORTABLE TOILETS
1 WASH STATION

PLACE NEAR THE EXISTING COMFORT STATION:
2 ACCESSIBLE PORTABLE TOILETS
2 NONACCESSIBLE PORTABLE TOILETS
2 WASH STATIONS

AS AN ALTERNATE TO, OR AS A SUPPLEMENT TO, THE PORTABLE TOILETS, THE TWO EXISTING SEPTIC TANKS SHOWN ON DRAWINGS C7.10 AND C7.33 MAY BE OPERATED AS HOLDING TANKS BY THE CONTRACTOR AND MAINTAINED TO THE SATISFACTION OF THE COR AT CONTRACTORS'S SOLE EXPENSE, INCLUDING SEPTAGE PUMPING AND DISPOSAL. THIS SCHEME ALSO REQUIRES THE CONTRACTOR TO COORDINATE WITH CONSTRUCTION OF THE PROPOSED WATER SYSTEM IMPROVEMENTS AND THE PROPOSED FIXTURE RETROFIT AT THE EXISTING COMFORT STATION AND WITH THE PARK'S COMFORT STATION O&M STAFF.

THE CONTRACTOR SHALL COORDINATE SHUT-DOWN OF THE EXISTING LEACHFIELD IN FOUR MILE ROAD SUFFICIENTLY IN ADVANCE OF 6" SEWER PIPELINE CONSTRUCTION TO PROVIDE FOR "DRYING" OF THE SOIL ALONG THE LEACH TRENCHES AS MAY BE NEEDED TO ACCOMMODATE 6" SEWER INSTALLATION. COORDINATE WITH THE COR.

SEE SPECIFICATIONS SECTION 01 11 00, "SUMMARY OF WORK".

B. THE EXISTING SOUTH ENTRANCE OWTS COMPONENTS, INCLUDING SEPTIC TANKS, MANHOLES, PIPELINES, AND APPURTENANCES, SHALL BE ABANDONED IN PLACE OR REMOVED AS SPECIFIED BELOW. SEE SUB-SHEET C7.33.

ABANDONMENT OF EXISTING UNDERGROUND PIPELINES LARGER THAN 2" NOMINAL DIAMETER SHALL CONSIST OF: (1) PIPELINE CLEANING; (2) COMPLETE REMOVAL OF THE PIPELINE, INCLUDING FITTINGS AND APPURTENANCES; (3) BACKFILLING PER PROJECT SPECIFICATIONS; (4) SURFACE RESTORATION PER PROJECT SPECIFICATIONS; AND (5) EROSION CONTROL BMPS PER PROJECT SPECIFICATIONS. PIPELINES THAT CANNOT BE REASONABLY REMOVED, IN THE OPINION OF THE COR, SHALL BE SLURRY FILLED PER PROJECT SPECIFICATIONS.

ABANDONMENT OF EXISTING UNDERGROUND PIPELINES WITH NOMINAL DIAMETERS OF 2" OR LESS SHALL BE BY ABANDONMENT IN PLACE. ANY CUT ENDS THAT ARE THE RESULT OF THE CONTRACTOR'S ACTIVITIES SHALL BE SECURELY PLUGGED TO THE SATISFACTION OF THE COR.

EXISTING SEPTIC TANKS SHALL BE PUMPED CLEAN IN ACCORD WITH COUNTY EHS REQUIREMENTS, AND THEN BE COMPLETELY REMOVED AND DISPOSED OF TO THE SATISFACTION OF THE COR. BACKFILLING, SURFACE RESTORATION AND EROSION CONTROL BMPS SHALL BE IN ACCORD WITH PROJECT SPECIFICATIONS.

EXISTING SEWER MANHOLES SHALL BE CLEANED AND THEN COMPLETELY REMOVED AND DISPOSED OF TO THE SATISFACTION OF THE COR. BACKFILLING, SURFACE RESTORATION AND EROSION CONTROL BMPS SHALL BE IN ACCORD WITH PROJECT SPECIFICATIONS.

C. DESTRUCTION OF EXISTING PIEZOMETER B-14 THAT IS LOCATED WITHIN THE PROPOSED LEACHFIELD FOOTPRINT WILL BE IN ACCORD WITH THE STATE OF CALIFORNIA WELL STANDARDS (BULLETIN 74-90) FOR MONITORING WELLS. THIS WORK WILL ALSO BE REGULATED BY THE MARIPOSA COUNTY ENVIRONMENTAL HEALTH SERVICES (EHS). THE COUNTY WILL NOT REQUIRE A PERMIT FOR DESTRUCTION OF THE EXISTING PIEZOMETER.

DESTRUCTION SHALL INCLUDE FILLING THE WELL CASING WITH NEAT CEMENT, OR OTHER APPROVED MATERIAL, IN A CONTINUOUS OPERATION THAT PREVENTS BRIDGING AND VOIDS. THE VOLUME OF FILL MATERIAL PLACED SHALL BE VERIFIED AS BEING EQUAL TO OR MORE THAT THE VOLUME OF THE CASING.

D. SEE SPECIFICATIONS SECTION 02 41 00, "DEMOLITION" FOR ADDITIONAL REQUIREMENTS. COORDINATE WITH COUNTY EHS.

3. EXISTING UTILITIES AND COORDINATION OF WORK

A. THE TYPES, LOCATIONS, SIZES AND/OR DEPTHS OF EXISTING UNDERGROUND UTILITIES AS SHOWN ON THESE IMPROVEMENT PLANS WERE OBTAINED FROM SOURCES OF VARYING RELIABILITY.

THE CONTRACTOR IS CAUTIONED THAT ONLY ACTUAL EXCAVATION WILL REVEAL THE TYPES, EXTENT, SIZES, LOCATIONS AND DEPTHS OF SUCH UNDERGROUND UTILITIES. LAUGENOUR AND MEIKLE, HEREINAFTER DESIGNATED AS THE ENGINEER, ASSUMES NO RESPONSIBILITY FOR THE COMPLETENESS OR ACCURACY OF ITS DELINEATION OF SUCH UNDERGROUND UTILITIES, NOR FOR THE EXISTENCE OF OTHER BURIED OBJECTS OR UTILITIES WHICH ARE NOT SHOWN ON THESE DRAWINGS.

EXISTING BURIED UTILITIES ARE INDICATED ON THE DRAWINGS WHERE SUCH UTILITIES ARE KNOWN. THE LOCATION AND EXTENT OF SUCH UTILITIES ARE APPROXIMATE ONLY. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO LOCATE, PROTECT AND MAINTAIN ALL EXISTING UTILITIES WHETHER OR NOT SHOWN ON THE DRAWINGS. THE NPS, THE UTILITY OWNER AND THE CONTRACTING OFFICER CAN ASSUME NO RESPONSIBILITY FOR THE COMPLETENESS OR ACCURACY OF THE DELINEATION OF SUCH UNDERGROUND UTILITIES WHICH MAY BE ENCOUNTERED.

B. CONTRACTOR SHALL BE RESPONSIBLE FOR LOCATING AND PREVENTING DAMAGE TO EXISTING UTILITIES. THE GOVERNMENT HAS ATTEMPTED TO SHOW THE APPROXIMATE LOCATION OF MAJOR UTILITIES ON THE DRAWINGS. SOME UTILITIES MAY NOT BE SHOWN. THE CONTRACTOR SHALL SWEEP ALL SITES AND ALIGNMENTS AFFECTED BY CONSTRUCTION OPERATIONS WITH SUITABLE DETECTION DEVICES.

C. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO VERIFY THE LOCATION OF ALL UNDERGROUND UTILITIES PRIOR TO ANY EXCAVATION. IN AREAS TO BE EXCAVATED, THE LOCATION OF ALL EXISTING UNDERGROUND UTILITIES OR KNOWN OBSTRUCTIONS SHALL BE CLEARLY MARKED AHEAD OF EXCAVATION FOR CONSTRUCTION.

THE CONTRACTOR IS RESPONSIBLE FOR CONTACTING THE UTILITY COMPANIES INVOLVED AND REQUESTING A VISUAL VERIFICATION OF THE LOCATIONS OF THEIR UNDERGROUND FACILITIES.

D. THE COUNTY IS A MEMBER OF THE UNDERGROUND SERVICE ALERT (U.S.A.) ONE-CALL PROGRAM. THE CONTRACTOR SHALL NOTIFY MEMBERS OF U.S.A. AT LEAST 2 WORK WEEKS IN ADVANCE OF PERFORMING EXCAVATION WORK BY CALLING THE TOLL-FREE NUMBER 800-227-2600 FOR COORDINATING AND MARKING THE LOCATION OF UNDERGROUND UTILITIES.

THE AREAS PLANNED FOR EXCAVATION SHALL BE OUTLINED BY THE CONTRACTOR IN WHITE MARKING PAINT WITH "USA" WRITTEN AT A HIGHLY VISIBLE LOCATION ADJACENT TO THE EXCAVATION AREA. ALL UTILITY LOCATION MARKINGS SHALL BE MAINTAINED BY THE CONTRACTOR UNTIL EXCAVATION IS COMPLETED, AT WHICH TIME THE CONTRACTOR SHALL REMOVE ALL MARKINGS. IF DAMAGE OCCURS, REPAIR OF UTILITY SHALL BE PERFORMED BY THE CONTRACTOR AT NO ADDITIONAL COST TO THE NPS.

E. AFTER THE UTILITY SURVEY IS COMPLETED, COMMENCE "POTHOLING" TO DETERMINE THE ACTUAL LOCATIONS AND ELEVATIONS OF ALL UTILITIES WHERE CROSSINGS, INTERFERENCES, OR CONNECTIONS TO NEW PIPELINES OR OTHER FACILITIES ARE SHOWN ON THE DRAWINGS, MARKED BY THE UTILITY COMPANIES, OR INDICATED BY SURFACE SIGNS OR FEATURES.

PRIOR TO THE PREPARATION OF PIPING SHOP DRAWINGS, OR THE EXCAVATING FOR ANY NEW PIPELINES OR STRUCTURES, THE CONTRACTOR SHALL LOCATE AND UNCOVER THESE EXISTING UTILITIES, INCLUDING SERVICES AND LATERALS, TO A POINT 1 FOOT BELOW THE UTILITY.

THE CONTRACTOR SHALL THEN MEASURE THE EXISTING LOCATIONS AND ELEVATIONS FOR POSSIBLE CONFLICTS OR OTHER INCONSISTENCIES WITH PLANS. IF THESE MEASUREMENTS REVEAL CONFLICTS OR OTHER INCONSISTENCIES WITH THE PLANS, THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) AND REQUEST DESIGN OR OTHER MODIFICATIONS TO RESOLVE THE CONFLICT OR INCONSISTENCIES.

F. EXCAVATIONS AROUND UNDERGROUND ELECTRICAL DUCTS AND CONDUITS SHALL BE PERFORMED USING EXTREME CAUTION TO PREVENT INJURY TO WORKMEN OR DAMAGE TO ELECTRICAL DUCTS OR CONDUITS. SIMILAR PRECAUTIONS SHALL BE EXERCISED AROUND GAS LINES, TELEPHONE, DATA AND TELEVISION CABLES. EXCAVATIONS SHALL HAVE A SURFACE DIMENSION OF NO MORE THAN 18 INCHES X 18 INCHES. AIR SPADES AND VACUUM EXCAVATORS SHALL BE USED TO LIMIT THE SIZE OF THE EXCAVATION AND DAMAGE TO ADJACENT FACILITIES. BACKFILL AFTER COMPLETING POTHOLING. IN EXISTING STREETS TEMPORARILY PAVE WITH 3" OF COLD MIX ASPHALT CONCRETE.

CONTRACTOR SHALL HAND DIG OR POTHOLE ALL EXISTING PRIMARY AND SECONDARY ELECTRICAL AND COMMUNICATIONS CIRCUITS PRIOR TO DIGGING WITH AN EXCAVATOR OR BACKHOE. CONTRACTOR SHALL BE RESPONSIBLE FOR REPAIRING ALL DAMAGED CONDUITS AND CONDUCTORS AT NO ADDITIONAL COST TO THE GOVERNMENT. THE YOSEMITE HIGH VOLTAGE SHOP WILL ASSIST THE CONTRACTOR WITH LOCATING EXISTING UNDERGROUND CIRCUITS AS BEST THEY CAN, BUT THE CONTRACTOR BEARS ULTIMATE RESPONSIBILITY FOR LOCATING ALL AFFECTED UTILITIES.

G. THE CONTRACTOR SHALL EXPOSE EXISTING UTILITIES, INCLUDING ELECTRICAL, TELECOMM, WATER, SEWER AND DRAIN LINES, AT CROSSINGS OF EXISTING AND PROPOSED UTILITIES, AND PRIOR TO THE STAKING OF THE PROPOSED UTILITIES AND PRIOR TO ANY EXCAVATION FOR PROPOSED UTILITY INSTALLATION. CONTRACTOR SHALL MEASURE THE ELEVATION OF THE EXISTING UTILITIES AND CHECK FOR CONFLICTS WITH THE PROPOSED UTILITIES. IF A GRADE CONFLICT DOES OCCUR, THE CONTRACTOR SHALL INFORM THE CONTRACTING OFFICER'S REPRESENTATIVE. THE CONTRACTING OFFICER'S REPRESENTATIVE SHALL RESOLVE ANY GRADE ADJUSTMENTS NECESSARY.

H. ANY NECESSARY RELOCATIONS OF UTILITIES, WHETHER SHOWN ON THE DRAWINGS OR NOT, SHALL BE COORDINATED WITH THE AFFECTED UTILITY. THE CONTRACTOR SHALL PERFORM RELOCATION ONLY IF INSTRUCTED TO DO SO IN WRITING FROM THE UTILITY AND THE COR.

I. THE CONTRACTOR SHALL COORDINATE ALL HIS WORK WITH PROJECT RELATED WORK TO BE PERFORMED BY UTILITY COMPANIES AND BY OTHER PROJECT CONTRACTORS (INCLUDING EARTHWORK, GRADING, PAVING, BUILDING, LANDSCAPE, PLUMBING, ELECTRIC AND TELECOMM). THE CONTRACTOR SHALL AFFORD THESE UTILITY COMPANIES AND CONTRACTORS REASONABLE OPPORTUNITY FOR THE EXECUTION OF THEIR WORK AND SHALL COORDINATE HIS WORK WITH THEIRS. IN THE EVENT OF DELAYS OR CHANGES IN THE WORK BEYOND THE CONTROL OF THE CONTRACTOR, TIME EXTENSIONS AND NECESSARY CHANGES SHALL BE MADE AS PROVIDED IN THE CONTRACT.

J. IF THERE ARE ANY DISCREPANCIES BETWEEN DIMENSIONS IN DRAWINGS AND EXISTING CONDITIONS WHICH WILL AFFECT THE WORK, THE CONTRACTOR SHALL BRING SUCH DISCREPANCIES TO THE ATTENTION OF THE CONTRACTING OFFICER'S REPRESENTATIVE FOR ADJUSTMENT BEFORE PROCEEDING WITH THE WORK. THE CONTRACTOR SHALL BE RESPONSIBLE FOR THE PROPER FITTING OF ALL WORK AND FOR THE COORDINATION OF ALL TRADES, SUBCONTRACTORS AND PERSONS ENGAGED UPON THE CONTRACT.

IF INTERFERENCES OCCUR AT LOCATIONS OTHER THAN SHOWN ON THE DRAWINGS, THE CONTRACTOR SHALL NOTIFY THE CONTRACTING OFFICER'S REPRESENTATIVE, AND A METHOD FOR CORRECTING SAID INTERFERENCES SHOULD BE SUPPLIED BY THE CONTRACTING OFFICER'S REPRESENTATIVE.

PAYMENT FOR INTERFERENCES THAT ARE NOT SHOWN ON THE PLANS, NOR WHICH MAY BE INFERRED FROM THE SURFACE INDICATIONS, SHALL BE IN ACCORDANCE WITH THE PROVISIONS OF THE GENERAL CONDITIONS. IF THE CONTRACTOR DOES NOT EXPOSE ALL REQUIRED UTILITIES PRIOR TO SHOP DRAWING PREPARATION, HE SHALL NOT BE ENTITLED TO ADDITIONAL COMPENSATION FOR WORK NECESSARY TO AVOID INTERFERENCES, NOR FOR REPAIR TO DAMAGED UTILITIES.

K. OVERHEAD FACILITIES: THERE MAY BE EXISTING OVERHEAD ELECTRIC AND TELEPHONE LINES AT THE SITE. THESE OVERHEAD UTILITIES MAY NOT BE SHOWN ON THE DRAWINGS. EXTREME CAUTION SHALL BE USED WHEN WORKING IN THE VICINITY OF OVERHEAD UTILITIES SO AS TO PREVENT INJURY TO WORKMEN OR DAMAGE TO THE UTILITIES. THE CONTRACTOR SHALL BE REQUIRED TO COMPLY WITH THE APPLICABLE PROVISIONS OF THE CALIFORNIA CONSTRUCTION SAFETY ORDERS WHEN WORKING ANYWHERE ON THIS PROJECT.

PHASE 5

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REGISTERED PROFESSIONAL ENGINEER
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SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
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SUB SHEET NO.
C7.60

TITLE OF SHEET
SOUTH ENTRANCE OWTS PLANS GENERAL NOTES
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
PMIS/PKG NO. 206431
SHEET 112 OF 291

GENERAL NOTES CONTINUED

L. ANY EXISTING UTILITY STRUCTURE, INCLUDING, BUT NOT LIMITED TO VALVE BOXES, SEWER MANHOLES AND CLEANOUTS, STORM DRAIN MANHOLES AND INLETS, ELECTRICAL, WATER, GAS AND TELEPHONE BOXES AND VAULTS, OCCURRING IN THE LOCATION OF PROPOSED IMPROVEMENTS SHALL BE RESET TO THE NEW FINISH GRADE.

M. NOTE THAT NPS UTILITIES REQUIRES 14 CALENDAR DAYS ADVANCE NOTICE TO LOCATE UTILITIES IN THE PARK.

4. UTILITY DISRUPTIONS

NOTIFY COR NOT LESS THAN 7 WORKING DAYS IN ADVANCE OF PROPOSED UTILITY DISRUPTIONS.

5. SURVEY CONTROL POINTS AND MONUMENTS

THE CONTRACTOR SHALL BE RESPONSIBLE FOR PRESERVING ALL EXISTING SURVEY MONUMENTS AND SURVEY CONTROL POINTS. ANY MONUMENTS OR SURVEY CONTROL POINTS OBLITERATED OR DAMAGED DURING CONSTRUCTION, WILL BE REPLACED AT THE CONTRACTOR'S EXPENSE BY A CALIFORNIA LICENSED LAND SURVEYOR.

6. PERMITS

A. THE GOVERNMENT WILL OBTAIN THE REQUIRED OWTS PERMIT FROM THE CVRWQCB. MARIPOSA COUNTY EHS WILL NOT ISSUE ANY PERMITS FOR THIS WORK – SEE NOTE 1 ABOVE.

B. THE GOVERNMENT DOES NOT REQUIRE ANY PERMITS FOR THE MECHANICAL, ELECTRICAL OR PLUMBING CONSTRUCTION WORK.

7. INSPECTIONS

A. ALL WORK SHALL BE SUBJECT TO THE INSPECTION AND APPROVAL OF THE NPS. WORK SHALL NOT NOT BE COVERED PRIOR TO INSPECTION AND APPROVAL. REJECTED WORK SHALL BE REDONE AT THE CONTRACTOR'S EXPENSE.

B. THE CONTRACTOR SHALL COMPLY WITH ALL EHS AND CVRWQCB INSPECTION REQUIREMENTS. FAILURE TO COMPLY WITH THE INSPECTION REQUIREMENTS COULD DELAY FINAL APPROVAL AND MAY REQUIRE EXCAVATION TO VERIFY THAT THE CONSTRUCTION WAS PERFORMED ACCORDING TO DESIGN. REJECTED WORK SHALL BE REDONE AT THE CONTRACTOR'S EXPENSE.

C. CONSTRUCTION STAKING OR MARKING THE LAYOUT OF ALL COMPONENTS OF THE SYSTEM SHALL OCCUR PRIOR TO COMMENCEMENT OF CONSTRUCTION SO THAT CONFIGURATION, LOCATION, AND SYSTEM DETAILS MAY BE VERIFIED.

D. THE FOLLOWING MEETINGS AND INSPECTIONS SHALL BE REQUIRED PRIOR TO COMMENCING CONSTRUCTION OR COVERING ANY PORTION OF THE SYSTEM. ALL MEETINGS AND INSPECTIONS SHALL BE SCHEDULED WITH THE COUNTY EHS, COR AND DESIGN ENGINEER AT LEAST 48 HOURS IN ADVANCE AND SHALL OCCUR DURING NORMAL BUSINESS HOURS:

1. OWTS PRE-CONSTRUCTION SITE MEETING
   - (a) VERIFICATION THAT THE SOIL MOISTURE IS ACCEPTABLE AND THAT LONG-TERM WEATHER CONDITIONS WILL ALLOW INSTALLATION OF THE SYSTEM.
   - (b) VERIFICATION THAT THE CONTRACTOR'S CONSTRUCTION STAKES OR MARKS ARE SET TO PROPERLY DELINEATE ALL COMPONENTS OF THE SYSTEM PER APPROVED PLANS. (THIS STAKING MUST BE COMPLETED PRIOR TO THE PRE-CONSTRUCTION SITE MEETING.) REVIEW STAKED LAYOUT FOR COORDINATION WITH THE SITE CONDITIONS, INCLUDING TOPOGRAPHIC FEATURES.
   - (c) REVIEW CONTRACTOR'S PROPOSED SEQUENCE OF WORK AND CONSTRUCTION SCHEDULE.
   - (d) REVIEW INSPECTION REQUIREMENTS, INCL. SCHEDULING.
   - (e) MATERIALS AND EQUIPMENT SUBMITTALS REVIEW.
   - (f) COORDINATION WITH OTHER PHASE 5 CONSTRUCTION ACTIVITIES.
   - (g) COORDINATION WITH TESCO PUMP CONTROL PANEL AND SCADA EQUIPMENT.
   - (h) COORDINATION WITH TREE, STUMP, DUFF AND TOPSOIL REMOVAL.
   - (i) COORDINATION WITH LEACHFIELD AREA GRADING.
   - (j) COORDINATION WITH PIEZOMETER DESTRUCTION & MONITORING WELL CONSTRUCTION.
2. GENERAL CONSTRUCTION INSPECTIONS
   - (a) OPEN TRENCH INSPECTIONS FOR ALIGNMENT ON CONTOUR, DEPTH AND SOIL CONDITION.
   - (b) DRAIN ROCK GRADATION AND WASHED CONDITION.
   - (c) DEPTH AND PLACEMENT OF DRAIN ROCK.
   - (d) PIPELINE ALIGNMENT AND GRADE IN TRENCHES.
   - (e) PD ORIFICE SIZE, CONDITION, SPACING AND ORIENTATION.
   - (f) HYDRAULIC TESTING (SQUIRT TEST) OF PUMPS AND DISTRIBUTION SYSTEM TO ENSURE THAT THE PUMP IS ADEQUATE FOR DESIGN FLOWS AND DELIVERS EFFLUENT EQUALLY TO ALL ORIFICES. SEE GENERAL NOTE 27/C7.63.
   - (g) ORIFICE SHIELDS AND INSTALLATION.
   - (h) GEOTEXTILE FABRIC PLACEMENT ON TOP OF DRAIN ROCK.
   - (i) TRENCH BACKFILL MATERIAL AND PLACEMENT.
   - (j) INSPECTION PORT MATERIALS AND INSTALLATION.
   - (k) MONITORING WELL MATERIALS, DRILLING AND CONSTRUCTION.
   - (l) DISTRIBUTING VALVE ASSEMBLY INSTALLATION AND OPERATION, INCLUDING VALVE MONITORS.
   - (m) INSTALLATION AND WATERTIGHTNESS OF SEPTIC TANKS, DOSING TANKS AND PUMP BASINS, INCL. PROTECTIVE COATINGS AND ALL APPURTENANCES.
   - (n) PUMP INSTALLATION AND PERFORMANCE.
   - (o) FUNCTION AND SETTINGS OF ANY CONTROL DEVICES INCLUDING, BUT NOT LIMITED TO, FLOATS, TIMERS, SWITCHES, PANELS, COUNTERS, ALARMS, VALVES, CHECK VALVES, AIR VALVES, PRESSURE GAUGES AND FLOW METERS.
   - (p) ALL THE REMAINING ELEMENTS FOR COMPLETION OF THE SYSTEM SHALL BE ON SITE AT TIME OF THE GENERAL CONSTRUCTION INSPECTION FOR VERIFICATION OF CONFORMANCE WITH THE PLANS AND SPECIFICATIONS.
3. FINAL INSPECTION
   - (a) ALL CONSTRUCTION ELEMENTS ARE IN GENERAL CONFORMANCE WITH THE APPROVED PLANS AND SPECIFICATIONS. ALL INSPECTION PORTS AND MONITORING WELLS ARE INSTALLED AND EROSION CONTROL HAS BEEN COMPLETED.
   - (b) A LETTER FROM THE DESIGN ENGINEER STATING THAT OWTS CONSTRUCTION AND OPERATION ARE IN SUBSTANTIAL COMPLIANCE WITH THESE PLANS AND SPECIFICATIONS SHALL BE PROVIDED TO EHS AND CVRWQCB.
4. FINAL VEGETATIVE COVER INSPECTION
   - (a) SEE NOTE 7/C7.51 FOR 1" DUFF LAYER PLACEMENT BY OTHERS AND SURFACE RESTORATION AND PLANTING WORK BY OTHERS.
   - (b) FINAL VEGETATIVE COVER MATERIALS AND INSTALLATION.

8. CONSTRUCTION SCHEDULE

SEE SPECIFICATIONS SECTIONS 01 11 00 AND 01 32 16.

9. SUBMITTALS

A. PREPARE AND SUBMIT COMPREHENSIVE SHOP DRAWINGS, MATERIALS AND EQUIPMENT TECHNICAL DATA SHEETS, PRODUCT DATA, SAMPLES, AND OTHER SUBMITTALS IN ACCORD WITH SPECIFICATIONS SECTION 01 33 23, FOR ALL CONSTRUCTION MATERIALS AND EQUIPMENT, INCLUDING, BUT NOT LIMITED TO, PIPING, FITTINGS, COUPLINGS, VALVES, DISTRIBUTING VALVES AND MONITORS, METERS, UTILITY BOXES AND EXTENSIONS, UTILITY ENCLOSURES, TANKS AND APPURTENANCES, MANHOLES, CLEANOUTS, PUMPS, CONTROLS, ELECTRICAL EQUIPMENT, CONDUIT, CONDUCTORS, APPURTENANCES, EFFLUENT FILTERS, PROTECTIVE COATINGS, BEDDING AND BACKFILL, LEACHFIELD ROCK, GEOTEXTILE FABRICS, MARKER POSTS, ORIFICE SHIELDS, TRACER WIRE, AND UTILITY MARKING TAPE.

B. WHENEVER REQUESTED BY THE COR OR THE ENGINEER, SUBMITTALS SHALL INCLUDE EVIDENCE OF THE NUMBER OF YEARS ANY MANUFACTURER HAS BEEN IN THE BUSINESS OF MANUFACTURING RELEVANT PRODUCTS/SYSTEMS.

C. CONTRACTOR SHALL FURNISH DOCUMENTATION RELATED TO PRESSURE DISTRIBUTION SYSTEM, INCLUDING CONTROLS SYSTEM, MANUFACTURERS (OR REPRESENTATIVES) SUPPORT SERVICES, INCLUDING, BUT NOT LIMITED TO;

1. INSTALLATION TRAINING PROGRAM AND SUPPORT MATERIAL
2. INSTALLATION OVERSIGHT PROGRAM AND SUPPORT MATERIAL
3. OPERATOR TRAINING PROGRAM AND SUPPORT MATERIAL
4. STARTUP SERVICES PROGRAM AND SUPPORT MATERIAL

10. QUALITY AND EXPERIENCE REQUIREMENTS

A. SEE SPECIFICATIONS SECTION 01 40 00.

B. THE OWTS SYSTEM, INCLUDING ALL MATERIALS AND EQUIPMENT, SHALL BE INSTALLED BY A COMPANY EXPERIENCED IN THE INSTALLATION OF ONSITE WASTEWATER TREATMENT SYSTEMS. INSTALLERS SHALL DEMONSTRATE AT LEAST 5 YEARS OF SUCCESSFUL EXPERIENCE IN THE CONSTRUCTION AND/OR OPERATION OF LARGE COMMERCIAL ONSITE WASTEWATER TREATMENT SYSTEMS (E.G., ORENCO PRESSURE DOSING WASTEWATER TREATMENT SYSTEMS, ETC.), SPECIFICALLY PRESSURE DOSING SYSTEMS. PROVIDE EXPERIENCE HISTORY FOR SIMILAR TYPE AND SIZE PROJECTS INCLUDING CONSTRUCTION OF THE PRESSURE DOSING TANKS AND SEPTIC TANKS, WET WELLS, BIOTUBE FILTERS, LEACH LINE INSTALLATION, PIPING INSTALLATION (ENSURING ORIFICES ARE PROPERLY FABRICATED & CONSTRUCTED), FLOW METER & DISTRIBUTING VALVE INSTALLATION, TESTING ONSITE WASTEWATER TREATMENT SYSTEMS, INSTALLATION OF THE ELECTRONIC CONTROL SYSTEM AND PROGRAMMING, ALONG WITH FAMILIARITY WITH SCADA AND TRAINING THAT HAS BEEN TAKEN OR PROVIDED RELATED TO PRESSURE DOSING ONSITE WASTEWATER TREATMENT SYSTEMS.

IF ANY SUBCONTRACTORS ARE INVOLVED, PROVIDE SUMMARIES AND DETAILED INFORMATION THAT DEMONSTRATES SIMILAR EXPERIENCE APPLICABLE TO THEIR SPECIFIC PLANNED ACTIVITIES.

PROVIDE CONTRACTOR'S HISTORY ASSOCIATED WITH THE REQUESTED SUCCESSFUL WORK EXPERIENCE NOTED ABOVE.

C. THE CONTRACTOR SHALL PROVIDE THE SERVICES OF OWTS EQUIPMENT MANUFACTURERS' TRAINED REPRESENTATIVES FOR A MINIMUM OF ONE VISIT EACH FOR THE PURPOSE OF QUALITY CONTROL DURING CONSTRUCTION.

11. TEMPORARY FACILITIES AND CONTROLS

SEE SPECIFICATIONS SECTION 01 50 00. SEE GEN. NOTE 2.

12. PRODUCT REQUIREMENTS

SEE SPECIFICATIONS SECTION 01 67 00.

13. OPERATION AND MAINTENANCE DATA

SEE SPECIFICATIONS SECTION 01 78 23. THE CONTRACTOR SHALL PROVIDE COMPLETE O&M MANUALS, PREVENTATIVE MAINTENANCE PLANS, AND COMMISSIONING AND DECOMMISSIONING PLAN FOR THE COMPLETE OWTS, INCLUDING ALL COMPONENTS.

14. DEMONSTRATION, TRAINING AND COMMISSIONING

A. SEE SPECIFICATIONS SECTION 01 79 00. SEE GENERAL NOTE 2.

B. TRAIN DESIGNATED PERSONNEL IN SYSTEM OPERATION AND ADJUSTMENT, INCLUDING SEASONAL AND EMERGENCY OPERATIONS, IF APPLICABLE; MAINTENANCE; AND SAFETY REQUIREMENTS OF EQUIPMENT AND SYSTEMS. INSTRUCTORS SHALL BE THOROUGHLY TRAINED IN OPERATING THEORY AS WELL AS PRACTICAL OPERATION AND MAINTENANCE WORK FOR EACH TYPE OF EQUIPMENT OR SYSTEM. THE SEQUENCE OF THE TRAINING SHALL FOLLOW THE APPROVED OUTLINE OF THE TRAINING GUIDE; I.E. OPERATING AND MAINTENANCE DATA. INDIVIDUAL SECTIONS SPECIFY THE DURATION OF TRAINING REQUIRED. IF NO DURATION IS LISTED, PROVIDE TRAINING OF SUFFICIENT DURATION TO ADEQUATELY COVER THE SUBJECTS.

C. THE CONTRACTOR SHALL PROVIDE THE SERVICES OF OWTS EQUIPMENT MANUFACTURERS' TRAINED REPRESENTATIVES FOR A MINIMUM OF ONE VISIT EACH FOR THE PURPOSE OF OWTS COMMISSIONING.

15. CORRECTIVE WORK

A. THE COST OF CORRECTIVE WORK REQUIRED FOR COMPLETION AND/OR ACCEPTANCE OF THE WORK NECESSITATED BECAUSE OF UNSATISFACTORY WORKMANSHIP OR MATERIALS SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR.

B. PRIOR TO THE CONCLUSION OF THE WORK, THE CONTRACTOR SHALL REPAIR OR REPLACE IN KIND ALL DAMAGED PROPERTY FOR WHICH HE IS RESPONSIBLE (INCLUDING CURB, GUTTER AND SIDEWALK). THE WORK SITE AND ADJACENT WORK SHALL BE CLEANED UP. THIS REPAIR AND REPLACEMENT, AND CLEAN UP WORK SHALL BE SUBJECT TO THE APPROVAL AND ACCEPTANCE OF THE CONTRACTING OFFICER'S REPRESENTATIVE.

16. RECORD DRAWINGS

A. THE CONTRACTOR SHALL SURVEY THE NORTH AND EAST COORDINATES AND ELEVATIONS OF ALL ANGLE POINTS AND FITTINGS ON NEW UTILITIES PLUS ALL EXPOSED EXISTING UTILITIES DISCOVERED DURING TRENCHING OR POTHOLING. THIS INFORMATION SHALL BE INCORPORATED INTO THE RECORD DRAWINGS. SEE SPECIFICATIONS SECTION 01 77 00 FOR RECORD DRAWING REQUIREMENTS AND CLOSEOUT PROCEDURES.

B. A COMPLETED CONSTRUCTION RECORD SUBMISSION MUST BE PREPARED TO THE SATISFACTION OF THE COR AND MUST CONTAIN, AT A MINIMUM, THE FOLLOWING ITEMS: ALL THE ITEMS CONTAINED IN THE DESIGN SUBMITTAL, AS INSTALLED, IDENTIFYING ANY CHANGES FROM THE APPROVED PLAN, THE MEASURED DRAWDOWN PER DOSE CYCLE; PLC SETTINGS; TIMER FUNCTIONS; PUMP RUN TIME AND PUMP TIME OFF; AND RESIDUAL PRESSURE AND/OR SQUIRT HEIGHT AT THE END OF EACH LATERAL, AS INSPECTED.

17. SITE PREPARATION, CLEARING, EARTHWORK AND GRADING

A. ALL EARTHWORK SHALL BE IN CONFORMANCE WITH THE SOILS REPORT PREPARED BY SHANNON & WILSON, INC., TITLED "GEOTECHNICAL DESIGN REPORT, YOSEMITE NATIONAL PARK, MARIPOSA GROVE RESTORATION" DATED MARCH 12, 2014, AND THE NOV. 5, 2015, DATED ADDENDUM NO. 1. THE CONTRACTOR SHALL OBTAIN A COPY OF THE SOILS REPORT FROM THE CONTRACTING OFFICER'S REPRESENTATIVE (COR), FAMILIARIZE HIMSELF WITH ITS CONTENTS AND PERFORM THE EARTHWORK/TRENCHING/BACKFILL IN ACCORDANCE WITH THE RECOMMENDATIONS CONTAINED THEREIN.

B. THE CONTRACTOR SHALL NOTIFY CONTRACTING OFFICER'S REPRESENTATIVE (COR) OF PROPOSED LOCATIONS AND TIMES FOR EXCAVATION.

C. SEE DRAWINGS C5.11, C5.12, C5.13, C7.50 AND C7.51 FOR ADDITIONAL TREE REQUIREMENTS. NOT ALL EXISTING TREES AND STUMPS ARE SHOWN ON C7.50.

D. LEACHFIELD AREA EXISTING "GROUND SURFACE" ELEVATIONS AND CONTOURS SHOWN ON THE PLANS WERE OBTAINED BY TOPOGRAPHIC SURVEYING PERFORMED PRIOR TO TREE CUTTING AND REMOVAL OF TREE STUMPS AND DUFF.

CONSEQUENTLY, THE ELEVATIONS AND CONTOURS SHOWN WILL PROBABLY NOT REPRESENT TOP OF TOPSOIL ELEVATIONS AFTER STUMP AND DUFF REMOVALS ARE COMPLETED.

E. FINISH GRADING AROUND OWTS COMPONENTS: GRADED SURFACES ARE TO DRAIN AWAY FROM STRUCTURES. THE FINISHED SURFACES SHALL BE SMOOTH AND COMPACTED, UNLESS SPECIFIED OTHERWISE.

GRADE WHERE NECESSARY TO ACHIEVE SMOOTH CONTOURS, ELIMINATE LOCAL PONDING, AND PROVIDE POSITIVE DRAINAGE. BLEND SLOPES INTO RELATIVELY LEVEL AREAS. MAKE GRADE CHANGES GRADUAL, WITH NO DEPRESSIONS OR RAISED AREAS.

PHASE 5

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MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM

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DESIGNED: TGH
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TITLE OF SHEET
SOUTH ENTRANCE
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YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 127662
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SHEET 113 OF 291

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GENERAL NOTES CONTINUED

F. THE PROPOSED LEACHFIELD AREA SHALL BE GRADED TO: ACHIEVE SMOOTH CONTOURS; ELIMINATE EXISTING SURFACE IRREGULARITIES AND THOSE RESULTING FROM CONSTRUCTION ACTIVITIES SUCH AS TREE, STUMP, DUFF AND TOPSOIL REMOVAL; PROVIDE POSITIVE SURFACE DRAINAGE; AND FACILITATE LEACH TRENCH INSTALLATION.

IN GENERAL, "SMOOTHING" SHALL BE ACCOMPLISHED BY EXCAVATION RATHER THAN FILLING, WITH FILL DEPTHS LIMITED TO 6" AND EXCAVATION DEPTHS NOT EXCEEDING 2'. ALL SUCH WORK SHALL BE CAREFULLY COORDINATED WITH THE COUNTY EHS, DESIGN ENGINEER AND COR.

G. LISTED BELOW IS A POSSIBLE SEQUENCE OF "GROUND WORK" RELATED TO THE FOUR MILE ROAD SITE. COORDINATE WITH THE COR.

1. ADDITIONAL TREE CUTTING AND REMOVAL WILL BE PERFORMED AS NEEDED BY THE PARK.
2. STUMP REMOVAL SHALL BE PERFORMED BY THE PHASE 5 CONTRACTOR. ALL STUMPS WITHIN THE WORK ZONE ARE TO BE REMOVED.
3. ALL SLASH (WOOD GREATER THAN NOMINAL 2" DIAMETER) IS TO BE REMOVED FROM THE SITE BY THE PHASE 5 CONTRACTOR.
4. ALL WOODY DEBRIS, INCLUDING DEBRIS FROM THE FALLING OF TREES, SHALL BE REMOVED FROM THE SITE BY THE PHASE 5 CONTRACTOR.
5. ROOT REMOVAL IN TRENCH CROSS-SECTIONS AND TANK EXCAVATION AREAS AND IN GRADED LEACHFIELD, YARD AND DRIVEWAY AREAS SHALL BE BY THE PHASE 5 CONTRACTOR. NO OTHER ROOT REMOVAL IS REQUIRED, BUT THE CONTRACTOR MAY ELECT TO DO MORE IF APPROVED BY THE COR. SEE NOTE 3/C5.12.
6. FOREST DUFF (ALSO KNOWN AS FOREST LITTER, CONSISTING OF LEAVES AND NEEDLES THAT HAVE FALLEN OFF THE TREES AND BUSHES) SHOULD ONLY BE REMOVED FROM GRADING, TRENCHING AND EXCAVATION AREAS. THE EXTENT OF DUFF REMOVAL, STOCKPILING AND REPLACEMENT NEEDS TO BE COORDINATED WITH THE PARK'S REVEGETATION PLAN. THE CONTRACTOR MAY ELECT TO DO MORE DUFF REMOVAL IF APPROVED BY THE COR. DUFF IS TO BE EXCLUDED FROM ALL BACKFILLING AND FILLING OPERATIONS. THE CONTRACTOR WILL NEED TO COORDINATE HIS DUFF REMOVAL, STOCKPILING AND REPLACEMENT PLANS WITH THE COR.
7. GRADING (1) TO FILL AND SMOOTH STUMP EXCAVATIONS, (2) TO SMOOTH LEACHFIELD CONTOUR IRREGULARITIES TO BETTER ACCOMMODATE LEACH TRENCH INSTALLATION, (3) TO PREPARE DRIVEWAY AND YARD AREAS, AND (4) TO CREATE TEMPORARY WORK AREAS, AS MAY BE NEEDED.
8. TOPSOIL REMOVAL IS NOT REQUIRED.

   IF THE CONTRACTOR CHOOSES TO REMOVE TOPSOIL, LIMIT TOPSOIL REMOVAL TO TRENCHING AND EXCAVATION AREAS. THE CONTRACTOR WILL NEED TO COORDINATE HIS TOPSOIL REMOVAL, STOCKPILING AND REPLACEMENT PLANS WITH THE COR.
9. DRAINAGE INTERCEPTOR DITCH CONSTRUCTION
10. TRENCHING AND EXCAVATION
11. BACKFILLING AND FILLING
12. TOPSOIL REPLACEMENT WHERE IT'S BEEN REMOVED. SEE ITEM 8 ABOVE.
13. DUFF REPLACEMENT: THE STOCKPILED DUFF IS TO BE REPLACED ONTO DUFF REMOVAL AREAS. SEE ITEM 6 ABOVE.
14. REMOVAL OF EXISTING UNDERGROUND UTILITIES IN FOUR MILE ROAD AREA.
15. INSTALLATION OF NEW UNDERGROUND UTILITIES IN FOUR MILE ROAD.
16. GRADING FOR FOUR MILE ROAD RECONSTRUCTION.
17. REVEGETATION BY PARK, INCLUDING COORDINATION WITH SWPPP BMPS BY CONTRACTOR.

18. <u>SANITARY SEWER SYSTEMS</u>

SEE SPECIFICATIONS SECTION 33 30 00.

19. <u>SEWER MANHOLES, FRAMES AND COVERS</u>

SEE SPECIFICATIONS SECTION 33 39 13.

20. <u>SEWER PIPING</u>

A. UNLESS SPECIFIED OTHERWISE ON THE PLANS, 4" AND 6" GRAVITY-FLOW PIPE & FITTINGS SHALL CONFORM TO ASTM D3034, SDR 26 PVC PIPE & FITTINGS, WITH ASTM D3212 ELASTOMERIC GASKET JOINTS AND PUSH-ON ASTM F477 GASKETS.

B. 3" AND SMALLER PRESSURE SEWER PIPING SHALL CONFORM TO ONE OF THE FOLLOWING, AS SPECIFIED ON THE PLANS:

1. PVC PIPE: ASTM D1785/D2466 SCH. 40 PVC PIPE AND SOCKET-TYPE FITTINGS WITH ASTM D2564 SOLVENT WELD CEMENT CONNECTIONS. USE SOLVENT WELD JOINTS, EXCEPT FLANGED OR THREADED JOINTS SHALL BE PERMITTED WHERE REQUIRED AT EQUIPMENT CONNECTIONS AND WHERE SPECIFIED ON THE PLANS.
2. PVC PIPE: ASTM D1785/D2467 SCH. 80 PVC PIPE AND SOCKET-TYPE FITTINGS WITH ASTM D2564 SOLVENT WELD CEMENT CONNECTIONS. USE SOLVENT WELD JOINTS, EXCEPT FLANGED OR THREADED JOINTS SHALL BE PERMITTED WHERE REQUIRED AT EQUIPMENT CONNECTIONS AND WHERE SPECIFIED ON THE PLANS.

   THREADED CONNECTIONS SHALL BE SCH. 80 PER ASTM D2464. FOR THREADED CONNECTIONS ON PVC PIPE, PROVIDE A SHORT NIPPLE, THREADED AT ONE END, WITH A SOCKET FITTING AT THE OPPOSITE END. PROVIDE THREAD SEALANT IN ACCORDANCE WITH THE PIPE MANUFACTURER'S RECOMMENDATIONS. TAKE CARE NOT TO OVERTIGHTEN THE CONNECTION.
3. HDPE PIPE:

(a) AWWA C901, PE 4710, SDR11, PC 200, HDPE PIPE. PIPE SHALL HAVE A RESIN CELL CLASSIFICATION PER ASTM D3350 OF PE445474C OR HIGHER. PIPE OUTSIDE DIAMETER SHALL BE IRON PIPE SIZE (IPS) PER ASTM D3035. SUBMIT CERTIFICATION THAT THE PIPE COMPLIES WITH THE SPECIFICATIONS.

(b) MANUFACTURER: PERFORMANCE PIPE DRISCOPLEX 4100, OR APPROVED EQUIVALENT.

(c) THE PIPE SHALL CONTAIN NO RECYCLED COMPOUNDS EXCEPT THAT GENERATED IN THE MANUFACTURER'S OWN PLANT FROM RESIN OF THE SAME SPECIFICATION FROM THE SAME RAW MATERIAL. ALL PIPES SHALL BE SUITABLE FOR USE AS PRESSURE CONDUITS.

(d) INSTALL PIPE AND FITTINGS PER THE PIPE MANUFACTURER'S RECOMMENDATIONS AND THE REQUIREMENTS OF AWWA C901. HDPE PIPE INSTALLATION SHALL COMPLY WITH ASTM D2321, "STANDARD PRACTICE FOR UNDERGROUND INSTALLATION OF THERMOPLASTIC PIPE FOR SEWERS AND OTHER GRAVITY FLOW APPLICATIONS"; AND ASTM D2774, "STANDARD PRACTICE FOR UNDERGROUND INSTALLATION OF THERMOPLASTIC PRESSURE PIPING."

MINIMUM COLD-BENDING RADIUS FOR HDPE PIPE IS 100 TIMES THE OUTSIDE DIAMETER OF THE PIPE.

(e) BUTT HEAT FUSION JOINING: THE ASSEMBLY OF ALL HDPE PIPE JOINTS, INCLUDING PIPE ENDS AND PIPE FITTINGS, SHALL BE BY CONVENTIONAL BUTT HEAT FUSION JOINING, EXCEPT WHERE JOINING BY OTHER MEANS IS REQUIRED TO CONNECT WITH OTHER PIPE, VALVE, OR EQUIPMENT MATERIALS. BUTT HEAT FUSION JOINING SHALL BE IN ACCORDANCE WITH THE PIPE MANUFACTURER'S RECOMMENDATIONS AND WITH ASTM D2657, "STANDARD PRACTICE OF HEAT FUSION JOINING OF POLYOLEFIN PIPE AND FITTINGS."

THE CONTRACTOR SHALL ENSURE THAT PERSONS MAKING HEAT FUSION JOINTS HAVE RECEIVED TRAINING IN THE MANUFACTURER'S RECOMMENDED PROCEDURE. THE CONTRACTOR SHALL MAINTAIN RECORDS OF TRAINED PERSONNEL, AND SHALL CERTIFY THAT TRAINING WAS RECEIVED NOT MORE THAN 12 MONTHS BEFORE COMMENCING CONSTRUCTION. EXTERNAL AND INTERNAL BEADS SHALL NOT BE REMOVED.

(f) BUTT FUSION FITTINGS SHALL BE PE4710 HDPE, COMPATIBLE FOR HEAT FUSION WITH THE PROJECT-SPECIFIED PIPE AND FITTINGS, AND APPROVED FOR AWWA USE. BUTT FUSION FITTINGS SHALL HAVE A MANUFACTURING STANDARD OF ASTM D3261. MOLDED & FABRICATED FITTINGS SHALL BE MADE BY HEAT FUSION JOINING SPECIALLY MACHINED SHAPES CUT FROM PIPE, POLYETHYLENE SHEET STOCK OR MOLDED FITTINGS. FABRICATED FITTINGS SHALL BE RATED FOR INTERNAL PRESSURE SERVICE AT LEAST EQUAL TO THE FULL SERVICE PRESSURE RATING OF THE MATING PIPE. FABRICATED FITTINGS SHALL BE TESTED IN ACCORDANCE WITH AWWA C906. FABRICATED FITTINGS ARE TO BE MANUFACTURED USING DATA LOGGERS. TEMPERATURE, FUSION PRESSURE AND A GRAPHIC REPRESENTATION OF THE FUSION CYCLE SHALL BE PART OF THE QUALITY CONTROL RECORDS.

(g) JOINING BY OTHER MEANS: HDPE PIPE AND FITTINGS MAY BE JOINED TOGETHER OR TO OTHER MATERIALS BY MEANS OF (A) FLANGED CONNECTIONS (FLANGE ADAPTERS AND BACK-UP RINGS), (B) MECHANICAL COUPLINGS DESIGNED FOR JOINING POLYETHYLENE PIPE OR FOR JOINING POLYETHYLENE PIPE TO ANOTHER MATERIAL, (C) MJ ADAPTERS OR (D) ELECTROFUSION. WHEN JOINING BY OTHER MEANS, THE INSTALLATION INSTRUCTIONS OF THE JOINING DEVICE MANUFACTURER SHALL BE OBSERVED.

(h) FLANGE ADAPTERS WITH DUCTILE IRON BACK-UP RINGS SHALL BE USED WHERE HDPE PIPE MATES TO DISSIMILAR MATERIALS, AND TO PUMPS, VALVES AND OTHER APPURTENANCES. FLANGE ADAPTERS SHALL BE PE4710 HDPE, COMPATIBLE FOR HEAT FUSION WITH THE PROJECT-SPECIFIED PIPE AND FITTINGS. FLANGE ADAPTERS SHALL HAVE A MANUFACTURING STANDARD OF ASTM D3261.

POLYETHYLENE FLANGE ADAPTERS: FLANGE ADAPTERS SHALL BE MADE WITH SUFFICIENT THROUGH-BORE LENGTH TO BE CLAMPED IN A BUTT FUSION-JOINING MACHINE WITHOUT THE USE OF A STUB-END HOLDER. THE SEALING SURFACE OF THE FLANGE ADAPTER SHALL BE MACHINED WITH A SERIES OF SMALL V-SHAPED GROOVES (SERRATIONS).

DUCTILE IRON BACK-UP RINGS, AND ALL METAL FITTINGS AND APPURTENANCES, SHALL BE DOUBLE-WRAPPED WITH 8-MIL PE ENCASEMENT PER AWWA C105. TAPE THE EDGES OF THE ENCASEMENT WITH PVC TAPE.

BACK-UP RINGS & FLANGE BOLTS: FLANGE ADAPTERS SHALL BE FITTED WITH BACK-UP RINGS THAT ARE PRESSURE RATED EQUAL TO OR GREATER THAN THE MATING PIPE. FLANGES SHALL CONFORM TO ANSI B16.1 DIMENSIONAL STANDARDS. THE BACK-UP RING BORE SHALL BE CHAMFERED OR RADIUSED TO PROVIDE CLEARANCE TO THE FLANGE ADAPTER RADIUS. FLANGE BOLTS AND NUTS SHALL BE TYPE 304SS WITH DIELECTRIC WASHERS AND CHLOROPRENE GASKETS.

(i) COMPRESSION FITTINGS SHALL BE FORD METER BOX, OR APPROVED EQUIVALENT, BRASS, PACK JOINT FITTINGS WITH 304SS SOLID TUBULAR INSERT STIFFENERS, DIMPLED AND FLANGED TO RETAIN POSITION WITHIN THE PIPE.

ID STIFFENER AND RESTRAINT: A STIFFENER SHALL BE INSTALLED IN THE BORE OF THE POLYETHYLENE PIPE WHEN AN OD COMPRESSION MECHANICAL COUPLING IS USED AND WHEN CONNECTING PLAIN END PE PIPE TO A MECHANICAL JOINT PIPE, FITTING OR APPURTENANCE. EXTERNAL CLAMP AND TIE ROD RESTRAINT SHALL BE INSTALLED WHERE PE PIPE IS CONNECTED TO THE SOCKET OF A MECHANICAL JOINT PIPE, FITTING OR APPURTENANCE EXCEPT WHERE AN MJ ADAPTER IS USED.

(j) MJ ADAPTERS: MJ ADAPTERS 4" THRU 16" SHALL BE PROVIDED WITH OPTIONAL STAINLESS STEEL STIFFENER. MJ ADAPTERS 14" AND ABOVE SHALL BE PROVIDED WITH HEAVY DUTY BACK-UP RING KITS.

(k) ELECTROFUSION JOINING, IN ACCORD WITH ASTM F1290, "STANDARD PRACTICE FOR ELECTROFUSION JOINING POLYOLEFIN PIPE AND FITTINGS", MAY BE USED UPON APPROVAL BY THE COR AND THE ENGINEER. ELECTROFUSION FITTINGS SHALL COMPLY WITH ASTM F1055.

(l) BRANCH CONNECTIONS TO THE MAIN SHALL BE MADE WITH SADDLE FITTINGS OR TEES.

C. ALL BURIED PIPING, EXCEPT LEACHFIELD PIPING, SHALL BE INSTALLED WITH A COPPER LOCATING WIRE PER DETAILS 3/D5.11 AND 4/D5.21.

D. UNLESS SPECIFIED OTHERWISE, ALL SEWER PIPING SHALL BE INSTALLED WITHOUT DEPRESSIONS OR HIGH POINTS THAT COULD TRAP AIR. VERIFY PIPELINE PROFILES WITH ENGINEER AND COR.

E. PROVIDE UNION CONNECTIONS TO EACH PIECE OF EQUIPMENT TO FACILITATE EASY REMOVAL AND REPLACEMENT.

F. ALL PIPE ENDS SHALL HAVE EXTERIOR AND INTERIOR SHARP EDGES REMOVED.

G. CAREFULLY FLUSH ALL SEWER PIPING.

H. PRESSURE PIPELINES SHALL BE HYDROSTATICALLY TESTED TO AT LEAST 100 PSI. HDPE PIPE SHALL BE HYDROSTATICALLY TESTED IN ACCORD WITH THE PLASTIC PIPE INSTITUTE HANDBOOK.

21. <u>OWTS TANKS</u>

A. ALL TANK CONSTRUCTION AND INSTALLATION, INCLUDING ALL APPURTENANCES AND ALL INSPECTION AND TESTING, SHALL BE IN ACCORD WITH THE MARIPOSA COUNTY SEWAGE DISPOSAL REGULATIONS (SEE GENERAL NOTE 1) AND WITH CPC APPENDIX H.

B. NEW AND REPLACEMENT SEPTIC TANKS SHALL BE LIMITED TO THOSE APPROVED BY THE INTERNATIONAL ASSOCIATION OF PLUMBING AND MECHANICAL OFFICIALS (IAPMO) OR STAMPED AND CERTIFIED BY A CALIFORNIA REGISTERED CIVIL ENGINEER AS MEETING THE INDUSTRY STANDARDS.

C. UNLESS SPECIFIED OTHERWISE ON THESE PLANS, ALL TANKS AND RISERS SHALL BE PRECAST REINFORCED CONCRETE PER ASTM C890 AND C1227-13. SEE PLANS FOR TANK SIZE. PLAN-SPECIFIED TANK SIZE REFERS TO MINIMUM OPERATING CAPACITY.

D. TANKS MUST BE DESIGNED AND CONSTRUCTED TO WITHSTAND ALL STRUCTURAL, HYDRAULIC, HYDROSTATIC, EARTH LOADS, AND ANY ANTICIPATED TRAFFIC LOADS. THEY MUST BE DESIGNED AND CONSTRUCTED SO THEY:

1. DO NOT COLLAPSE, DEFORM, OR CRACK WHEN SUBJECTED TO THE ANTICIPATED LOADS WHEN THE TANKS ARE EITHER ULL OR EMPTY;
2. SUPPORT A DEAD LOAD EQUIVALENT TO AT LEAST THREE FEET OF EARTH COVER WITH A UNIT DENSITY OF AT LEAST 110 PCF AND A 2,500 LB/WHEEL LOAD CONCENTRATION OVER THE CRITICAL ELEMENTS OF THE TANK;
3. ACCOUNT FOR MINIMUM HYDROSTATIC LOAD OF 62.4 PCF AND SUPPORT EARTH BACKFILL AND HYDROSTATIC PRESSURES. MINIMUM LATERAL LOAD CALCULATIONS MUST INCLUDE PRESSURES DUE TO EFFECTIVE WEIGHT OF ADJACENT EARTH BACKFILL AND HYDROSTATIC LOADS ASSUMING THE WATER TABLE IS 2 FEET BELOW GROUND LEVEL;

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MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM

PRIME:
DAVID EVANS AND ASSOCIATES, INC.
BELLEVUE, WA

SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.62

PHASE 5

TITLE OF SHEET

SOUTH ENTRANCE OWTS PLANS GENERAL NOTES

YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104 / 127662

PMIS/PKG NO. 206431

SHEET 114 OF 291

GENERAL NOTES CONTINUED

4. ALLOW FOR SEPTAGE PUMPING DURING HIGH GROUNDWATER CONDITIONS. INTERNAL HYDROSTATIC PRESSURES MUST BE INCLUDED IN THE CALCULATIONS TO ALLOW FOR SEPTAGE PUMPING DURING HIGH GROUNDWATER CONDITIONS ASSUMING WATER TABLE IS 2 FEET BELOW GROUND LEVEL;

5. COUNTERACT BUOYANCY EFFECTS, ASSURING AN ADEQUATE FLOTATION SAFETY FACTOR IN HIGH GROUNDWATER CONDITIONS. THE CONTRACTOR'S TANK DESIGN ENGINEER SHALL SUBMIT TO THE COR CALCULATIONS TO DEMONSTRATE THE TANK'S ABILITY TO COUNTERACT BUOYANCY EFFECTS AND INCLUDE THIS INFORMATION AS PART OF THE SEWAGE TANK INSTALLATION INSTRUCTIONS; AND

6. UNLESS SPECIFIED OTHERWISE ON THESE PLANS, TANKS, INCLUDING MANWAY RISERS, FRAMES AND COVERS, SHALL BE DESIGNED FOR AASHTO HS20-44 HEAVY-TRAFFIC LOADING PER ASTM C890, WITH A WHEEL LOAD OF 16,000 LB/WHEEL AND FOURTEEN FEET AXLE SPACING.

E. STRUCTURAL DESIGN SHALL COMPLY WITH THE CBC. STRUCTURAL PLANS AND CALCULATIONS, PREPARED BY AN ENGINEER LICENSED IN CALIFORNIA, ARE REQUIRED TO BE PROVIDED BY THE CONTRACTOR, TO THE SATISFACTION OF THE COR AND THE PROJECT ENGINEER.

DRAWINGS OF THE TANKS MUST BE COMPLETE AND SHOW ALL DIMENSIONS, CAPACITIES, REINFORCEMENT, STRUCTURAL CALCULATIONS, AND OTHER DATA REQUESTED BY THE COR. THE DRAWINGS MUST BE DRAWN TO SCALE AND SHOW:

1. A SIDE SECTION VIEW OF THE TANK WITH DETAILS ON INLETS, OUTLETS, AND ANY INTERCOMPARTMENTAL DEVICES;
2. MATERIAL SPECIFICATIONS;
3. A PLAN AND SIDE SECTION VIEW OF THE TANK SHOWING THE DIMENSIONS, INCLUDING THICKNESS OF VARIOUS PORTIONS OF THE TANK;
4. REINFORCEMENT DETAILS;
5. THE SIZE AND LOCATION OF ALL INSPECTION AND MAINTENANCE ACCESS, AND INLET AND OUTLET OPENINGS IN THE TANK;
6. THE NUMBER OF COMPARTMENTS;
7. THE CAPACITY OF EACH COMPARTMENT IN THE TANK;
8. THE EXCAVATION, BACKFILL, COMPACTION, DEPTH OF BURY, BEDDING AND INSTALLATION REQUIREMENTS; AND
9. THE PROVISIONS TO COUNTERACT BUOYANCY FORCES.

F. TANKS, INCLUDING RISERS (MANWAYS), FRAMES AND COVERS, AND PIPING CONNECTIONS, SHALL BE WATERTIGHT AND PREVENT SURFACE DRAINAGE AND GROUNDWATER FROM ENTERING THE TANK. SEALS AND GASKETS FOR INLET, OUTLET, AND INTERCOMPARTMENTAL FITTINGS MUST BE RESILIENT, WATERTIGHT, CORROSION-RESISTANT, AND FLEXIBLE. SEALS MUST BE USED TO JOIN THE TANK WALL AND THE PIPING TO PREVENT LEAKAGE AT THE WALL CONNECTION.

USE HENRY COMPANY'S BN109 BUTYL-NEK, OR APPROVED EQUIVALENT, ASTM C990 COMPLIANT JOINT SEALANT, CAREFULLY INSTALLED IN ACCORD WITH THE MFR'S INSTRUCTIONS.

ALL PIPE-TO-TANK CONNECTIONS SHALL BE WATERTIGHT WITH NPC KOR-N-SEAL, OR POLYLOK TYPE VI SEAL BOOT, OR APPROVED EQUIVALENT, CONNECTORS, SECURELY AND ACCURATELY INSTALLED PER MANUFACTURER'S RECOMMENDATIONS.

G. UNLESS NOTED OTHERWISE, MANWAY RISERS AND GRAY CAST-IRON FRAMES AND COVERS SHALL BE 30" MIN. INSIDE DIAMETER. COVERS SHALL BE NON-VENTILATED WITH 1/4" DIAMETER NEOPRENE O-RING GASKET SET INTO A GROOVE ON THE SIDE OF THE COVER FOR A GAS-TIGHT SEAL. THE UNDERSIDE OF THE COVER AND THE INTERIOR SURFACE OF THE FRAME SHALL BE COATED WITH A MINIMUM 8 MIL DFT COAL-TAR EPOXY. ALSO, SEE GENERAL NOTE 19 FOR OTHER FRAME AND COVER REQUIREMENTS. INSTALL MANWAYS ABOVE INLET AND OUTLET TEES, EFFLUENT FILTERS AND AS SHOWN ON THE PLANS. COVERS TO BE LABELED "SEWER". SEE PLANS FOR CONCRETE COLLARS AROUND MANWAY FRAMES.

H. ACCESS OPENINGS MUST BE LARGE ENOUGH FOR A PERSON WITH EQUIPMENT TO EASILY CLEAN, MAINTAIN, REMOVE, AND REPLACE SEWAGE TANK COMPONENTS. ACCESS OPENINGS MUST BE LOCATED ABOVE THE INLET AND THE OUTLET. ACCESS OPENINGS MUST BE LOCATED DIRECTLY ABOVE ANY PUMPING OR DOSING EQUIPMENT, OR EFFLUENT FILTER. CONNECTION OF THE RISER TO THE TANK AND THE CONNECTION OF ADDITIONAL RISER SECTIONS MUST INCORPORATE JOINT GROOVES OR ADAPTERS TO PREVENT LATERAL MOVEMENT AND TO REMAIN WATERTIGHT. ACCESS RISERS AND LIDS MUST BE STRUCTURALLY SOUND TO WITHSTAND THE ANTICIPATED SITE-SPECIFIC LOAD CONDITIONS OF THE RISER. MANWAY COVERS SET AT OR ABOVE GRADE SHALL BE SECURED WITH 304 STAINLESS STEEL BOLTS, OR APPROVED EQUIVALENT, TO PREVENT UNAUTHORIZED ACCESS.

I. ALL INLET AND OUTLET DEVICES MUST HAVE SANITARY TEES CONSTRUCTED OF PVC PER ASTM D3034 OR ABS PER ASTM D2680, AND SHALL EXTEND AT LEAST 4 INCHES ABOVE AND 12 INCHES BELOW THE WATER LEVEL IN THE TANK. SEWAGE TANKS MUST MAINTAIN AT LEAST A ONE-INCH AIR SPACE BETWEEN THE UNDERSIDE OF THE TOP OF THE TANK AND THE TOP OF ANY OF THE INLET, OUTLET, OR INTERCOMPARTMENTAL FITTING TO VENT GASES. SEWAGE TANKS MUST PROVIDE AIR SPACE TO ALLOW GASES TO VENT THROUGH THE MAIN BUILDING SEWER VENT OR OTHER PLUMBING VENT STACKS TO THE ATMOSPHERE. AIR SPACE MUST BE ABOVE THE LIQUID SURFACE IN THE TANK BACK AND THROUGH THE TANK'S INLET.

THE OUTLET SANITARY TEE SHALL EXTEND BELOW THE LIQUID LEVEL AT LEAST 25 PERCENT, BUT NOT MORE THAN 35 PERCENT OF THE LIQUID DEPTH. THE OUTLET SANITARY TEE MUST EXTEND SUFFICIENTLY TO ALLOW SCUM STORAGE AND VENTING, AND TO A POINT NOT LESS THAN ONE INCH FROM THE UNDERSIDE OF THE TOP OF THE TANK. THE OUTLET TEE MAY EXTEND INTO THE RISER FOR VENTING.

THE SEPTIC TANK MUST HAVE AN INTERCOMPARTMENTAL WALL FITTING (TRANSFER PORT) INSTALLED SO THAT THE ENTRY INTO THE PORT IS PLACED 65% TO 75% IN THE DEPTH OF THE LIQUID, MEASURED FROM THE FLOOR OF THE COMPARTMENT.

J. ALL TANK INTERIOR CONCRETE SURFACES SHALL BE COATED WITH XYPEX CONCRETE WATERPROOFING, WITH AN INITIAL COAT OF XYPEX CONCENTRATE AND A SECOND COAT OF XYPEX MODIFIED. PREPARATIONS FOR COATING, COATING APPLICATION AND COATING CURING SHALL COMPLY WITH THE MANUFACTURER'S RECOMMENDATIONS. THIS COATING SHALL ALSO BE APPLIED TO MANWAY OPENINGS IN THE TOP OF THE SEPTIC TANK AND TO THE INTERIOR SURFACE OF THE PRECAST CONCRETE MANWAY RISERS.

CONTRACTOR TO FURNISH LETTER TO COR AND ENGINEER FROM XYPEX MANUFACTURER'S AUTHORIZED REPRESENTATIVE STATING THAT COATING APPLICATION, INCLUDING CURING, WAS PERFORMED SATISFACTORILY.

K. TANK INSTALLATION, INCLUDING EXCAVATION, SITE PREPARATION, SETTING, VENTING, SEALING, BACKFILLING AND TESTING, SHALL BE ACCORDING TO THE MANUFACTURER'S INSTRUCTIONS.

WATERTIGHTNESS TESTING MUST BE COMPLETED AND APPROVED AFTER TANK INSTALLATION, INCLUDING JOINT SEALANT CURING, BUT BEFORE THE TANK IS PUT INTO SERVICE. TESTING FOR WATERTIGHTNESS SHALL BE PERFORMED USING EITHER HYDROSTATIC WATER TEST OR VACUUM TEST PER ASTM C1227 OR ASTM C1613, EXCEPT THAT FOR WATER TESTING THE TANK SHALL BE FILLED WITH WATER 2 INCHES INTO THE RISER. THEN ALLOW THE WATER TO STAND FOR 24 HOURS. THEN REFILL THE TANK AS NEEDED TO THE ORIGINAL LEVEL AND ALLOW IT TO STAND FOR ONE HOUR. THE TANK PASSES IF THE WATER LEVEL IS UNCHANGED AFTER ONE HOUR. PRIOR TO TESTING, VERIFY TEST PROCEDURE WITH TANK MANUFACTURER.

THE CONTRACTOR SHALL BE SOLELY RESPONSIBLE FOR CONDUCTING THE TEST.

THE CONTRACTOR SHALL FURNISH TO THE COR AND ENGINEER A LETTER FROM THE TANK MANUFACTURER'S AUTHORIZED REPRESENTATIVE STATING THAT TANK ASSEMBLY AND INSTALLATION HAVE BEEN COMPLETED IN ACCORD WITH THE MANUFACTURER'S RECOMMENDATIONS.

L. SEPTIC TANKS MUST BE DESIGNED AND CONSTRUCTED TO ACCOMMODATE EFFLUENT FILTERS.

22. EFFLUENT FILTERS

A. EFFLUENT FILTERS SHALL PROVIDE 1/16" FILTRATION AND SHALL BE ORENCO BIOTUBE MDL. FTP 1260-36R, OR APPROVED EQUIVALENT, WITH SLIDE RAIL AND EXTENDED HANDLE TO FACILITATE FILTER CARTRIDGE REMOVAL.

B. FILTERS SHALL BE CONSTRUCTED TO PREVENT SCUM OR OTHER FLOATABLE SOLIDS FROM DISCHARGING FROM THE TANK.

C. PER THE ORENCO GUIDELINES, THE APPROXIMATE MEAN TIME BETWEEN VCC SEPTIC TANK EFFLUENT FILTER CLEANINGS (MTBC) IN YEARS = A/(CxQ), WHERE A=FILTER AREA IN SF, C=FILTER COEFFICIENT 0.002 SF/GPD/YR, AND Q=DESIGN FLOW IN GPD:

VCC: MTBC = (4x30)/(0.002x15,830) = 3.8 YEARS.
ECS: MTBC = (1x30)/(0.002x3,890) = 3.9 YEARS.
RO: MTBC = (1x30)/(0.002x200) = 75 YEARS.

THE CONTRACTOR SHALL FURNISH THE COR AND ENGINEER WITH A COPY OF THE FILTER MANUFACTURER'S CALCULATIONS OF THE APPROXIMATE MTBC FOR THE DESIGN FLOWS SHOWN IN TABLE 1/C7.01.

D. THE CONTRACTOR SHALL PROVIDE A COMPLETE O&M MANUAL FOR THE FILTERS.

E. THE GOVERNMENT SHALL MAINTAIN THE FILTER IN GOOD OPERATING CONDITION IN ACCORD WITH THE MFR'S INSTRUCTIONS. PRIOR TO FILTER REMOVAL FOR CLEANING, THE TANK SHALL BE PUMPED DOWN ADEQUATELY TO PREVENT PASSAGE OF SCUM AND SOLIDS THROUGH THE TANK OUTLET PIPING WHILE FILTER CLEANING IS PERFORMED. THE FILTER "WASH-OFF" MATERIAL SHALL BE PLACED INTO THE PRIMARY COMPARTMENT OF THE SEPTIC TANK.

F. THE CONTRACTOR SHALL FURNISH TO THE COR ONE ORENCO BIOTUBE CLEANING TRAY AND TWO BIOTUBE CLEANING BRUSHES, OR APPROVED EQUIVALENT TRAY AND BRUSHES.

23. DOSING TANK

DOSING TANK SHALL COMPLY WITH THE SPECIFICATIONS FOR SEPTIC TANKS EXCEPT THAT THIS TANK SHALL SHALL BE SINGLE-COMPARTMENT TYPE.

24. DOSING PUMP BASIN

DOSING PUMP BASIN SHALL COMPLY WITH THE SPECIFICATIONS FOR SEPTIC TANKS EXCEPT THAT THIS BASIN SHALL SHALL BE A 72" DIAMETER MANHOLE - SEE GENERAL NOTE 19 AND PLANS.

25. FLOWMETERS

A. THE 2" FLOWMETER SHALL BE ENDRESS + HAUSER PROMAG W400, MODEL NO. 5W4C50-AAALIN5DHA1KGA+CBZ1 (NEMA 6P/IP68)-VERIFY MODEL NO. WITH MFR-, MICROPROCESSOR-BASED ELECTROMAGNETIC METER WITH FLANGED, STAINLESS STEEL FLOW TUBE, EPOXY-COATED CARBON STEEL ASME B16.5 CL. 150 FLANGES, HARD RUBBER LINER, TYPE 316SS ELECTRODES PROTECTED FROM CONTACT WITH THE FLUID, GASKETS, AND CARBON STEEL MOUNTING BOLTS. THE SENSOR SHALL HAVE AN ACCIDENTAL SUBMERGENCE PROOF RATED CONNECTION USING THE FACTORY POTTING KIT. THE FLOWMETER SHALL BE CAREFULLY MOUNTED BETWEEN ANSI CLASS 150, VAN STONE TYPE SCH. 80 PVC FLANGES, OR OTHER APPROVED FITTINGS, IN SUCH A MANNER AS TO FACILITATE EASE OF METER REMOVAL AND REPLACEMENT. FURNISH AND INSTALL GROUNDING RINGS PER MANUFACTURER'S RECOMMENDATIONS.

B. EACH NEMA 4X ENCLOSED TRANSMITTER SHALL BE REMOTE-MOUNTED INSIDE THE NEARBY TESCO PUMP CONTROL PANEL (PCP) - VERIFY LENGTH OF CABLE REQUIRED TO CONNECT THE SENSOR TO THE TRANSMITTER PRIOR TO ORDERING THE METER. TRANSMITTER HOUSING SHALL BE POLYCARBONATE OR ALUMINUM. COORDINATE ALL WORK WITHIN THE PCP WITH TESCO. POWER SUPPLY IS 120VAC.

THE UNIT SHALL HAVE PROGRAMMABLE LOW AND HIGH FLOW ALARMS WITH RELAY CONTACTS. CAREFULLY INSTALL THE TRANSMITTER IN THE PUMP CONTROL PANEL - COORDINATE WITH THE PANEL MANUFACTURER, TESCO. ELECTRONIC OUTPUTS SHALL INCLUDE 4-20mA HART, 2 x PULSE/FREQUENCY/SWITCH OUTPUT, STATUS INPUT, AND SHALL BE COMPATIBLE WITH THE PUMPING EQUIPMENT SCADA SYSTEM. FLOWRATE UNITS SHALL BE GPM AND TOTAL FLOW SHALL BE GALLONS.

C. PROVIDE O&M MANUAL, INSTRUCTIONS, INSTALLATION, TESTING, CALIBRATION, STARTUP AND VERIFICATION TO THE SATISFACTION OF THE CONTRACTING OFFICER'S REPRESENTATIVE AND IN ACCORD WITH THE MANUFACTURER'S RECOMMENDATIONS. VERIFICATION TO INCLUDE A LETTER FROM THE MANUFACTURER'S AUTHORIZED REPRESENTATIVE STATING THAT THE METER HAS BEEN PROPERLY INSTALLED AND TESTED, AND IS READY FOR FULL-TIME OPERATION.

INSTRUMENT STARTUP SERVICES BY A QUALIFIED E+H SERVICE TECHNICIAN SHALL INCLUDE: INSTALLATION CONFIRMATION; REVIEW OF PROCESS CONDITIONS TO ENSURE OPTIMAL SETUP; VERIFICATION OF ELECTRICAL CONNECTIONS; INSTRUMENT CONFIGURATION; FUNCTION TEST; TRAINING OF PARK PERSONNEL; ADVICE FOR FUTURE MAINTENANCE REQUIREMENTS; AND ISSUANCE OF SERVICE REPORT.

D. THE LOCAL E+H REP IS JPR SYSTEMS OF SF, CA, PHONE 800-478-1002 X105.

26. DOSING PUMPS, CONTROLS & APPURTENANCES

A. COMPLETE SYSTEMS: PROVIDE PUMPS, CONTROLS, PIPE, FITTINGS, CONDUIT, WIRING, CONNECTIONS AND SUPPORTS TO PRODUCE A COMPLETE, OPERABLE SYSTEM WITH ALL ELEMENTS PROPERLY INTERCONNECTED AND CAPABLE OF MEETING THE MINIMUM HYDRAULIC FLOW AND HEAD REQUIREMENTS.

IF A SPECIFIED DIMENSIONED LOCATION IS NOT SHOWN FOR INTERCONNECTIONS OR SMALLER SYSTEM ELEMENTS, SELECT APPROPRIATE LOCATIONS AND SHOW THEM ON SHOP DRAWING SUBMITTALS FOR REVIEW.

THE DOSING PUMP MANUFACTURER'S AUTHORIZED REP SHALL FURNISH THE CONTRACTING OFFICER'S REPRESENTATIVE (COR) WITH A COMPUTERIZED HYDRAULIC ANALYSIS OF THE COMPLETE DOSING SYSTEM, INCLUDING A GRAPHICAL SYSTEM FLOW-HEAD CURVE SUPERIMPOSED ON THE SELECTED PUMP'S PERFORMANCE CURVE. THE ANALYSIS SHALL BE PERFORMED TO THE SATISFACTION OF THE ENGINEER AND COR.

B. PROVIDE EQUIPMENT AND MATERIAL NEW AND WITHOUT IMPERFECTIONS. ERECT IN A NEAT AND WORKMANLIKE MANNER; ALIGNED, LEVELED, CLEANED AND ADJUSTED FOR SATISFACTORY OPERATION; INSTALLED IN ACCORDANCE WITH THE RECOMMENDATIONS OF THE MANUFACTURERS AND THE BEST STANDARD PRACTICES FOR THIS TYPE OF WORK SO THAT CONNECTING AND DISCONNECTING OF PUMP, PIPING AND ACCESSORIES CAN BE READILY MADE AND SO THAT ALL PARTS ARE EASILY ACCESSIBLE FOR INSPECTION, OPERATION, MAINTENANCE AND REPAIR. IN ORDER TO MEET THESE REQUIREMENTS WITH EQUIPMENT AS FURNISHED, MINOR DEVIATION FROM THE DRAWINGS MAY BE MADE AS FAVORABLY REVIEWED BY THE COR.

ALL PUMPS MUST BE INSTALLED SO THAT THEY CAN BE EASILY REMOVED AND/OR REPLACED FROM THE GROUND SURFACE. UNDER NO CIRCUMSTANCES SHALL PUMP REPLACEMENT AND/OR REPAIR REQUIRE SERVICE PERSONNEL TO ENTER THE PUMP TANK.

ALL PUMPS MUST BE FITTED WITH UNIONS, VALVES AND ELECTRICAL CONNECTIONS NECESSARY FOR EASY PUMP REMOVAL AND REPAIR.

C. THE RECOMMENDATIONS AND INSTRUCTIONS OF THE MANUFACTURERS OF PRODUCTS USED IN THE WORK ARE HEREBY MADE PART OF THESE SPECIFICATIONS, EXCEPT AS THEY MAY BE SUPERCEDED BY OTHER REQUIREMENTS OF THE SPECIFICATIONS.

D. PUMPS AND ELECTRICAL HOOK-UPS MUST CONFORM TO ALL STATE AND LOCAL ELECTRICAL CODES.

ELECTRICAL CONTROL AND OTHER ELECTRICAL COMPONENTS MUST BE APPROVED BY UNDERWRITERS LABORATORIES (UL), OR EQUIVALENT.

IN ADDITION, PUMPS AND CONTROLS SHOULD HAVE GAS-TIGHT JUNCTION BOXES OR SPLICES AND HAVE ELECTRICAL DISCONNECTS (AS PER NATIONAL ELECTRIC CODE) APPROPRIATE FOR THE INSTALLATION. THE BOXES SHOULD BE PLACED SO THAT THEY DO NOT INTERFERE WITH THE SERVICING OF OTHER COMPONENTS.

FLOAT SWITCHES MUST BE MOUNTED INDEPENDENT OF THE PUMP AND DISCHARGE LINE SO THAT THEY CAN BE EASILY REPLACED AND/OR ADJUSTED WITHOUT REMOVING THE PUMP. USE SINGLE FLOAT TYPE FOR ALL FUNCTIONS, AND TIME DELAYS TO PREVENT "CHATTERING".

E. MANUFACTURER OF THE PUMP CONTROL PANEL (PCP) AND TERMINAL CABINETS ENCLOSURE AND PROGRAMMABLE LOGIC CONTROLLER (PLC) SHALL BE TESCO, NO EQUAL. PCP SIZE, AS SHOWN ON THESE PLANS, IS APPROXIMATE. VERIFY SIZE WITH TESCO.

THE PROGRAMMABLE LOGIC CONTROLLER (PLC) SHALL BE SCADA READY AND MOUNTED INSIDE THE PUMP CONTROL PANEL.

PHASE 5

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MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: DAVID EVANS AND ASSOCIATES, INC. BELLEVUE, WA	TGH	C7.63	SOUTH ENTRANCE OWTS PLANS GENERAL NOTES	104 / 127662
SUBCONTRACTOR LAUGENOUR AND MEIKLE WOODLAND, CA	CADD: MSW		YOSEMITE NATIONAL PARK	PMIS/PKG NO. 206431
	TECH. REVIEW: TGH		MARIPOSA GROVE RESTORATION	SHEET 115 OF 291
	DATE: 2/26/2016			

GENERAL NOTES CONTINUED

F. PROVIDE A COMPLETE AND FUNCTIONING PUMP CONTROL SYSTEM COMPATIBLE FOR USE WITH A TELEPHONE-BASED TELEMETRY SYSTEM (BY OTHERS) TO IMPLEMENT THE DATA AND CONTROL LINKS AS SHOWN ON THE DRAWINGS AND AS SPECIFIED HEREIN. THE TELEMETRY SYSTEM (BY OTHERS) SHALL CONSIST OF THE FOLLOWING TYPICAL EQUIPMENT:

A DIGITAL REMOTE TELEMETRY UNIT (RTU) SYSTEM FOR PUMP STATION WHICH IMPLEMENTS A POLLING OR POLLED-QUIESCENT TYPE ARCHITECTURE IN WHICH DATA EXCHANGES ARE INITIATED BY AND CHANNELED THROUGH THE CENTRAL STATION EQUIPMENT.

TELEPHONE AND MODEM SYSTEM, AS MAY BE REQUIRED, TO MEET THE FUNCTIONALITY REQUIREMENTS OF THESE SPECIFICATIONS.

PUMP CONTROLS AS DESCRIBED HEREIN.

REVISE EXISTING SCADA PROGRAM TO INCLUDE REVISIONS NECESSARY TO REORGANIZE THE OVERALL NAVIGATIONAL SCHEME WHILE INCORPORATING THE NEW PROGRAMMING FOR NEW EQUIPMENT.

REFER TO TESCO'S SCADA SYSTEM INPUT/OUTPUT SUMMARY ON E6.05. FOR DESCRIPTIONS OF DIGITAL AND ANALOG RTU REQUIREMENTS FOR EACH SITE AND CONTROL STRATEGIES.

OWTS COMPONENTS TO BE MONITORED BY SCADA INCLUDE: FOUR DUPLEX DOSING PUMP UNITS, FOUR FLOW METERS, FOUR DISCHARGE PRESSURE TRANSDUCERS, TWO LIQUID LEVEL SENSORS IN DOSING PUMP BASINS, AND FOUR DISTRIBUTING VALVE MONITORS LOCATED IN THE LEACHFIELD.

G. SPARE PARTS: THE CONTRACTOR SHALL FURNISH TO THE COR ONE COMPLETE DUPLEX PUMPING UNIT, INCLUDING PUMP VAULT, FILTER CARTRIDGE, DISCHARGE ASSEMBLY WITH VALVING AND FLEX HOSES, AND FLOAT ASSEMBLY.

H. FOR SYSTEMS REQUIRING TIME-DOSED PRESSURE DISTRIBUTION, ACCESSIBLE CONTROLS AND WARNING DEVICES ARE REQUIRED, AND MUST:

MEET THE FUNCTIONAL REQUIREMENTS FOR PRESSURE DISTRIBUTION.

DELIVER PRESCRIBED DOSE SIZES UNIFORMLY TO THE ORIFICES IN THE DISTRIBUTION NETWORK.

DELIVER THE EFFLUENT TO THE DISTRIBUTION NETWORK IN EVENLY SPACED DOSES OVER A 24-HOUR PERIOD.

PROVIDE PRESCRIBED RESTING PERIODS BETWEEN DOSES.

ASSURE NO MORE THAN THE DESIGN VOLUME FOR EACH 24 HOUR PERIOD IS DELIVERED TO THE LEACHFIELD.

RECORD AND STORE THE PUMP RUN TIME AND NUMBER OF DOSE CYCLES FOR EACH PUMP.

I. THE PUMP CONTROL PANEL MUST:

1. HAVE CYCLE COUNTERS AND ELAPSED TIME METERS FOR EACH PUMP.
2. BE EQUIPPED WITH BOTH AUDIBLE AND RED VISUAL HIGH AND LOW LIQUID LEVEL ALARMS. PROVIDE ALARM TEST SWITCHES AND AUTOMATIC ALARM SILENCE RESET, AND REMOTE ALARM CONTACT.
3. HAVE BUILT-IN LCD SCREEN AND PROGRAMMING KEYS FOR FIELD ADJUSTMENTS.
4. HAVE ALARM CIRCUITS INSTALLED INDEPENDENT OF PUMP POWER CIRCUITS.
5. PROVIDE FOR ALTERNATION OF EACH PUMP IN THE DUPLEX PUMP UNITS, AND FOR ALTERNATION OF PUMP UNITS FOR UNIFORM DISTRIBUTION OF EFFLUENT INTO THE LEACHFIELD.
6. USE MOTOR CONTRACTORS TO POWER THE PUMPS.
7. PROVIDE SEPARATE ON & OFF FLOAT CAPABILITY.
8. PROVIDE VISUAL FLOAT STATUS INDICATORS.
9. LOG PUMP AND ALARM ACTIVITY DATA.
10. HAVE INTRINSICALLY SAFE CONTROL RELAYS.
11. HAVE CURRENT SENSOR.
12. HAVE DISCONNECT SWITCHES.
13. HAVE ACCURATE, MULTIPLE TIMER SETTINGS FOR OPTIMUM DOSING.
14. HAVE HOA SWITCH.
15. HAVE CABINET HEATER.
16. HAVE CABINET VENTILATION FAN.
17. HAVE FILTERED CABINET AIR INLETS.
18. HAVE STEEL SUNSHADE CANOPY MOUNTED ABOVE CABINET - SEE DTL. 1/C7.36 FOR FREE STANDING CANOPY.
19. HAVE PUMP RUN LIGHTS.
20. HAVE POWER LIGHT.
21. HAVE SURGE ARRESTOR.
22. HAVE REDUNDANT-OFF FUNCTION.
23. HAVE A 20 AMP GFI RECEPTACLE FOR EQUIPMENT MAINTENANCE.
24. HAVE DISPLAY FOR THE PRESSURE IN EACH OF THE FOUR DOSING PUMP DISCHARGE PIPELINES VIA THEIR TRANSDUCERS.
25. INCLUDE A GENERATOR RECEPTACLE MOUNTED ON THE OUTSIDE OF THE PCP TO ALLOW FOR EASY CONNECTION OF A TEMPORARY GENSET POWER SUPPLY FOR: (1) THE CONTRACTOR'S OPERATIONAL TESTING OF THE PUMP SYSTEMS, INCLUDING PUMP CONTROLS AND ALARMS, IN THE EVENT THAT THE PERMANENT POWER SERVICE IS NOT YET FUNCTIONAL; AND FOR (2) YNP'S OPTION TO USE THIS SAME TEMP POWER GENSET CONNECTION AT THE PUMP CONTROL PANEL FOR ANY EMERGENCY POWER SITUATION THAT MIGHT BE ENCOUNTERED IN THE FUTURE.

J. PUMPING AND CONTROL SYSTEMS:

PROVIDE O&M MANUALS, STARTUP INSTRUCTIONS, INSTALLATION TESTING, CALIBRATION AND VERIFICATIONS TO THE SATISFACTION OF THE CONTRACTING OFFICER'S REPRESENTATIVE AND IN ACCORD WITH THE MANUFACTURERS' RECOMMENDATIONS. VERIFICATIONS TO INCLUDE LETTERS FROM THE MANUFACTURERS' AUTHORIZED REPRESENTATIVES STATING THAT THE PUMPING AND CONTROL SYSTEMS ARE COMPLETE, HAVE BEEN PROPERLY INSTALLED AND TESTED, THAT ALL SEQUENCES HAVE BEEN CHECKED AND ARE FUNCTIONING PROPERLY, AND ARE READY FOR FULL-TIME OPERATION.

K. POSTED OPERATING INSTRUCTIONS: FURNISH OPERATING INSTRUCTION MOUNTED INSIDE THE PUMP CONTROL PANEL IN A WEATHER-PROOF ENCLOSURE. INCLUDE WIRING DIAGRAMS, CONTROL DIAGRAMS, CONTROL SEQUENCE, START-UP ADJUSTMENT, OPERATION, SHUTDOWN, SAFETY PRECAUTIONS, PROCEDURES IN CASE OF EQUIPMENT FAILURE AND OTHER ITEMS OF INSTRUCTION RECOMMENDED BY MANUFACTURERS.

L. CONTROL SYSTEMS INTEGRATION:

UNDER SUBCONTRACT TO THE GENERAL CONTRACTOR, TESCO CONTROLS, INC. SHALL PROVIDE PLC, SCADA PROGRAMMING AND SOFTWARE INTEGRATION SERVICES TO COMPLETE THE PROJECT. IN SUMMARY, THE CONTRACTOR SHALL AT A MINIMUM PERFORM THE FOLLOWING FOR THE PROJECT: PARTICIPATE IN DESIGN REVIEW AND CONTROL STRATEGY REFINEMENT WITH THE COR AND THE DESIGN ENGINEER; SELECT, FURNISH, CONSTRUCT, INSTALL, CONFIGURE, CALIBRATE, TEST, AND PLACE INTO OPERATION TRANSMITTERS, INSTRUMENTS, PROGRAMMABLE CONTROLLERS, CONTROL PANELS, MOTOR CONTROLS, ALARM EQUIPMENT, COMMUNICATIONS, MONITORING EQUIPMENT, AND ACCESSORIES; COORDINATE PROGRAMMING WORK WITH THE COR AND THE DESIGN ENGINEER; PROGRAM PLCs TO IMPLEMENT THE SIGNAL PROCESSING AND LOGIC REQUIRED FOR THE PROJECT, CONFIGURE THE SCADA HMI SYSTEM; CONFIGURE LOCAL OPERATOR INTERFACE PANELS (HMI); PROVIDE STARTUP AND TESTING SERVICES FOR THE ABOVE ITEMS; AND PROVIDE TRAINING, O&M MANUALS, AND RECORD DOCUMENTS FOR THE ABOVE ITEMS.

DUPLEX DOSING PUMP UNITS NOS. 1-4, AND EACH OF THE 2 PUMPS IN EACH DUPLEX UNIT, ARE TO ALTERNATE, WITH NO MORE THAN ONE PUMP OPERATING AT ANY TIME. COORDINATE PUMP CONTROL PROGRAMMING WITH THE ENGINEER AND COR.

27. <u>PRESSURE DISTRIBUTION SYSTEM OPERATIONAL TESTING</u>

A. THE INTENT OF PRESSURE DISTRIBUTION IS UNIFORM DISTRIBUTION OF EFFLUENT THROUGHOUT THE LEACHFIELD. WHEN THE DOSING TANK LIQUID LEVEL REACHES THE PREDETERMINED LEVEL, THE TIMER "ON" FLOAT SWITCH ACTIVATES A TIMER SEQUENCE IN THE PCP. THE TIMER CYCLES THE PUMPS "OFF" AND "ON" AT SET INTERVALS UNTIL THE LIQUID LEVEL DROPS ENOUGH TO DEACTIVATE THE TIMER "OFF" FLOAT SWITCH. SEE PLAN FOR TIMER "ON" AND "OFF" FLOATS AND SETTINGS.

B. THE TESTING SHOULD VERIFY THAT DISTRIBUTION IS UNIFORM WITH THE REQUIRED 5-FOOT MINIMUM RESIDUAL HEAD, THAT THE SYSTEM IS DOSED AT THE PROPER VOLUME AND FREQUENCY, AND THAT THE ALARMS ARE FUNCTIONING PROPERLY. GUIDELINES ARE PROVIDED BELOW. THE COUNTY MAY HAVE ADDITIONAL REQUIREMENTS. IF PROBLEMS ARE ENCOUNTERED DURING TESTING, THE INSTALLER SHOULD NOTIFY THE ENGINEER AND THE COR.

1. MEASURE SQUIRT HEIGHT PRIOR TO INSTALLATION OF ORIFICE SHIELDS AND ROCK BACKFILL. MINIMUM SQUIRT HEIGHT FOR 1/8 INCH ORIFICE SIZE IS 5'.

2. CHECK UNIFORMITY OF SQUIRT HEIGHT.

   THE VARIATION IN ORIFICE DISCHARGE RATES WITHIN ANY ONE LATERAL MUST NOT BE MORE THAN 10%. THE SQUIRT HEIGHT DIFFERENCE MUST NOT EXCEED 12.5 INCHES = 21% OF 5' (EQUIVALENT TO 10% FLOW DIFFERENCE) BETWEEN ORIFICES ON ANY ONE LATERAL.

   THE VARIATION IN ORIFICE DISCHARGE RATES OVER THE ENTIRE DISTRIBUTION SYSTEM MUST NOT BE MORE THAN 15%. THE SQUIRT HEIGHT DIFFERENCE OVER THE ENTIRE SYSTEM MUST NOT EXCEED 19 INCHES = 32% OF 5' (EQUIVALENT TO 15% FLOW DIFFERENCE).

   ADJUST BALANCING VALVES ON LATERALS AS REQUIRED. A MINIMUM RESIDUAL HEAD OF 5 FEET IS REQUIRED FOR SYSTEMS WITH 1/8 INCH DIAMETER ORIFICES.

   TO ESTABLISH A BASELINE FOR FUTURE ANNUAL SQUIRT HEIGHT MEASUREMENTS, THE CONTRACTOR SHALL FURNISH AND INSTALL A TEST ORIFICE AT THE OUTLET OF EACH PURGE VALVE AND SHALL RECORD THE SQUIRT HEIGHT MEASUREMENTS FOR EACH LATERAL.

3. CHECK FLOAT PLACEMENT. HIGH WATER ALARM, "ON" LEVEL, "OFF" LEVEL, AND "REDUNDANT OFF" ALARM MUST ACTIVATE OR DEACTIVATE AT THE ELEVATION CALLED OUT ON THE PLAN. (IT IS SUGGESTED THAT, FOR SIMPLICITY AND ACCURACY, THESE ADJUSTMENTS BE MADE WITH THE FLOAT TREE OUT OF THE WATER.)

4. ENSURE THAT EACH PUMP DELIVERS THE CORRECT DOSE TO THE DRAINFIELD.

5. DETERMINE THE TIME REQUIRED TO SEND A FULL DOSE TO THE DRAINFIELD. THIS CAN BE DONE BY RUNNING THE SYSTEM IN MANUAL. BE SURE THERE IS PLENTY OF WATER IN THE PUMP CHAMBER. TIMER RUN TIMES MUST BE FIELD VERIFIED.

6. USING THE TIME OBTAINED ABOVE, VERIFY THAT WHEN THE SYSTEM RUNS AUTOMATICALLY IT RUNS THE TIME REQUIRED TO SEND THE PROPER DOSE TO THE DRAINFIELD.

7. VERIFY THAT THE TIMER "OFF" TIME IS THE SAME AS THAT SPECIFIED ON THE PLAN OR WILL DOSE THE SYSTEM THE CORRECT NUMBER OF TIMES A DAY. CHECK THIS NUMBER IN MINUTES AND NOTE THE "OFF" TIME. ONE MAY VERIFY ACTIVATION LEVELS BY USE OF LIGHTS ON TIMER.

8. VERIFY THAT HIGH WATER ALARM DOES NOT TURN THE PUMP ON. IF HIGH WATER ALARM TURNS THE PUMP ON, THE SYSTEM WILL NOT BE APPROVED.

9. VERIFY THAT SYSTEM WILL DOSE THE CORRECT NUMBER OF TIMES PER DAY AND THAT NO FLOAT IN THE SYSTEM TURNS THE PUMP ON INDEPENDENT OF THE TIMER. A SYSTEM WITH A TIMER OVERRIDE FLOAT WILL NOT BE APPROVED.

28. <u>DOSING PUMP BASIN</u>

A. CONCRETE BASIN

1. 72-INCH SEWER MANHOLE PRECAST BARREL SECTIONS PER GENERAL NOTE 19/C7.62.
2. PROTECTIVE INTERIOR COATING PER GENERAL NOTE 21.J/C7.63.
3. SEE GENERAL NOTE 21/C7.62 - C7.63 FOR ADDITIONAL REQUIREMENTS, INCLUDING WATERTIGHTNESS TESTING, INTERIOR PROTECTIVE COATING AND BOUYANCY COUNTERACTION.
4. CAST-IN-PLACE, REINFORCED CONCRETE BASE.
5. PRECAST, REINFORCED CONCRETE FLAT-SLAB TOP WITH REINFORCED CONCRETE RISER, AND REINFORCED CONCRETE COLLAR AT TOP OF RISER TO PROVIDE HATCH ANCHORAGE.

B. ACCESS HATCH

1. DOUBLE LEAF BILCO MODEL JD-H20-R, OR APPROVED EQUIVALENT.
2. DOOR LEAVES: ¼-INCH THICK STEEL, DIAMOND PATTERN, REINFORCED AS REQUIRED TO AASHTO H-20 WHEEL LOADING.
3. FRAME: ¼-INCH THICK FORMED STEEL CHANNEL WITH ANCHOR FLANGE AROUND PERIMETER. PROVIDE CHANNEL TO COLLECT DRAIN WATER AND PROVIDE 1½-INCH DRAINAGE COUPLING FOR CONNECTION TO DRAINLINE.
4. DOORS: EQUIP WITH HEAVY FORGED STEEL HINGES, STAINLESS STEEL PINS, COMPRESSION SPRING OPERATORS AND AN AUTOMATIC HOLD-OPEN ARM WITH A POSITIVE AUTOMATIC LATCH THAT WILL SECURE THE DOOR IN THE OPEN POSITION UNTIL THE RELEASE HANDLE IS ACTIVATED. SUBMIT DETAILS OF LATCH FOR REVIEW. PROVIDE SNAP-LOCK WITH REMOVABLE HANDLE. PROVIDE STEEL RECESSED HASP TO DOOR AND FRAME FOR PADLOCK.
5. SAFETY GRATING: INSTALL HATCH WITH FALL PROTECTION GRATING SYSTEM.
6. ODOR CONTROL GASKET: A CONTINUOUS EPDM GASKET SHALL BE AFFIXED TO THE FRAME AND FORM AN ODOR-RESISTANT BARRIER AROUND THE ENTIRE PERIMETER OF THE COVER.
7. PROVIDE STAINLESS STEEL HOLD-OPEN PIN THROUGH HOLES IN HOLD-OPEN ARMS TO INSURE AGAINST ACCIDENTAL HATCH CLOSURE. ATTACH PIN TO HATCH WITH A SHORT STAINLESS STEEL CHAIN TO PREVENT LOSS.
8. SAFETY CHAIN: PROVIDE A STAINLESS STEEL SAFETY CHAIN BETWEEN DOUBLE LEAF DOORS AT OPPOSITE END FROM LATCH.
9. ALL HARDWARE: STAINLESS STEEL THROUGHOUT. PROVIDE FOUR (4) KEYS AS SPARES.
10. FINISH: FIELD APPLY TWO COATS OF AMERON AMERCOAT 385 OR TNEMEC SERIES 69 TWO-COMPONENT EPOXY COATING SYSTEM PER MANUFACTURER'S RECOMMENDATIONS FOR A TOTAL MINIMUM DRY FILM THICKNESS OF 12 MILS. COLOR WILL BE SELECTED BY THE COR.
11. WARNING SIGN: PROVIDE A 10-INCH X 12-INCH MINIMUM SIZE SIGN PERMANENTLY ATTACHED TO THE UNDERSIDE OF ALL HATCH DOORS READING: "DANGER: MAKE SURE HOLD-OPEN LATCH IS POSITIVELY ENGAGED BEFORE USING. INSERT PIN IN HOLES IN HOLD-OPEN ARMS TO HOLD DOOR OPEN."
12. DELIVER HATCHES TO JOB SITE IN TIME FOR INSTALLATION IN THE CONCRETE POUR.
13. COORDINATE CONNECTION OF DRAINAGE COUPLING TO PLUMBING DRAINLINE PRIOR TO THE CONCRETE POUR.
14. SET FRAME LEVEL AND TRUE TO PLANE AT ALL FOUR CORNERS, AND FLUSH WITH ADJACENT FINISHED SURFACES, AND WITH A SLIGHT PITCH TOWARD THE DRAIN CORNER. DOORS, WHEN CLOSED, SHALL BE FLUSH WITH FRAMES AND FLUSH WITH EACH OTHER.
15. SECURELY INSTALL AN EASY-TO-READ, CORROSION-PROOF, HEAVY-DUTY NAMEPLATE WITH PUMP NO. OVER RESPECTIVE PUMP. COORDINATE WITH ENGINEER.

29. <u>PLUG VALVES</u>

4" FLANGED, ALL STAINLESS STEEL (SS), DEZURIK PEC, OR APPROVED MILLIKEN OR CLOW EQUIVALENT VALVE, SUITABLE FOR SUBMERGED SERVICE IN SEPTIC TANK EFFLUENT, WITH WORM-GEAR MANUAL ACTUATOR AND 2" SQUARE OPERATING NUT. INSTALL SECURELY WITH EXTENSION ASSEMBLY, INCL. EXTENSION ROD, BEARING PLATES AND COUPLINGS - ALL STAINLESS STEEL. SECURELY INSTALL VALVE AND ANCHOR EXTENSION ASSEMBLY BEARING PLATES TO TANK WALL WITH SS HARDWARE TO SATISFACTION OF ENGINEER AND COR.

30. <u>PUMP BASIN WATER LEVEL TRANSMITTERS</u>

PMC VERSALINE MODEL VL2113, OR APPROVED EQUIVALENT, WITH SINK WEIGHT, CABLE HANGER, MFR'S CABLING, AND MP11 VENT SPHERE INSTALLED INSIDE A TE-11 ENCLOSURE UNIT, PER MFR'S RECOMMENDATIONS. COORDINATE WITH ENGINEER AND COR.

31. <u>DISCHARGE PIPELINE PRESSURE TRANSMITTER</u>

PMC VERSALINE MODEL VL1000, OR APPROVED EQUIVALENT WITH MFR'S CABLING, AND INSTALLED WITH ASHCROFT MODEL 101, OR APPROVED EQUIVALENT, DIAPHRAGM SEAL.

PHASE 5

Thomas G. Horgan

REGISTERED PROFESSIONAL ENGINEER
THOMAS G. HORGAN
No. 24772
CIVIL
STATE OF CALIFORNIA

MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM
PRIME:
DAVID EVANS AND ASSOCIATES, INC.
BELLEVUE, WA

SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO. C7.64

TITLE OF SHEET
SOUTH ENTRANCE
OWTS PLANS
GENERAL NOTES
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO. 104
127662
PMIS/PKG NO. 206431
SHEET 116 OF 291

1 FOUR MILE ROAD SITE SOILS INVESTIGATION PLAN

LEGEND

TP-25 TEST PIT DESIGNATION AND APPROXIMATE LOCATION

P-24 PERCOLATION TEST DESIGNATION AND APPROXIMATE LOCATION

B-14 BORING/PIEZOMETER DESIGNATION AND APPROXIMATE LOCATION

INFO ON THIS SHEET IS TAKEN FROM THE SHANNON & WILSON, INC. "GEOTECHNICAL DESIGN REPORT, YOSEMITE NATIONAL PARK, MARIPOSA GROVE RESTORATION," S&W PROJECT NO. 23-1-01328-004_R1, AND THE NOVEMBER 5, 2015, DATED ADDENDUM NO. 1. REFER TO THE ENTIRE REPORT FOR COMPREHENSIVE INFORMATION AND FOR REPORT LIMITATIONS.

50 0 50 100
SCALE OF FEET

PHASE 5

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA
MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: DAVID EVANS AND ASSOCIATES, INC. BELLEVUE, WA	TGH	C7.70	SOUTH ENTRANCE OWTS PLANS SOILS INFO	104 127662
SUBCONTRACTOR LAUGENOUR AND MEIKLE WOODLAND, CA	CADD: MSW		YOSEMITE NATIONAL PARK MARIPOSA GROVE RESTORATION	PMIS/PKG NO. 206431
	TECH. REVIEW: TGH			SHEET 117 OF 291
	DATE: 2/26/2016			

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TABLE 1
SUMMARY OF LABORATORY AND FIELD RESULTS

Boring	Depth (feet)	USCS				USDA				Natural Water Content (%)	Percolation Rate (min/in)	Percolation Hole Depth (feet)	Root Depth (feet)	Comments	Color	Boring
		Symbol & Classification	% Gravel	% Sand	% Fines	Classification	% Sand	% Silt	% Clay							
TP-23	3.5	ML-Sandy Silt				Sandy Loam					8	3.5	7			TP-23
	0.5 – 4	ML-Sandy Silt													Brown to Red Brown	
	4 – 7.5	ML-Sandy Silt													Light Gray-Brown	
TP-24	0.5 – 3	GM-Silty Gravel											3	Some Cobbles	Brown to Red Brown	TP-24
	3 – 7.5	SC-Clayey Sand												Trace Gravel	Light Gray-Brown	
TP-25	5.5	SM-Silty Sand				Sandy Loam	61.9	26	12.1	9	4	2				TP-25
	0.5 – 3	ML-Sandy Silt											3		Brown to Red Brown	
	3 – 8	SM-Silty Sand													Light Gray-Brown	
TP-26	2.5	SM-Silty Sand				Sandy Loam	64.2	29.4	6.3	10.7	60	3.5				TP-26
	0.5 – 6	SM-Silty Sand											3		Brown to Red Brown	
	6 – 7.5	ML-Sandy Silt													Light Gray-Brown	
TP-28	7	SM-Silty Sand				Sandy Loam	56.6	25.6	17.8	9.6						TP-28
TP-29	6.3	SM-Silty Sand				Sandy Loam	55.2	28.5	16.2	9.6						TP-29
TP-31	3.5	SM-Silty Sand				Sandy Loam	63.6	24.7	11.7	8.8						TP-31
TP-32	4.5	SM-Silty Sand				Sandy Loam	57.1	25.3	17.5	11.7						TP-32
	0.5 – 8	SM-Silty Sand				Sandy Loam							3	Cobbles at 3', Trace Gravel	Brown to Red Brown	
TP-35	2	ML-Sandy Silt				Sandy Loam					4	2	2			TP-35
	0.5 – 6	ML-Sandy Silt												2" – 4" Gravel, Cobbles	Brown to Red Brown	
	6 – 8	SM-Silty Sand													Light Gray-Brown	
TP-36	4.5	ML-Sandy Silt	0	46.5	53.5	Sandy Clay Loam	48.7	27.4	23.6	18.6	308	5	7			TP-36
	0.5 – 4	ML-Sandy Silt													Red-Brown to Dark Red-Brown	
	4 – 7	ML-Sandy Silt													Red-Yellow	
	7 – 9.5	SM-Silty Sand													Light Red	
TP-37	2.5	SM-Silty Sand	0.3	68.6	31.1	--	--	--	--	8	2.2	3.3	8.5			TP-37
	1 – 8	SM-Silty Sand												Cobbles at 4'	Yellow-Brown	
	8 – 10	SM-Silty Sand													Light Red to Yellow	
TP-38	4.5	SM-Silty Sand	0	70.9	29.1	--	--	--	--	7.7	2.9	4.4	8			TP-38
	1.5 – 7.5	SM-Silty Sand													Red-Yellow	
	7.5 – 9.5	SM-Silty Sand													Light Red	
TP-39	3.5	SM-Silty Sand	0	73.2	26.8	--	--	--	--	6.6	8.3	4.3	8			TP-39
	1 – 8	SM-Silty Sand													Red-Yellow	
	8 – 9.5	SM-Silty Sand													Light Red	
TP-40	4.5	SM-Silty Sand	0	53	47	Sandy Loam	55.1	32.7	12.1	18.4	75	3.6	8			TP-40
	1 – 3.5	ML-Sandy Silt													Red-Brown	
	3.5 – 9	SM-Silty Sand													Yellow-Brown	
	9 – 10	SM-Silty Sand													Light Red	
TP-41	3.5	SM-Silty Sand	0	65.3	34.7	--	--	--	--	11	5.6	4.6	6			TP-41
	0.5 – 3	SM-Silty Sand													Red-Brown	
	3 – 7	SM-Silty Sand													Red-Yellow	
	7 – 10	SM-Silty Sand												Cobbles at 9'	Red-Yellow	
TP-42	4.5	SM-Silty Sand	0	52	48	Sandy Loam	54.9	32.2	11.9	9.4	66.7	2.5	7.5			TP-42
	0.5 – 4	SM-Silty Sand												Some Rocks to 3'	Red-Yellow	
	4 – 8	SM-Silty Sand													Red-Yellow	
	8 – 9.5	SM-Silty Sand													Red-Yellow	
TP-43	2.5	SM-Silty Sand	0.1	60.7	39.2	Sandy Loam	64.5	26	9.5	8.1	33.3	3	7			TP-43
	0.5 – 7	SM-Silty Sand													Yellow-Brown	
	7 – 10	SM-Silty Sand												Some Cobbles	Red-Yellow	
TP-44	2.8	SM-Silty Sand	0	60.1	39.9	Sandy Loam	64.7	25.7	8.8	9.6	--	--	5			TP-44
	4.5	SM-Silty Sand	0	58.6	41.4	Sandy Loam	61.4	29.6	9	14.5	18.5	4.6				
	0.5 – 4	SM-Silty Sand													Yellow-Brown	
	4 – 10	SM-Silty Sand													Yellow-Brown	
TP-45	4.5	SM-Silty Sand	0	57.5	42.5	Sandy Loam	61.4	26.9	11.5	7.2	12.5	3.1	7.5			TP-45
	8.5	SM-Silty Sand	0	66.6	33.4	--	--	--	--	6.4	--	--				
	0.5 – 4	SM-Silty Sand													Red-Brown	
	4 – 8	SM-Silty Sand													Red-Yellow	
	8 – 10	SM-Silty Sand													Red-Yellow	

TABLE 1
SUMMARY OF LABORATORY AND FIELD RESULTS

Boring	Depth (feet)	USCS				USDA				Natural Water Content (%)	Percolation Rate (min/in)	Percolation Hole Depth (feet)	Root Depth (feet)	Comments	Color	Boring
		Symbol & Classification	% Gravel	% Sand	% Fines	Classification	% Sand	% Silt	% Clay							
TP-46	4.5	SM-Silty Sand	0	52.1	47.9		54.4	33.2	11.3	8.3	54.2	3.5	8			TP-46
	1 – 4	SM-Silty Sand													Yellow-Brown	
	4 – 8	SM-Silty Sand													Yellow-Brown	
	8 – 9	SM-Silty Sand													Red-Brown	
TP-47	5.5	SM-Silty Sand	0.4	61.7	37.9	Sandy Loam	64.8	24.8	10.4	8.3	0.6	3.4	7			TP-47
	0.5 – 5	SM-Silty Sand												Rocks to 3'	Yellow-Brown	
	5 – 7	SM-Silty Sand													Yellow	
	7 – 10	SM-Silty Sand													Light Red	
TP-48	2.5	SM-Silty Sand	0.4	65.8	33.8	Sandy Loam	69.2	19	11.8	7.9	62.5	4.2	7			TP-48
	5.8	SM-Silty Sand	0	77	23	--	--	--	--	5	--	--				
	0.5 – 5	SM-Silty Sand												Boulder at 3'	Yellow-Brown	
	5 – 8	SM-Silty Sand												Some Cobbles, Very Hard Digging at 8'	Light Red to Yellow	
TP-49	3.5	SM-Silty Sand	0.6	63.6	35.8	--	--	--	--	8.7	25	3.2	7			TP-49
	0.5 – 7	SM-Silty Sand													Red-Yellow	
	7 – 9	SM-Silty Sand													Red-Yellow	
	9 – 10	SM-Silty Sand													Light Red	
TP-50	0.5 – 5	SM-Silty Sand											4	Cobbles	Red-Brown	TP-50
	5 – 7	SM-Silty Sand												Cobbles	Light Red	
	7													Very Hard Digging		
TP-51	7.5	SM-Silty Sand	0	69.5	30.5	--	--	--	--	7.7	13.9	3.3	8			TP-51
	0.5 – 7	SM-Silty Sand													Red-Yellow	
	7 – 9.3	SM-Silty Sand												Boulder at 8'	Yellow	
TP-52	2.5	SM-Silty Sand	0	71.7	28.3	--	--	--	--	6.3	16.7	4	7			TP-52
	0.5 – 7	SM-Silty Sand													Yellow-Brown	
	7 – 10	SM-Silty Sand													Red-Yellow	
TP-53	2.5	SM-Silty Sand	0	53.3	46.7	Sandy Loam	57.1	28.2	14.2	8.5	16	3	7			TP-53
	0.5 – 7	SM-Silty Sand													Yellow-Brown	
	7 – 9	SM-Silty Sand												Boulder at 7.5'	Red-Yellow	
TP-54	5.5	SM-Silty Sand	0	53.7	46.3	Sandy Loam	57.8	26.1	16.1	9.1	52.8	4	7			TP-54
	0.5 – 5	SM-Silty Sand													Yellow-Brown	
	5 – 9.5	SM-Silty Sand												Very Hard Digging at 9.5'	Red-Yellow	
TP-55	6.5	SM-Silty Sand	0.1	58	41.9	Sandy Loam	62	24.1	13.9	8.8	5.7	3.4	9			TP-55
	0.5 – 6	SM-Silty Sand												Hard Digging	Red-Brown	
	6 – 8.5	SM-Silty Sand													Yellow	
	8.5 – 10	SM-Silty Sand													Light Red to Yellow	
TP-56	0.5 – 4	SM-Silty Sand											4	Boulders. Hard Digging.	Red-Yellow	TP-56
TP-57														Similar to TP-56		TP-57
TP-58	3.5	SM-Silty Sand	0.6	67.3	32.1	--	--	--	--	8	8.3	4.5	7			TP-58
	6.5	SM-Silty Sand	0	63.5	36.5	--	--	--	--	10.1	--	--				
	0.5 – 5	SM-Silty Sand												Some Cobbles to 4'	Yellow-Brown	
	5 – 8	SM-Silty Sand													Yellow	
	8 – 9	SM-Silty Sand												Hard Digging	Light Red	
TP-59													6	Rocky. Hard Digging.		TP-59
TP-60	3.8	SM-Silty Sand	0.5	59.2	40.3	Sandy Loam	64.1	24.9	11	16.8	--	--	5			TP-60
	0.5 – 3	SM-Silty Sand													Red-Brown	
	3 – 5	SM-Silty Sand													Red-Brown	
	5 – 8	SM-Silty Sand													Red-Yellow	
TP-61	4	SM-Silty Sand	0.3	60.8	38.9	Sandy Loam	64.6	21.9	13.6	17.3	--	--	6	Some Cobbles to 6'		TP-61
P-21		SM-Silty Sand				Sandy Loam					60	3.5		Perc Test		P-21
P-22		ML-Sandy Silt				Sandy Loam					4	2		Perc Test		P-22
P-23		ML-Sandy Silt				Sandy Loam					8	3.5		Perc Test		P-23
P-24		ML-Sandy Silt				Sandy Loam					4	2		Perc Test		P-24

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STATE OF CALIFORNIA

MARCH 14, 2016

Mark	Sheet	REVISION	Date	Initial

A/E FIRM
PRIME:
DAVID EVANS AND ASSOCIATES, INC.
BELLEVUE, WA

SUBCONTRACTOR
LAUGENOUR AND MEIKLE
WOODLAND, CA

DESIGNED: TGH
CADD: MSW
TECH. REVIEW: TGH
DATE: 2/26/2016

SUB SHEET NO.
C7.71

PHASE 5
TITLE OF SHEET
SOUTH ENTRANCE
OWTS PLANS
SOILS INFO
YOSEMITE NATIONAL PARK
MARIPOSA GROVE RESTORATION

DRAWING NO.
104
127662

PMIS/PKG NO.
206431

SHEET
118 OF 291

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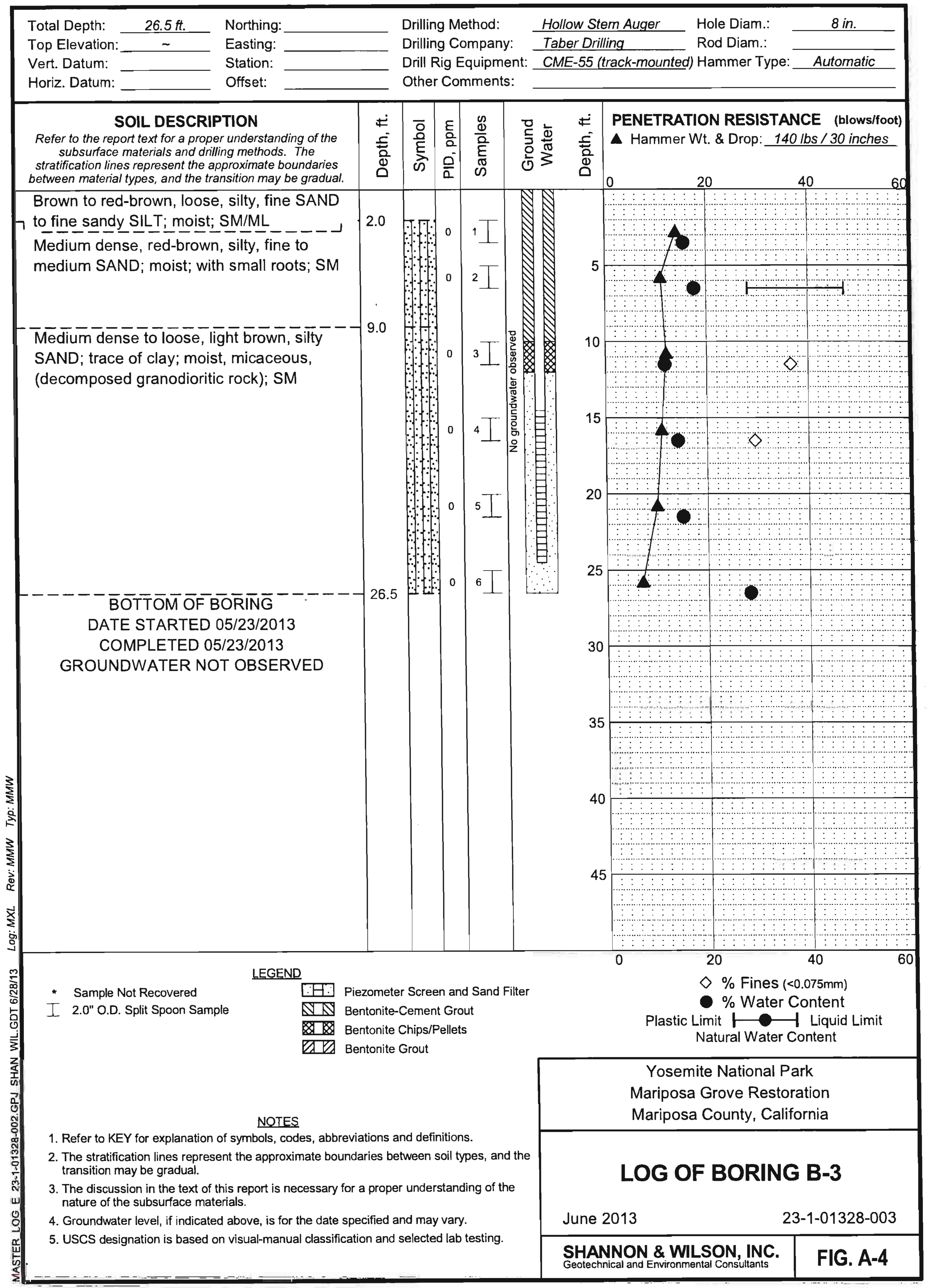

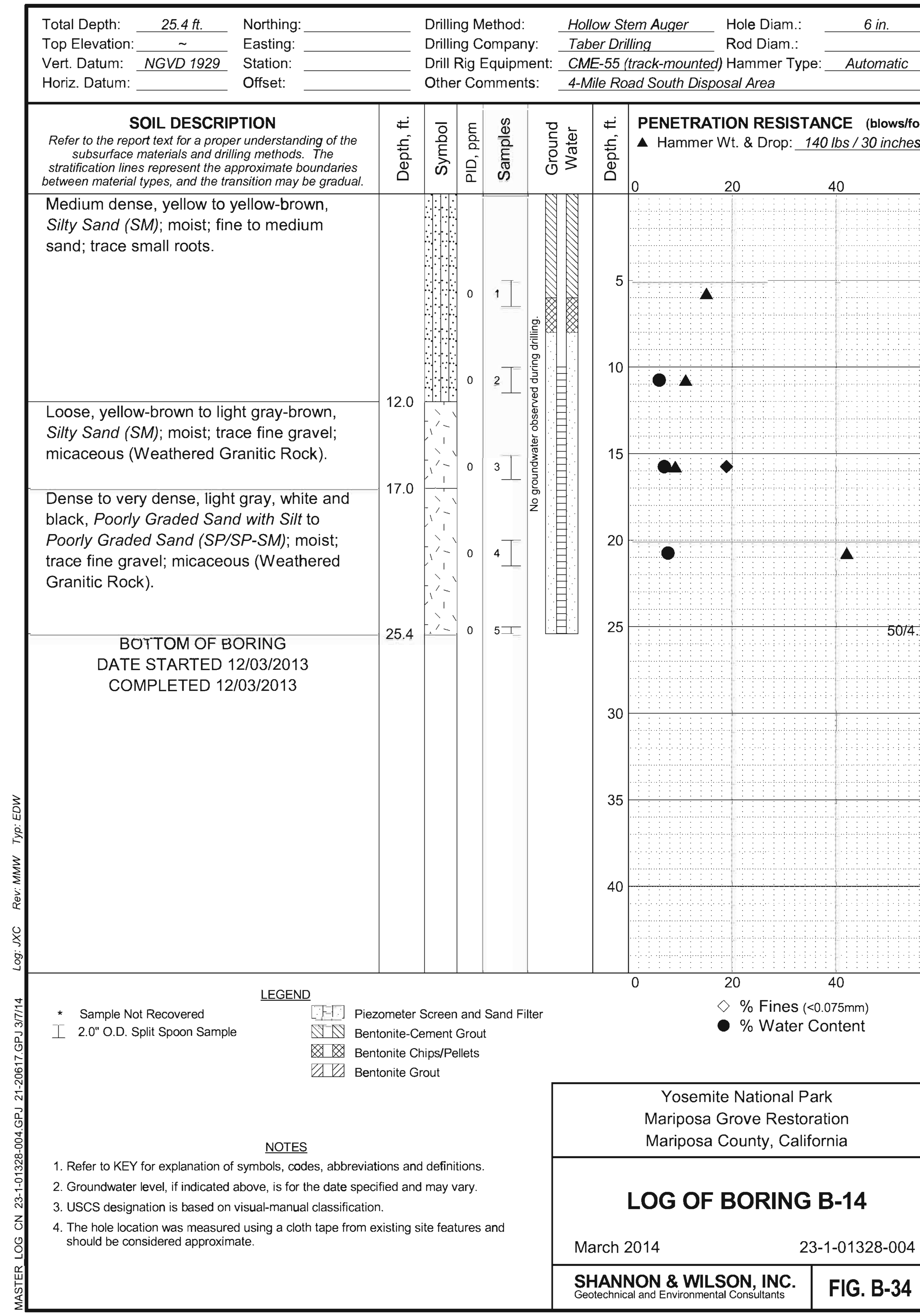

TABLE 2: GROUNDWATER MEASUREMENTS AT FOUR MILE ROAD

BORING	SAND PACK INTERVAL (FT)	DEPTH OF BORING OF WELL (FT)	TRANSDUCER DEPTH/ BOTTOM OF WELL (FT)	GROUNDWATER DEPTH (BELOW GROUND SURFACE) (FT) AT TIME OF DRILLING	NOV-22 (2013)	JAN-14 (2014)	MAY-8 (2014)	OCT-28 (2014)	FEB-10 (2015)	JUNE-18 (2015)	DATE OF DRILLING COMPLETION
B-3	12 TO 26.5	26.5	24.5	NGWO	NGWO	NGWO	NGWO	NGWO	NGWO	NGWO	05/23/2013
B-14	8 TO 25	25.4	25.4	NGWO	NA	NGWO	NGWO	NGWO	NGWO	NGWO	12/03/2013

NA = NOT APPLICABLE AT TIME (PIEZOMETER NOT YET INSTALLED)   NGWO = NO GROUND WATER OBSERVED

INFO ON THIS SHEET IS TAKEN FROM THE SHANNON & WILSON, INC. "GEOTECHNICAL DESIGN REPORT, YOSEMITE NATIONAL PARK, MARIPOSA GROVE RESTORATION," S&W PROJECT NO. 23-1-01328-004_R1, AND THE NOVEMBER 5, 2015, DATED ADDENDUM NO. 1. REFER TO THE ENTIRE REPORT FOR COMPREHENSIVE INFORMATION AND FOR REPORT LIMITATIONS.

PHASE 5

Thomas G. Horgan
REGISTERED PROFESSIONAL ENGINEER THOMAS G. HORGAN No. 24772 CIVIL STATE OF CALIFORNIA
MARCH 14, 2016

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SUBCONTRACTOR LAUGENOUR AND MEIKLE WOODLAND, CA

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SUB SHEET NO. C7.72

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MARIPOSA GROVE RESTORATION

DRAWING NO. 104 / 127662
PMIS/PKG NO. 206431
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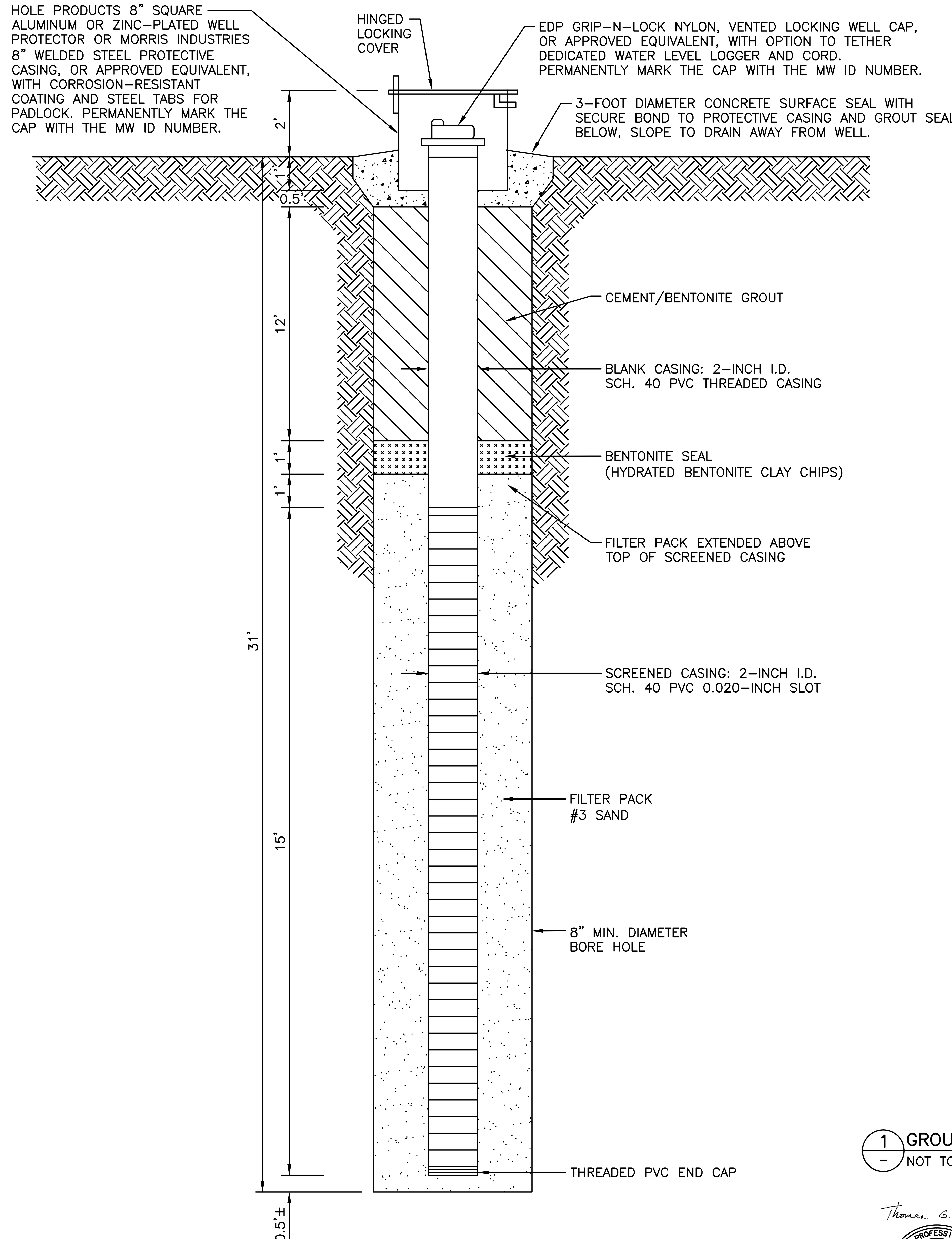

1 GROUNDWATER MONITORING WELLS
– NOT TO SCALE

MONITORING WELL CONSTRUCTION NOTES

A. REGULATIONS:

CONSTRUCTION OF THE NEW MONITORING WELLS THAT ARE TO BE LOCATED NEAR THE PROPOSED LEACHFIELD WILL BE IN ACCORD WITH THE STATE OF CALIFORNIA DEPARTMENT OF WATER RESOURCES (DWR) WELL STANDARDS (BULLETIN 74-90) FOR MONITORING WELLS. THIS WORK WILL ALSO BE REGULATED BY THE MARIPOSA COUNTY ENVIRONMENTAL HEALTH SERVICES (PHONE 209-966-2220). THE COUNTY WILL NOT REQUIRE A MONITORING WELL WORKPLAN OR WELL CONSTRUCTION PERMIT AS THESE COUNTY-APPROVED OWTS PLANS INCLUDE MONITORING WELL REQUIREMENTS. THESE WELLS WILL BE USED FOR MONITORING FLUCTUATIONS IN GROUNDWATER LEVELS; NO GROUNDWATER QUALITY MONITORING IS BEING REQUIRED.

B. DRILLING:

A C-57 LICENSED DRILLING CONTRACTOR WILL CONSTRUCT THE WELLS. THE BORINGS WILL BE DRILLED USING A DRILL RIG EQUIPPED WITH HOLLOW STEM AUGERS. WHILE DRILLING, A GEOLOGIST WILL OBSERVE THE WORK, COMPILE A RECORD OF EVENTS AS THEY OCCUR, AND CLASSIFY THE SOIL AND LOG THE BOREHOLES. SOIL SAMPLES WILL BE OBSERVED FOR LITHOLOGIC CLASSIFICATION. THE DEPTH TO FIRST ENCOUNTERED GROUNDWATER, ALONG WITH THE DATE AND TIME OBSERVED, WILL BE RECORDED ON THE DRILLING LOG. SUBSEQUENT GROUNDWATER LEVEL MEASUREMENTS WILL BE MADE FOLLOWING A PERIOD OF TIME NOT LESS THAN 15 MINUTES AFTER FIRST ENCOUNTER, AND RECORDED ON THE LOG.

WATER FROM DRILLING WILL BE DISCHARGED TO THE GROUND SURFACE AFTER THE DRILL CUTTINGS ARE FILTERED OUT TO THE SATISFACTION OF THE COR. CONTRACTOR TO LEGALLY DISPOSE OF CUTTINGS OUTSIDE OF THE PARK.

DURING DRILLING, SOIL SAMPLES WILL BE COLLECTED AT DEPTHS OF EVERY 5 FEET FOR LOGGING PURPOSES ONLY. SAMPLES WILL BE COLLECTED THROUGH THE HOLLOW STEM AUGER VIA A STANDARD PENETRATION TEST (ASTM D1586). ALL SAMPLING EQUIPMENT WILL BE PROPERLY CLEANED AND RINSED PRIOR TO USE. THE GEOLOGIST WILL RECORD SAMPLE DEPTHS AND OTHER INFORMATION ACQUIRED DURING THE DRILLING, INCLUDING A DESCRIPTION OF THE SOIL AND ITS UNIFIED SOIL CLASSIFICATION SYSTEM (USCS) SYMBOL.

SOIL SAMPLES WILL NOT BE RETAINED FOR LABORATORY ANALYSIS.

C. WELL INSTALLATION:

THE BORINGS WILL BE CONVERTED TO GROUNDWATER MONITORING WELLS UPON COMPLETION OF THE DRILLING. THE DEPTH TO GROUNDWATER WILL BE MEASURED AND RECORDED IMMEDIATELY PRIOR TO INSTALLING THE WELLS. EACH MONITORING WELL WILL BE CONSTRUCTED OF ASTM F480 COMPLIANT, 2-INCH INSIDE DIAMETER, SCHEDULE 40 PVC CASING WITH FLUSH-THREADED JOINTS. 0.020-INCH FACTORY MACHINE SLOTTED WELL SCREEN WILL BE USED.

WHEN THE SCREEN IS HANGING FREE AND POSITIONED AT THE PROPER DEPTH, THE MATERIALS THAT FILL THE ANNULAR SPACE SURROUNDING THE CASING WILL BE ADDED THROUGH THE OPEN BOREHOLE. A FILTER PACK OF SAND WILL BE PLACED AROUND THE SCREEN TO APPROXIMATELY 1 FOOT ABOVE THE TOP OF THE SCREEN SLOTS. FOLLOWING PLACEMENT OF THIS SAND, THE WELL WILL BE SURGED TO ASSURE THAT THE SAND IS SETTLED AND THAT THERE ARE NO VOIDS IN THE SAND COLUMN.

A 1-FOOT THICK LAYER OF HYDRATED BENTONITE CLAY CHIPS WILL BE PLACED ABOVE THE SAND PACK. AFTER WAITING AT LEAST 45 MINUTES FOR THE BENTONITE TO HYDRATE, GROUT WILL BE PLACED DOWN-HOLE, COMMENCING AT THE TOP OF THE FILTER PACK. THE REMAINING ANNULAR SPACE OF THE BOREHOLE WILL BE SEALED WITH A BENTONITE-CEMENT GROUT MIX TO PREVENT SURFACE WATER INFILTRATION INTO THE WELL. THIS BENTONITE-CEMENT GROUT WILL BE PLACED FROM THE TOP OF THE BENTONITE LAYER UPWARD TO WITHIN 1.5 FEET OF THE GROUND SURFACE, AND CONSIST OF 5 TO 7 GALLONS OF POTABLE WATER AND 5 POUNDS NON-BENEFICIATED BENTONITE TO EACH 94-POUND SACK OF TYPE I-II PORTLAND CEMENT.

ANY VEGETATION OR SURFACE COVER DISTURBED DURING INSTALLATION WILL BE RESTORED TO PREVIOUS CONDITION BY THE CONTRACTOR.

D. DEVELOPMENT:

AFTER INSTALLATION, THE WELLS SHALL BE DEVELOPED TO REMOVE SEDIMENT-LADEN WATER BY SURGING AND BAILING. FIRST, THE SCREENED INTERVAL OF THE WELLS SHALL BE SURGED WITH A SURGE BLOCK. SURGING DRAWS SEDIMENT INTO THE WELL FROM THE SAND PACK. THEN, THE SEDIMENT LADEN WATER SHALL BE PURGED FROM THE WELLS USING A PUMP OR TEFLON OR NEW PVC BAILERS. DEVELOPMENT SHALL CONTINUE UNTIL AT LEAST 5 WELL VOLUMES ARE REMOVED FROM THE WELL. DEVELOPMENT WATER AND SEDIMENT SHALL BE DISCHARGED TO THE GROUND SURFACE.

FOLLOWING SURGING, THE DEPTH TO TOP OF SAND WILL BE MEASURED AND ENOUGH SAND ADDED TO AGAIN BE APPROXIMATELY 1-FOOT ABOVE THE TOP OF THE SCREEN SLOTS.

E. TRANSDUCER INSTALLATION:

IN DECEMBER 2013 SHANNON & WILSON (S&W), GEOTECHNICAL AND ENVIRONMENTAL CONSULTANTS WORKING FOR THE PARK, INSTALLED SOLINST LEVELOGGER TRANSDUCER DEVICES IN PIEZOMETERS B-3 AND B-14, AND A BAROLOGGER IN PIEZOMETER B-3. THESE INSTRUMENTS WERE REMOVED BY S&W IN 2015, AND THEN REINSTALLED BY THE PARK IN 2016.

THE LEVELOGGER IN PIEZOMETER B-14 SHALL BE REMOVED BY THE PARK AND THE DATA DOWNLOADED BY THE PARK PRIOR TO PIEZOMETER DESTRUCTION (SEE GENERAL NOTE 2/C7.60).

EACH NEW MONITORING WELL SHALL BE FITTED WITH A NEW TRANSDUCER SUITABLE FOR MONITORING SEASONAL HIGH GROUNDWATER LEVELS. NOTE – FOR EXTENDED PERIODS THERE MAY BE NO GROUNDWATER SHALLOW ENOUGH TO BE MONITORED BY THE MONITORING WELLS. EACH TRANSDUCER IS TO BE CONNECTED TO THE PLC IN THE THE DOSING PUMP CONTROL PANEL.

EACH TRANSDUCER SHALL BE A PMC SUBMERSIBLE MODEL VL4513, OR APPROVED EQUIVALENT, WITH AN SP10 SURGE PROTECTOR AND MP11 VENT SPHERE INSTALLED IN A TE-11-A ENCLOSURE UNIT, PER MFR'S RECOMMENDATIONS. COORDINATE WITH ENGINEER AND COR. CONNECT TRANSDUCERS TO THE PCP WITH 1" SCH. 40 PVC CONDUIT AND THE MFR'S RECOMMENDED CABLING.

INSTALL TRANSDUCERS PER THE MANUFACTURER'S RECOMMENDATIONS, INCLUDING CALIBRATION AND VERIFICATION WITH MANUALLY MEASURED WATER LEVELS IF GROUNDWATER IS PRESENT IN THE WELL.

F. WELL INSTALLATION REPORT:

UPON COMPLETION OF THE FIELD WORK, A REPORT OF INSTALLATION WILL BE PREPARED BY THE CONTRACTOR AND FURNISHED TO THE COR. THE REPORT WILL FOLLOW THE GUIDELINES CONTAINED IN THE CALIFORNIA WELL STANDARDS. AT A MINIMUM THE REPORT WILL INCLUDE:

- COMPLETION AND FILING OF THE REQUIRED DWR REPORTING FORMS
- GENERAL INFORMATION
- DRILLING DETAILS
- WELL CONSTRUCTION DETAILS
- WELL DEVELOPMENT INFORMATION
- TRANSDUCER INSTALLATION & CALIBRATION INFORMATION

G. EXISTING GROUNDWATER LEVEL INFO:

SEE TABLE 2/C7.72 FOR RECORD OF EXISTING PIEZOMETER GROUNDWATER LEVEL DATA.

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